KENYATTA UNIVERSITY
INSTITUTE OF OPEN LEARNING

SZL 201
INVERTEBRATE ZOOLOGY

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MODULE ON INVERTEBRATE ZOOLOGY

Introduction to the module

The purpose of this module is to direct the attention of the beginning students to the profusion of invertebrate wildlife that surrounds them.

You will find this information to have value in other numerous courses, e.g.Parasitology, Protozoology, Biochemistry, Entomology,Ecology,Palaentology and even Molecular genetics etc because these subdisciplines allow little time for the study of these myriads of animals without backbones.

The animal world in terms either of numbers and individuals or a number of species is mostly invertebrates- **lacking a backbone**. There are probably nearly 2 million living species of animals, approximately half of which have been formally described – and some 96% of these are "invertebrates". This module gives a survey of this diversity of animals. These invertebrates in their huge numbers, but often invisible lives exert a profound influence on humans.They keep the planet healthy. They are part of the great tapestry of life.Their beneficial effects both to our ancestors and to our busy modern technological living include:

1) pollination
2) processes taking place in agricultural soil
3) scavengering
4) biological control of pests
5) part of the complex food webs of higher animals
6) purifying both our air and our water
7) Many are beautiful and of great scientific interest , helping us gain important insights into how the world works.

The invertebrates in being multicellular organisms, represent several steps along the road to the organizational complexity that makes us what we are.

Their detrimental effects which are of great consequences to us all include:

1) parasitism
2) disease carrying
3) pests of food

In the lectures that follow, the invertebrates are surveyed phylum–by-phylum.Within this framework the representative classes and genera are selected.

PHYLA THAT MAKE UP INVERTEBRATES

Protozoa – unicellular animals; comprised of 8 phyla

1. Rhizopoda
2. Foraminifera (Granuloneticulosa)
3. Acrasiomycota
4. Myxomycota
5. Zoomastigina
6. Apicomplexa
7. Ciliophora
8. Oomycota

Metazoa - multicellular animals; comprised of 34 phyla

- **Simple metazoans**

9. Placozoa
10. Porifera

Radiate animals

11. Cnidaria
12. Ctenophora

Acoelomates

13. Mesozoa
14. Platylelminthes
15. Gnathostomukda
16. Nemertea
17. Gastrotricha

Pseudocoelomates.

18. Kinorhyncha
19. Rotifera
20. Nematomorpha
21. Acanthocephala
22. Nematoda
23. Entoprocta

Coelomates

Lower protostomes

24. Priapulida

25. Echiura
26. Sipuncula
27. Pogonophora
28. Onychophora
29. Tardigrada
30. Pentastomida

Higher Protostomes

31 Annelida
32.Arthropoda
33.Mollusca

Lophophorates

34.Phoronida
35.Bryozoa (Ectoprocta)
36.Branchiopoda

Lower Deuterostomes

37.Chaetognatha
38.Hemichordata

Higher Deuterostomes

39.Echinodermata
40.Urochordata
41.Cephalochordata

Corset bearers

42.Loricifera (Noum)

LESSON ONE

Structure

Introduction to invertebrates

Objectives

The evolution of animals
Salient features of invertebrates
Types of classification systems
Definition of terms
Further reading
Questions

Welcome to Lesson One. This is the first lesson of invertebrate zoology. In this lesson you will study the evolution of animals and to give important characteristic features of invertebrates. There are over a million described species of animals.

The first animals appeared in Precambrian sea and it is from this that evolution led to a vast diversity of species. The animal kingdom displays a seemingly endless variety of shapes, colours and sizes ranging from simple body plans to the most complex forms of life on earth .Several types of classification have been adopted to keep track of the enormous numbers of organisms on earth. Depending on which groups are lumped together in taxa (taxonomic groups), the living members of animal the animal kingdom comprise of up to approximately 42 phyla, distinguished mainly by their body plans.

One subphylum of the phylum Chordata (Vertebrata) contains all the vertebrates, animals with backbones. The rest of the chordates, and the members of all other animal phyla, are invertebrates, animals without backbones. Vertebrates comprise only about 5% of the known animal species while invertebrates comprise the other 95% of which the vast majority are insects.

Objectives

After reading this lesson you should be able to:

1) Outline how invertebrates evolved from the sea/oceans to colonize new habitats (fresh water, land)
2) Explain the salient features of invertebrates
3) List the five kingdoms of organisms used in this lesson and state the criteria used to assign species to each kingdom

Evolution of animals

Animals evolved from the sea/ocean. Competition for food or predation pressure eventually drove many animals into new habitats. Some moved from sea bays into rivers as adaptations to the hypotonic environment of fresh water environment evolved. Other animals, living on the shore adapted to even higher and drier sites, and became terrestrial (land dwelling). The sea is a more stable and hospitable environment for life than either fresh water or land. For instance, most marine invertebrates have no problem of gaining or losing water by osmosis because the total salt concentration of the sea is very similar to that of their cells. Although light temperature varies largely with depth, the temperature in any one area of the sea changes slowly within fairly narrow limits. Furthermore, algae and protozoa thrive in areas of the sea with abundant nutrients, providing a constantly renewed source of food.

In contrast, fresh water makes up less than 5% of the earth's surface water and usually contains lower concentration of nutrients than the sea water. It supports less life and many fewer species. Because fresh water is hypnotic to living cells, a fresh water animal must constantly expend energy to retain its salts and expel the excess water that enters the body through osmosis. Land is an even more difficult environment because water is often in short supply, and it readily evaporates into dry air, making desiccation a constant danger for many land animals.

Despite its advantages, life in the sea poses some difficulties. Marine animals depend on algae as the ultimate source of food, and the algae exist only near the surface, where there is enough light for photosynthesis. However, the ocean surface is constantly tossing about. Fish, which are vertebrates, can swim at will despite water movements, but few invertebrates are strong enough swimmers to do this.

Invertebrates have several types of adaptations that permit them to cope with this problem. One is to be small enough to float around as members of zooplankton, close to their food supply of other members of the plankton. The other is to live on the bottom (**benthic**) usually in water shallow enough for photosynthetic algae to live and supply food for other forms of life. A bottom living animal (bentho) may burrow in the sea floor, cling to rocks or other stable objects (substratum) using structures such as claws or suckers, or be sessile ("sitting") on any available hard surface. Most sessile animals are filter feeders. They may sit in a certain of water and trap passing food in a net of tentacles or mucus, or they may use cilia or muscles to create water currents past or through their bodies and filter out or seize any food that the current brings. A sessile organism needs protection from mobile predators. It may have active protection such as stingers or passive protection such as a thick shell, a burrow, or a coating of toxic mucus.

Sexual reproduction poses problems for sessile animals because they cannot move around and find a mate. Instead, most of them have behavioural adaptations in which all members of a species release their gametes into the water at the same time (Synchronization). This occurs in responses to environmental cues (stimuli), such as water temperature and phase of the moon. The gametes then must find one another.

Most sessile or slow-moving marine invertebrates develop from a larva that is motile, able to move around. In many species, the larva undergoes several changes of shape as it develops, and the different larval stages may have different names. In marine invertebrates, the larva is usually the dispersal stage. It swims or is swept away, often as part of the plankton, and eventually settles in a new location. Like other members of the plankton, most larvae are transparent, making them difficult to see. In addition, all have adaptations that prevent them from sinking. Some remain afloat due to possession of large surface area. For example, they may be flat and thin, or have long tentacles or flaps. Others swim using cilia or flagella.

The disadvantage of the larvae is their high mortality predators eat many and many others do not end up in a suitable habitat. This has selected for the production of vast numbers of offspring in many invertebrates. The danger of being carried into unfavourable habitats by currents is especially great in streams and rivers, and in most freshwater invertebrates the larval stage has been suppressed. Thus, marine snails produce larvae but fresh water ones do not and instead have direct development into a miniature adult.

Many sessile animals can reproduce asexually. Fission occurs by dividing into usually two parts (binary fission) or many parts (multiple fission). Budding is also common and involves production of a smaller individual, which may later become independent, or may remain attached as part of a colony. A sexual reproduction avoids the problems of synchronizing gamete release and also permits an animal that has found a good spot/habitat to populate the area with a clone of individuals containing its own genes.

1.4. Animal structure.

Animals generally have oogamous sexual reproduction, in which large eggs are fertilized by tiny sperm. The zygote develops into a solid ball of cells (morula) and then a hollow ball (blastula) containing a cavity called the blastocoel. Embryonic development, with cell differentiation and specialization is the hallmark of higher eukaryotes (plants and animals). In all but the simplest animals, cells organize themselves into tissue. Tissues are arranged into organs, and organs into organ systems which form the organism. The multicellular architecture of animals is different from that of plants, mainly because animal’s cells do not have rigid cells centred to their neighbours. The absence of cell walls permits more intimate contact between cell surfaces, greater flexibility of cells, tissues and organs, and movement of cells within the body. The absence of stiff cell walls also permitted the evolution of muscle tissue. All animals are motile in at least one stage of the life cycle, and animals with muscles can produce stronger, more rapid movements than organisms that depend solely on cilia and flagella. The development of muscles led to the evolution of a wide range of animal structures and behaviours used in hunting, capturing and eating food, movement, finding and courting mates and reproduction. Muscle tissue also speeds up internal functions such as digestion, circulation and gaseous exchange.

1.5. Salient features of Invertebrates.

1.5.1. Body symmetry.

Animals tend to look symmetrical (having both or more sides exactly alike). The concept of symmetry allows for the division of a whole body into two or more equal portions, by separation along lives or planes. Four kinds of symmetry are possible, and all are found amongst invertebrates.

a) **Asymmetry** - there are no lines/planes of regular symmetry - occurs in animals with amorphous shape (without definite shape) e.g. Rhizopod protozoa such as amoeba. (Figure).
b) **Spherical symmetry** - the body is divisible into equal halves in all directions and planes. Found in rhizopod protozoas commonly called radiolarieus e.g Acenthometra (figure).
c) **Radial symmetry** - lines of symmetry exist, but in he vertical planes only. Found in animals with a cylindrical body with parts arranged around a central axis. Any of several planes passing along this axis divides the body into mirror-image halves. Radial symmetry permits an animal to detect food or danger from any side, an advantage of sessile or passively drifting animals e.g. cridaria and Echinodrmeta (figure).
d) **Bilateral symmetry** - found in freely mobile animals with consequent development of dorsal and ventral surfaces, and of anterior and posterior ends. Only one plane of symmetry divides the animal into equal halves; includes the majority of active invertebrates. Bilateral symmetry allows the evolution of a streamlined shape and concentration of the power of muscles and appendages into producing motion in one direction e.g. arthropods. (Fig).

1.5.2. Cephalisation

Development of Bilateral symmetry in active invertebrates was concomitant with movements concentrated in a particular direction, hence the animals body became polarized resulting with one and leading (anterior end). The concentration of sense organs, and usually the feeding apparatus, occurred at the leading end, and this is the process of cephalisation (kephale = head - hence cephalization means development of a head).

It involved the specialization of a portion of the nervous system known as the brain, and the formation of the head. The latter has sensory and nervous tissue that monitors the area the animal is entering.

1.5.3. Coelom

In most metazoan animals, the body plan is made up of a tube within a tube. The inner tube is the digestive system and the outer tube deals with the external environment. In the latter, we find sense organs and protective structures such as skin, shells, spines, and slime glands. Between the inner and outer body layers, most animals have a middle layer containing muscles and structures for internal functions such as circulation, excretion and reproductive organs. The body plan is laid from early in development. Gastrulation converts the blastula into an embryo with three germ layers (triploblastic). The inner layer, the endoderm, develops into the living that surrounds the digestive cavity. The outer layer, the ectoderm, with form external structures. The third germ layer, the nesoderm, lies between these layers (Endon = within; Ektos = outside, mesos = middle; derma = skin).

Some animals have two germ layers and are called diploblastic. Animals with three body layers can be divided into three categories. In the primitive forms, there is no space between the layers, and they are acoelomate (a= without; koilos = hollow). The most advanced animals are coelomate. They contain a coelom, a fluid -filled space that develops in the mesoderm. A coelom permits body organs to move independently and this in turn permitted the evolution of rapid locomotion in coelomates such as insects and vertebrates. Some lower invertebrates are pseudocoelomates, with a fluid filled pseudocoel between the endosperm an mesoderm. A pseudocoel is the remains of the embryonic blestocoe, whereas a coelom opens up in the mesoderm later in development.

Four main body plans of animals (Figure).

NB. Embryologically the coelom is derived either from

a) The cavitation of blocks of mesoderm: to form a schizocoel.
b) The formation of sacs of mesoderm from the archeuteron: giving rise to an enterocoel.

1.5.4. Metamerism - can be defined as repeated segmentation. Each segment (metamere) contains one pair of some or all of the organs of the body (nephridia, coelomoduets, gonads, gauglia). Metamerism appears for instance in the arrelids, arthropods, cestodes and chordates (not invertebrates). Most cestodes (tapeworms) form many identical segments that are shed terminally when the reproductive capacity is fulfilled. In anaelids, the space between the gut and body wall (coelom) is divided into repeated segments (metamenes) by partitions called septa (singular = septum). Annelids have developed septate segmentation and compartmentation of the nervous and muscular systems under the adaptive stress of locumotory and burrowing needs. Externally each segment bears setae (Seta= singular), or bristls composed of the nitrogenous polysaccharide called chitin.

1.5.5. Cleavage - the division of fertilized egg or zygote in to many cells i.e. 2, 4, 8, 16, 32, 64 etc. the embryology and development of invertebrates takes place from a variety of egg types, through a number of styles of division to form a number of larval types of juveniles. There are however, certain strong lines of similarity between groups.

a) **Egg types**. Cleavage pattern depend upon the nature, size and relative proportions of the eggs, as shown in the table below.

Types of eggs and cleavage

Egg type	Amount of yolk	cleavage	Type
Isolecithal telolecithal	Small or none Large	Holoblastic Meroblastic	Total cleavage Only animal Pole divides

Cleavage follows different patterns in different animals, and the pattern can generally be predicted from the amount of yolk in the egg. Yolk is a store of energy and nutrients for the embryo, containing lipids, carbohydrates, and proteins. In eggs with little yolk, such as those of mammals and many invertebrates, cleavage produce many cells of roughly the same size. In eggs with a lot of yolk (e.g. frogs), the cleavage furrow, formed as the cells divides, passes more slowly through the yolk at the bottom of the egg than through the clearer area above. As a result, the top (animal pole) of the morula gets a head of the bottom (vegetal pole) and many small cells are formed in the top half of the embryo while the bottom has few larger ones. Birds and reptile eggs contain so much yolk that the cleavage furrow cannot pass through the yolk at all, and the egg cleaves only in a small area on top of the yolk.

b) **Cleavage patterns** - The division of the fertilized egg follows two major patterns amongst metazoan invertebrates, both are holoblastic (total cleavage) and involve the whole egg; they are termed radial and spiral cleavage. In radial cleavage, the cleavage furrow divides the egg into radio segments like those of an orange. In spiral cleavage, the newly formed cells lie in a spiral pattern. Another radical difference exists between these two cleavage patterns. In spirally cleaving eggs, by the time 32 cells have been formed, the future position and function of each blastomese (cells formed) is fixed. This is determinate development. Radially cleaving eggs do not show such fixation of cellular fate, each cell retaining the ability to form various kinds of organs and tissue until late in development. This is indeterminate development.

Figure (   )

NB: Cleavage in insects shows another variation. The zygote nucleus divides many times before any plasma membranes are laid down between the nuclei. e.g. After 3 hours 2 development a fruit fly (*Drosophila melanogaster*) embryo contains about 5000 nuclei. Then, membranes grow in from the plasma membrane and separate all these nuclei and the cytoplasm around them into cells lying at the surface of the egg and surrounding the yolk in the centre.

1.5.6. Protostomia - Denterostomia evolutionary lines.

The coelemate animals may be divided into two evolutionary lines, largely on the basis of their embryonic development, a feature that often provides evidence about relationship. It must, however be stressed that many small phyla have intermediate characters falling between the two major projected lines, and that not all representatives lying within these two lines in fact exhibit all the characteristic features.

One line contains the protostomes, which include Annelida, Mollusca and Arthrooda. The other, called deuterostome has three major phyla: Echinodermata, Hemichordata and Chordata (including the vertebrates).

Line	**Protosomia**	**Deuterostomia**
Phyla	Annelida	Echinodermata
	Arthropoda	Hemichordata
	Mollusca	Chordata

Characteristics

1. Spiral cleavage	1. Radial cleavage
2. Blastopore forms mouth	2. Blastopore forms anua
3. Schizocaelic coelom	3. Enterocoelic coelom
4. CNS ventral	4. CNS dorsal
5. Trochophore-like larva	5. Dipleurula - like larva
6. Determinate development	6. Indeterminate development.

NB. Modern research has shown that many of the above characteristics are wrong. It has been shown that many protomes do not have spiral cleavage e.g. *D. melanogaster* has neither spiral nor radial cleavage. In humans and other vertebrates the coelom is schizocoelic and not enterocoelic and the mouth and anus arise in many positions relative to the blastopore in members of both groups.

At the molecular level, there are more similarities than differences in development between protostomes and deuterostomes. Nevertheless, protostomes and deuterostomes do form distinct evolutionary lines. Echinoderms and chordates belong to one natural group, and their embryonic development is quite similar. But the protostomes are a heterogeneous group.

1.6. Classification.

Living and fossil organisms show certain consistent features of which they may be identified and assigned to taxonomic groups or taxa of similar organisms which are distinct from other groups. The process of defining such taxonomic/systematic groups is called classification. The process of classification is based on the study of the variation of organisms and aims at establishing taxa whose members show the greatest number of common features and thus overall similarity. All living and extinct organisms are related to one another more or less closely by way of evolutionary descent/kinship. Taxonomic groups are arranged in ranks or categories of increasing exclusivity (or decreasing inclusivity) i.e they are arranged into kingdom, phylum, classes, orders, families, genera and species (sometimes with further intermediate ranks such as superclass, sub-family etc). This gives a hierarchical system of classification, with kingdom being the most inclusive category, and species the most exclusive. Taxa should be monophyletic (i.e) all members of the taxon should have descended from a common ancestor of the same or lower categorical rank. Polyphyletic taxa, whose members have evolved from two or more ancestors of the same categorical rank, are to be avoided, and should be broken down into monophyletic taxa.

The process of classification and naming of organisms is essential for their correct identification, and for storage, retrieval and communication of information concerning them now and even in future generations. The allocation of names to taxa by classification (e.g. Insecta, Protozoa etc) is nomenclature. The latter was first established in its modern form by Carl Linnaeus in the mid 18^{th} century and allocates two latin names to every organism, with the exception of viruses. The first is the generic name (or genus) and the second is the specific name (or species).

In the past, classification was based largely on a study of the phenotypic or physical appearance of the organism (phenetic classification), but nowadays many other types of evidence are used including chemcials, physiological, ultrastructural, genetic, embryological, developmental, behavioural ,ecological etc. Often this has called for the revision of the older classification.

Taxonomy, a term often used synonymously with systematic, has been defined as the scientific or theoretical study of the classification of organisms; but systematic also includes their identification, nomenclature and phylogeny (evolutionary history and relationship). Modern taxonomy or classification is divided into three main schools of thought. Underling all the three schools is the theory of evolution and accepted mechanisms of evolutionary change. The three schools differ in their basic assumptions, methods and the classification avoided at:

a) Evolutionary taxonomy or classification - organisms are grouped using both phenetic resemblance's and evolutionary/phytogenetic relationships.
b) Numerical or phenetic classification - organisms are grouped according to the degree of physical or phenetic resemblance between them. it is based on the mathematical analysis of overall similarity of a large number of taxonomic character. Phenetic criteria are used to classify organisms whose phylogeny is not known e.g. bacteria.
c) Cladism or cladistic classification - organisms are grouped according to their genetic or phylogenetic (phyletic) relationships, or the degree to which they are related to one another genetically. Cladistus emphasize that each taxa should be monophyletic, containing one ancestor and its descendants, and polyphyletic taxa should be broken up.

1.6.1. Types of classification.

i) **Two-kingdom classification system** - This is again called linnaeus system of classification and had two kingdoms, the plants (plantae) and animals (Animalia). This seemed reasonable during his days, since plants and animals were the most familiar organisms, and they were clearly very different. Plants did not move. Around, they did not 'eat' but seemed to need only water in order to grow. Animals were motile; that is, they could move from one place to another. Animals had eaten plants or each other in order to stay alive. On a microscopic level, plant cells could be seen to have cell walls, which animal cells lacked. Fungi seemed to be aberrant plants, since they had cell walls and root-like structures that absorbed food from living or dead organisms but lacked the green pigment of the other plants.

However, as more organisms were discovered the division into plant and animal kingdoms seemed to become more and more artificial, and the boundary between the tow became more and more confusing. Some organisms, such as the one-celled Euglena, seemed to fit both descriptions. Euglena has a rather stiff covering (Pellicle), not as thick as a plant cell will but certainly affording more protection than an animal's plasma membrane. Euglena also has chloroplasts and can carry out photosynthesis when exposed to light. However, it has a flagellum with which it can swim - an animal like characteristic. It can also engulf other organisms and digest them as food, just like an animal. Bacteria presented another problem since they have cell walls, but often also posses flapella that make them motile; most cannot make their own food, but some carry out photosynthesis.

Recent attempts to revise biological classification at the kingdom level have been many and varied, and there are several conflicting schemes in use.

2. Five - kingdom classification system.

This was popularized by ecologist Robert Whittaher of Cornell University in the 1970's. The first organisms on earth probably belonged to kingdom Prokaryotae (or Monera), which contains all species of organisms with prokaryotic cells. These are Archaeobacteria (bacteria inhabiting hot springs, acidic or salty environments), Eubacteria (true bacteria) and cyanobacleria (blue green algae). The other four kingdoms, are separated according to two main criteria; complexity of tissues and mode of nutrition. Kingdom protista (or protoctista) is composed of unicellular enkaryotes and of comparatively simple multicellular forms that do not qualify for one of the other three kingdoms. Kingdoms plantae, Animalia and fungus are distinguished largely by their nutrition. Members of plantae are phtosynthetic autotrophs whereas members of Animalia are ingestive heterotrophs, that is the engulf or swallow their food and digest it internally. The fungi absorptive heterotrophs, they absort organic molecules form outside their bodies directly through their exterior plasma membranes.

NB: Organisms in monera and protista kingdoms may be autorophs ("self nourishing"), synthesizing their own food from inorganic molecules with the aid of solar or chemical energy. They may also be hetrotrophs ("nourished by others"), obtaining than food from the bodies of other organisms or they may be both.

The main problem with the fine-kingdom system is that the protists (and possibly plants and monera) are polyphyletic. Breaking then up into truly monophyletic groups would result in more different kingdoms. Despite its limitations, the five-kingdom systems is widely used today, not because it is natural but because it is convenient.

3. Three kingdom classification system.

In recent years, a classification containing only three kingdoms has gained popularity. The kingdoms are Archaea, bacteria and eukarya. Archaea contains the Archaeobacteria, a group that includes bacteria adapted to life in hot springs or in extremely acidic or salty environments. Bacteria contains all other bacteria whereas eukarya contains all eukaryotes (protists, fungi, animals and plants).

This new classification resulted from the influence of cladistics and from studies on ribosomal RNA and mechanisms of protein synthesis during the 1980s. Some cladisties argue that the five kingdom system contain two main polyphyletic groups. Other biologists concluded that archaeobacteria are so different from other prokaryots and form eukaryotes that they belong to a separate kingdom. Some authorities who believe tat the five kingdom are too useful to abolish choose to treat Archaea, bacteria and eukarya as superkingdoms, or domains. Others argue that if the goal is to produce clades, there is evidence that both bacteria and eukarya are polyphyletic and there is no point in replacing five polyphyletic kingdoms with three.

Further reading

1) Ruppert, E. E. and Barnes, R.D. (1994) Invertebrate Zoology. 6th Edition.
2) Moore, J. (2001) An Introduction to the invertebrates.
3) Anderson, D.J.(2001) Invertebrate Zoology.2nd Edition
4) Barrington, E.J.W. (1979) Invertebrate structure and function. 2nd Edition.Thomas Nelson and sons Ltd

Questions

1) Describe Linnaeus contribution to taxonomy and the basis for the classification.
2) List the five kingdoms used in this module, and state the citeria used to assign species to each kingdom.
3) What is the adaptive advantage to segmentation?
4) What are the differences between the protostomes and the deuterostomes?
5) What are the basic characteristics of the coelomates?
6) What is a coelom? Why are some invertebrates described as acoelomates or pseudocoelomates?

LESSON TWO

Protozoa

Protozoa remain to this day as representatives of simple animals which play an important role in the economy of aquatic habitats. They provide the basic food material on which many higher animals ultimately depend for their existence. There are many examples of protozoans**.**In old literature protozoa was recognized as a phylum, but in the classification adopted in this module, protozoa will be divided into eight (8) phyla namely **Rhizopoda, Foraminifera, Acrasiomycota,Myxomycota, Zoomestigina, Ciliophora, Apicomplexa and Oomycota**.

Protozoa are generally motile, unicellular or symcytial, wallless heterotrophic protists. They may be freelining, predators or scavengers, ingesting other organisms or bits of organic matter, or parasitic or mutualistic symbionts.

2.2. Rhizopoda.

These are amoebas and their allies. The phylum **Rhizopoda** "root foot" contains about 200 species and the freshwater Amoeba profeus which has been observed and studied in the laboratories is one of the most thoroughly studied protist. It is easy to rear on a diet of bacteria and is large enough to be used to study amoeboid locomotion and endocytoses. It is used in experiments on nucleus -cytoplasm interaction, since its nucleus can be removed or transplanted by microsurgery to find out how the cytoplasm reacts. Although members of the genus Amoeba are naked (have no cell wall), many other rhizopods have tests, which they construct from bits of solid matter. Rhizopods use their pseudopodia for both locomotion and feeding. Rhizopods live worldwide (are cosmopolitan), in soil, in salt or freshwater, and in the bodies of animals. Two species of Entamoeba live harmlessly in the human mouth (*E. gingivalis*) and intestines, but a third species, *E. histolytica*, causes devastating amoebic dysentry. Amoebas never have cilia or flagella, nor meiosis and sex. They do form mitotic spindles, but their mitosis is unusual, with the nuclear envelope persisting during most or all of the processes.

Other examples of rhizopods include the genera Arcella, chaos, Difflugia.

Figure (   )

2.3. Foraminifera (Graniloreticulosa)

These comprise about 4000 species. Foraminiferans have slender, granular pseudopodia used more for food capture than for locomotion (reticulopodia). These pseudopods poke out through holes in a test made up of calcium carbonate or of sand grains cemented by organic secretions. Outside the test, the pseudopods may branch and join, farming a net that traps and digests prey. These features are described by the phylum's two proposed names: foraminifera (foramen = an opening; ferre = to carry) and Granuloreticulosa (granulum = little grain; reticulum = little net). Forams, as they are nicknamed, contain one to many nuclei. The larger forms may exceed as millimeter across, and one fossil species grew larger than 10 cm. Nearly all forams are marine (Ocean-dwelling). Most live in the sand or attached to other organisms, but planktonic forms occur in large numbers and provide food for many invertebrate animals. Some forams retain chloroplasts from algal prey, thereby gaining extra food photosynthesis. Most shallow water species contain symbiotic algae and respond to light by moving towards it.

The fossilized members of this group are more than the living ones and are represented by fossilized tests, which sank to the bottom after their occupants died. Millions of year remains of these for animal feran debris formed chalk rock or limstone. Fossil forams are also common in deposits of oil, which forms form the remains of ancient floating algae. These fossils provide chees to where oil is likely to be found. Some forams have never been seen to reproduce sexually, but others have both mitosis and meiosis in complex life cycles.

Figure

Other examples include the genera Fusulina, Globigerina, Nodosaria, Textularia.

2.4. Acrasiomycota.

These are commonly called cellular slime moulds and are approximately 50 species. Both members of this phylum and phylum myxomycota (Plasmodial slime moulds) combine amoeba and fungus like features. Because the fungus - like stage is more obvious, these organisms were first studied by mycologists who gave them the common name "slima moulds" and claimed they were fungi.

Cellular slime moulds spend the active part of their lives in a mobile, naked, amoemba like state engulfing organic matter and bacteria. Individual cells remain discrete or separate, but when conditions become adverse, they for fungus like reproductive structures called fruiting bodies (sporangia): Some cells form a stalk, and others become spones with cellulose cell walls. This cooperation and division of labour shows a primitive degree of multicellularity.

Acrasiomycota (akrasia = bad mixture, mykes = fungus) are called cellular slime moulds because the feeding stage is a uninucleate amoeba. These amoebas, and even the reproductive structures they form, occur widely in the soil but are so small that they are seldom observed. When their food supply is exhausted, some of the amoebas secrete a chemical that attracts others. The amoebas crawl along the chemical's concentration gradient and aggregate into a slug-like chemp, a pseudoplasmodium, which may crawl around for a while. When it stops, it forms a fruiting body topped by spones. After maturation, spores are released and germinate to form amoebas. The formation of fruiting bodies by cellular slime moulds provides a simple and useful model for the experimental study of how cells differentiate. Sexual reproduction is rare.

Several species of cellular slime moulds may occur in the same soil and form fruiting bodies in response to the same environmental cues. In some cases, different species secrete different chemical attractants and therefore congregate at different centres. However, some species appear to have the same aggregation chemical, and so all the amoebas come together in the same place. Here they sort themselves out into species specific clumps based on the compatibility of cell surface molecules.

Figure - Dictyostelium.
Other examples include the genera Acrasia, polysphondyhim.

2.5. Myxomycota
They are plasmodial slime moulds and comprise of about 550 species. Plsmodial slime moulds are placed in phylum Myxomycota (myxa=mucus). The feeding stage is a plasmodium, an amoeba - like syncytium. A plasmodium usually forms by the fusion of many small plasmodia. The plasmodium streams around in soil, wood, dung or decayed vegetation, engulfing bacteria or particles of organic food. When conditions become adverse, the plasmodium forms a fruiting body with cell walls spones are produced by meiosis. Germinating spores release haploid amoebas (myxamoeba) or cells which may develop flapella. Two compatible amoebas or flagellated (myxoflapellates) fuse and form a diploid

cell. This develops into a young plasmodium as the diploid nucleus divides but the cytoplasm does not.

Figure (*Phyparum*)

Other examples include the genera.

2.6.Phylum Zoomastigina

Commonly called zooslagellates and there are about 1500 species. The phylum Zoomastigina is a plyphyletic collection of heterotrophic protozoa with flagella, including the group that is probably ancestral to animals and possibly fungi (Choanoflagellates). Zooflagellates lack chloroplasts and contain one to many flagella. Some zooflapellates are free-living, either freshwater or marine, whereas others are symbionts or parasites. Sexual reproduction is known in only a few groups. The zooflagellate symbionts (*Triconymphs*) found in the guts of termites and wood roaches (engulf and digest the wood eaten by their insect hosts. Because the insects cannot make their own wood-digesting enzymes (cellulase), they are completely dependent on their symbionts.

Of great medical and veterinary importance are parasitic zooflagellates of the genus *Tyrispanosoma,* which live in the blood of vertebrates, mainly mammals. Human disease caused by trypanosome infection include sleeping sickness (Apuca) and chaga's disease (South and central America). Nagana is also a trypanosome disease that kills 3 million cattle a year and makes cattle farming impossible in much of sub-Saharan Africa. Both sleeping sickness and Nagana are transmitted to their mammalian hosts by bites of tse tse fly. On the other hand, chaga's disease is transmitted by blood-sucking reduviid bugs (*Triatoma*) which hide in walls and thatched roofs, and crawl out as night to feed on the people, dogs, cats and Guinea pigs. Species of *Triatoma* transmit the disease not through bite, but when the victim rubs the bug's faecel material into the eyes or mucous surfaces. Trypanosomes are especially difficult for the body to conquer because they conquer because they continuously change their cell-surface proteins to avoid being recognized by host's immune system. Other examples are *Leishmama, Glardia* and *Trichomones.*

Figure(   )

2.7. Phylum Apicomplexa

There are about 2400 species of apicomplexaus. All are parasities and most have complicated life cycles. The phylum is named for the "apical complex", a distinctive arrangement of organelles and cytoskeletal fibres at one end of he infective motile stage, used to enter host cells. The feature is thought to indicate that all apicaomplexans share a common lineage. The apicomplexan *Toxoplasma gondii* may well be the most common. This protozoan invades body cells and usually causes mild symptoms (enlarged lymph nodes) that pass unnoticed. However, it can cause serious diseases in a foetuses, newborn or a person weakened by AIDS, leading to blindness, mental retardation, or death. Toxoplasma is usually spread in the faeces of cats or in undercooked meat from infected pigs. Human malaria, caused by four species of the genes plasmodium, illustrates the complex life cycles typical of parasites. Malaria parasites require two different hosts to complete the life cycle: humans and female mosquitoes of the gemes Anopheles, which is the vector. About 270 million people worldwide are infected at any one time, and approximately 2.5 million die each year, mainly young children, plasmodium is yet another parasite that is hard for the body to combat because of rapid mutation of cell surface proteins. Other examples include *Eimeria, Babesia, Anaplasma, Theileria*, *Monocystis Gaegarina* etc.

Figure (   )

2.8. Phylum Cillophora or Ciliata

They comprise of about 8,000 species. All protests with cilia belong to one highly successful lineage, placed in phylum ciliophora, "eye-lash bearers". They have rows of cilia either all over the body or in specialized areas of the cell surface.

Ciliates have a very complex organization. The cell covering, the pellicle, consists of two layers of membrane saudwiching a layer of vesicles between them. The outermost layer of cytoplasm, the cortex, contains a network of protein fibres correcting the basal bodies of cilia. It may also contain many trichosysts, barbed or poisoned thread-like organelles that can be discharged to the outside. Trichocysts serve for anchorage, defense, or prey capture (offence). Most ciliates prey on bacteria, small animals, fellow protests, some eat organic particles from the surrounding water, some are symbionts, and a few are parasites. Specialized cilia around the mouth region (cytostome) sweep food into a gullet (cytopharynx). From here food enters a food vacuole which fuses with a lysosome full of digestive enzymes. Food is digested and products of digestion are digested into the surrounding cytoplasm, and undigested remains are discharged at a specific, site on the cell surface (cytoproct). The contractile vacuole, which discharges excess water and salts, also expel their contents at specific sites.

A unique feature of ciliates is that they have two different kinds of nuclei. Each cell has one or more small diploid micronuclei and a large macronucleus. The macronucleus controls the cell's growth and metabolism (somatic functions) whereas the micronucleus is involved in sexual reproduction. When the ciliate reproduces asexually, the micronuleus divides by mitosis, with the spindle forming inside the nuclear envelop. However, the macronucleus divides into two roughly equal parts without mitosis, a process that reduces its quality over many generations. Sexual reproduction restores genetic vigour. During conjugation two cells come together and exchange haploid gamete nuclei resulting from meiosis of their micronuclei. After a long complex process, both cells end up with genetically identical micronuclei, which divide and form new macronuclei. Meanwhile the old macronuclei have degenerated.

Praramenum is the most familiar genes of freshwaters ciliates. Some Paramecia are hosts for other organisms. For example *Chlovela*, a unicellular green algae, sometimes lives in the cytoplasm of *Paramecium bursaria*. In this symbiosis, the alga gains protection, and the ciliate has captive photosynthesis. Strains of *Paramecium* containing kappa particles (gram negative bacteria with bacteriophages) are called "killers" because the produce a toxin lethal to paramecia that do not contain the particles. Lambda particles, found in other strains of paramecia, have been grown outside their ciliate hosts and have also been identified as gram-negative bacteria.

Figure (   )
Other examples include Vorticella, Stentor, Didinium, Stylonychia, Acineta etc.

2.9. Phylum Oomycota
These are commonly referred to as water moulds.
NB: Three phyla acrasiomycota, myxomycota and Oomycota were once classified as fungi because of their resemblance to fungi which seems to have been as a result of convergent evolution. The Oomycota resemble fungi in having bodies made up of thread like filaments called hyphae and in reproducing and dispersuing by means of spores. Oomycotes live as

saprobes or parasites. They feed by growing hypae into a food source, releasing digestive enzymes, and absorbing the resulting small molecules. The vegetative hyphae are coenocytic, with diploid nuclei and cellulose walls. They form asexual spores, which swim through water by means of two dissimilar/unlike flagella, one hairy and forward pointing, the other smooth and trailing. These distinctive different flagella are also found in several other phyla of protests, including photosynthetic golden algae (Chlorophyta) and brown algae (Phaeophyta). Ribosomal RNA sequencing confirms that all these phyla are closely related. One theory holds that Oomycotes evolved from coenocytic, diploid algae that lost their diloroplasts.

Oomycota are commonly called water moulds because their phyphae form a white fuzz on aquarium fish or on organic matter sitting in water. However, many oomycotes are terrestrial, including some very damaging plant parasites such as downy mildews and root rotting forms. Oomycota means "egg fungi" referring to the large eggs formed in the female reproductive structures. *Phytophthora infestaus*, the parasitic Oomycote that causes late blight of potatoes/tomatoes was responsible for Irish potato famine. Other examples include *Sapolegnia* (mould on living or dead animals in water), *Plasmopara* (downy mildew), Albugo (downy mildew) and peronosphora (downy mildew) and peronospora (downy mildew).

Figure (   )

2.10 Physiology of protozoaus

2.10.1 Feeding

The group contains heterotrophic species. Heterotrophs utilize only materials manufactured by other organisms, either by absorbing them through the body surface (osmotrophs – most parasites) or ingesting solid pasticles (phagotrophs e.g most Rhizopodsand ciliates). Digestion of solid food takes place in cytoplasmic food vacuoles. The food vacuoles undergo acid and alkaline phases during digesting, with enzymes being secreted into the vacuoles. Absorbable materials pass into the cytoplasm.

2.10.2 Osmoregulation/excretion

In many fresh water species, osmoregulation is carried out by contractile vacuoles. The vacuoles are cavities which undergo cyclic processes of filling and emptying/draining to the exterior. In species with a pellicle the ejection spot for solid wastes is the cellamis or cytoproct. Soluble waste products are removed across the general body surface. The main nitrogeneous waste product is ammonia with urea and urates occurring less often.

2.10.3. Movement

Locomotion may be by swimming (ciliates & flapellates), slow gliding/gregarine movement or body flexion (some apicomplexans), stepping (using cirri), rolling (using axopodia) or by peristaltic waves (englenoids, or amoeboid movement (by means of pseudopodia).

2.10.4. Respiration

Respiratory/gaseous exchange occurs across the body surface. Most protozoans show aerobic respiration.

2.10.5 Circualtion

There is no elaborate circulatory system as in metazoans. Internal transport of materials is by diffusion but may be aided by endoplasmic streaming in some species.

2.10.6. Reproduction

2.10.6.1. **Sexual reproduction** – involves fusion of gametes which are produced by modification or division of single cells. It may involve gametogamy (fusion of free gametes), gamontogamy (fusion of gamont

2.10.6.2 **Asexual reproduction** - occurs in all protozoans and in some it is the only mode of reproduction. Usually it takes the form of:

i) **Binary fission** - the cell divides into two similar daughter cells. In flagellates, it is usually symmetrogenic (longitudinal) whereas in ciliates, it is homothetogenic (Transverse).

ii) **Budding** - a mother cell produces a much smaller daughter cell which may remain attached or detaches from the mother.

iii) **Schizogony -** this is multiple fission which leads to production of several daughter cells e.g. schizozoites or merozoits of *Plasmodium.* In most apicomplexans, sporogony (Spore formation) is a form of multiple fission which occurs after sexual reproduction, giving rise to sporozoites.

Questions

1) Describe the various ecological niches inhabited by protozoa.
2) Describe “conjugation” as it occurs in Paramecium.
3) Are sponges metazoans? Discuss.
4) Describe polymorphism in the phylum Coelenterata.
5) Desribe the life cycle of a typical schisstosome as S. haematobium.In what was does this life cycle differ from that of a tape worm?
6) Briefly describe the male reproductive system of a named cestode.
7) What are the unique features of coelenterates?
8) Discuss “ Nutient acquisition” in protozoans.
9) How do coelenterates show advancement over the sponges?

METAZOA

Multicellular animals comprised of 34 phyla.

Simple Metazoa

Phylum - Placozoa

Discovered, described and named in 1883.

Trichoplax adhaerens was at first speculated to be a larval stage of a Cnidarian (coelenterates) and was subsequently forgotten. It was rediscovered in early 1960's and was cultured in the laboratory. At preset it is widely accepted as a simple metazxoan (multicellular animal) and is the only know member of the phylum placozoa.

Trichoplax may be found gliding over the walls of a sea aquarium containing corals or other tropical marine organisms, the only kind of habitate in which it has so far been encountered. It looks like a giant, flattened amoeba, measuring up to 2 or 3 mm across, and takes a variably lobed shapes as it moves this way and that in a strikingly amoeboid manner. When examined more closely, it is found to consist of many tiny cells organized into two epithelia around a fluid middle layer of mesenchyme cells, like a filled pancake. Functionally, the lower epithelium corresponds to the inner or gut epithelium of other animals, while the upper epithelium corresponds to their outer or epidermal epithelium. Asexual reproduction is by fission, fragmentation and budding. There is also a form of sexual reproduction observed when two different clones are mixed (not fully understood because no sperms were observed but gametes appear to take the form of single huge cells, 2 or 3 per organism).

NB: The phylum is made up of organisms without any symmetry, distinct tissues or organs, body cavity or nervous system.

Phylum - Porifera

These are commonly called sponges. They are simple, sessile filer -feeders shaped like vases, tubes, cups, or crusts with porous walls. The tern porifera means "pore-bearing". Sponges grow singly or in colonies and range from a few millimeters to the size of a barrel. As filter feeders, sponges must live in water. They abound in all seas, usually in shallow water, on bubstrates such as rocks or the shells of other invertebrates such as rocks or the shells of other invertebrates. About 100 of the estimated 500 species live in fresh water (comprised of two families).

Sponges differ from true metazoans in having several types of specialized cells but no tissues. their cell layers do not correspond to the true germ layers of other animals and their cells are remarkably independent. This was shown in 1907 by embryologist H.V Wilson, who discovered that he could push a living sponge through fine silk mesh, breaking it up into individual cells and cell debris. This would bring doom to any adult animal, but on standing in sea water, the sponge cells aggregated into large masses, and after about three weeks, they had formed a functional sponge. This is a proof that sponges have high rates of regeneration.

The simplest sponge body plan is like a rase with porous sides. Some sponges however are more complex, with the inner walls of the vase. Folded such that pores open into internal canals or chambers, which inturn lead to the central cavity. Most sponges lack symmetry, although some do have a very regular shape and radial symmetry.

A sponge body has two layers of cells sandwiching a gelationous matrix. This contains mobile amoeboid cells (amoeboytes) as well as a skeleton composed of resilient fibrous protein, spongin, or of sharp spicules (little spikes) made of calcium carbonate or silica. The natural sponges are classified largely by the chemical composition and shape of the skeletal structures. The pores of a sponge are tunnels through porocytes, ring-shaped cells that span the layers of the body wall.

Sponges have no muscle cells and therefore move very little. A nervous system is also absent. People assumed they were plants until 18th century discovery of flagella hat propel a current of water into the body via the pores and cut through the large opening at the top (Osculum). The water current serves for gaseous exchange and waste removed as well as carrying food particles into the body. The flagella propelling this current are part of specialized cells, called **choanocytes** ("collar cells") because of the ring of microvilli encyicling the base of their flagellum. Choanocytes are similar to choanoflagellates, evidence that sponges probably evolved from complex colonial members of this protistan group. The microvilli trap food particles swept into the sponge's body with the feeding current. Choanocytes then ingest the food by endocytosis and pass it onto amoeboid cells (amoeborytes) which digest it and distribute it to other cells. Amoeboryte cells involved in digestion are sometimes called archaeocytes. Individual cells are involved in osmonegulation and excretion (but is mainly carried out by archaeocytes). The cells of fresh water sponges possess contractile vacuoles.

Sponges might be called the ocean's intrafiltration system because they ingest very small particles. As little as 20% of their food consists of plankton large enough to be seen with a light microscope (>0.4 μm). The rest of their food is smaller than this, mostly fine debris from dead organisms. Sponges success is partly due to their ability to feed on these tiny particles, which many other animals cannot trap. Because they are sessile, sponges can live only in clear water, free of detritus that would burry them or clug their pores.

Inorder to define themselves, more than half the species of sponges produce toxic chemicals, which prevent larvae of other sessile animals from settling on them and enter predators. The sharp, brittle specules also make them noxious to predators. These defenses make sponges a safe haven for photosynthetic symtionts such as cyanobacteria or algae and for other animal tenants within the central cavity.

Like other marine invertebrates, many sponges are brightly coloured, either by the photosynthetic pigments of their symbiouts or by their own pigments. Purple, yellow and orange species abound. Most sponges are hermaphroditic, producing both sperm and eggs, but at different time. Hence, cross-fertilization s usual. Sex cells are generally derived form archaeocytes or chroanocytes. Flagellated sperm leave the parent's body and enter another sponge with the feeding current. After fertilization, the zygote usually remains inside the parent until it develops into a ciliated, free-swimming amphiblastide larva, which lives in the plankton before settling to the bottom and forming a young sponge. Sponges also reproduce asexually by budding. Many freshwater and a few marine species form germnules, little balls of amoeboid cells surrounded by skeletal spicules. Gemmules can survive drying or freezing and develop into a new sponge under favourable conditions.

Figure( ).

RADIATE ANIMALS

Phylum-Cnidaria

The word Cridaria is derived from the word (knide= 'netlle' or sting). Originally they were called Coelentrata (Coel=hallow; enteror – gut, but he name was discarded because sponges could also qualify to be members. This Phylum contains hydras, sea anemones, corals, and jelly fish. The adults usually have radial symmetry, although the larvae and some adults have bilateral symmetry. These animals are metazoans, with cells forming two germ laryers (Diploblastic) in he embryo and tissues in the adult. The endovernial and Ectodermal body layers are well developed, but the middle layer (mesogloea = middle glue) between them ranges from a non-cellular membrane to masses of jelly-like substance containing scattered ectodermal cells. Due to lack of a mesoderm, these animals are acoelomates. The gut is a blind sac, with one opening serving both to ingest food and to expel indigestible remains. This gastrorascular cavity (gastro=stomach, vascular=vessel) doubles as a circulatory system, distributing food around the body cnidarians have a network nervous system which is not centralized.

Cnidarians exist in two essentially similar body forms: medusa and polyp. In both forms, a ring of tentacles surrounds the mouth, which leads to the gastrovascular cavity. Polypsare sessile, and most medusas are free-living, moving by contraction of heir bell-shaped bodies. Many cnidarians are colonial, consisting of numerous polyps and/or medusas attached to one another (some are solitary). Individual polyps may be microscopic but most species are macroscopic.

Most cnidarians are carnivores. They do not chase their prey but sit (or float) and wait. Although cnidarians are too weak to subdue animals with well developed musculatere, they have special organs: nematocysts, organelles used for offence and defence. These are mostly/concentrated on the tentacles. A nematocyst is the business and of a cell called a cnidocyte, which also usually possesses a trigger-like cnidocil. The cnidoil responds to touch or chemical stimuli, causing the cnidocyte to evert its thread-like nematocyst. The nematocyst may twist about bristles on the body of he prey, entangling it, or may secrete a sticky or paralytic substance. Nematocyst toxins can cause a nasty sting, occasionally fatal to swimmers. Some predators are immune to the toxins of cnidarians. Leatherback sea turtles eat jelly fish and several coral reef living fishes, such as parrotfish and butterfly fish eat sea anemones. Starfish also feed on corals.

In most cnidarians, the tentacles pull prey trapped by the nematocysts and stuff it into the mouth. Having nematocysts, a mouth and a gastro-vascular cavity permits cnidarians to eat much large prey than a protozoan or spingecavage. Small particles of food are engulfed form he gastro-vascular cavity by amoeboid cells, which complete digestion and distribute food to other cells. Most cnidarians are marine but there are a few freshwater forms (mainly hydrozoa).

Classification

Phylum Cnidaria has three classes: **Hydrozoa, Schyphozoa** and **Anthozoa.**

1. **Class Hydrozoa** – Hydrozoan life cycle combines the polyp and medusa body form in different ways. Usually polyps, either alone or in colonies, are the feeding stage and reproduce asexually, giving rise to other polyps or to medusas. The medusas swim off and then reproduce sexually. Eggs and sperm fuse to dorm a zygote that develops into a ciliated plamula larva, which gives rise to a new polyp generation. Many hydrozoans have a reduced medusa stage. Hence, it is not surprising that the mdedusa as well as the larvae are absent in the commonly studies freshwater genus Hydra – Hydra, like most hydrozoans, and Sayphozoaus, have separate sexes, whereas authozoans are hermaphroditic. Examples include Hydra, physaria, portita, velella, Tubularia.
2. **Class Scyphozoa –** commonly referred to as jelly fishes. They have a greatly reduce polyp stage and a prominent planktonic medusa. Some species of scyphozoans can swim quite actively by contraction of muscle fibers arranged around the bell. The almost transparent body is nearly invisible to prey, predators, and unwary swimmers alike.
3. **Class Anthozoa** – These are the sea anemones and corals which apparently lost the medusa stage during evolution and a tiny ciliated plamila larva is the dispersal stage

sea anemones are large anthozoaus attached to rocks and other surfaces, mainly between high and low water marks around the ocean shore. On the other hand, corals are tiny micorsocpic colonial polyps that build and inhabit a common sheleton of calcium carbonate. The latter has formed rock layers (coral reefs) and islands in the warmer seas of the world. The Australia's Great Barrier Reef, move than 2000 km long, is the world's largest structure built by living organisms, a process that has taken an estimated ½ to 1 million years.

Corals contain symbiotic algae, which colour the transport polyps yellow or brown. The alga's photosynthesis often supplies more than a half of the coral's food and the coral's prey supplies nitrogen and phosphorous for both partners. Warmer than normal water temperatures during some years in the 1980's and 1990's are blamed for several episodes of widespread coral bleaching, when the polyps expelled their algae, leaving only the transparent animals in their white skeletons. This disrupted the coral's food supply and hence their growth and reproduction. It also drastically reduced the deposition of calcium carbonate to only 4-30% of the normal rat, which may not keep pace with desruciton of a reef by waves and human activity. Because a coral reef serves as a "high rise apartment complex" for diverse plant and animal populations, coral bleaching threatens many forms of tropical marine life.

Phylum - Ctenophora - comb-jellies
(Ctene=Comb, Phora = bearers).

Are about 90 species and were once classified as cnidaris, which they closely resemble. However, only one ctenophore species has nematocysts. The name of the phylum ctemphora, as well as the common name "Comb-jelly" refers to the 8 vertical rows of ciliary combs that radiate over the surface of the animal used for swimming. Warm water species (tropical and subtropical, appear yellow-brown or golden from the photosynthetic dioflagellate cells that live in their tissues.

Characteristics.

1. All are marine, pelagic animals, most-free swimming, but a few creeping or sessile forms.
2. The larva (cydippid) is free swimming.
3. Body spherical to ovoid, or flattened and elongated.
4. Most orders tentaculate (bear tentacles)
5. Tentacles, where present bear spherical adhesive collablast cells used in prey capture.

6. Transparent, gelatinous and luminiscent (radiate light).
7. Body layer of 2 layers - ectoderm, mesogloea and endorerm.The bulk of the body is made of a gelatinous,transparent mesogloea which is considered to be unusual connective tissue, rich in acidic polysaccharides. Mesogloea contains mesenchyme cells and true muscle cells.
8. Possess 8 radially arranged ciliated bonds (comb rows or costae). Each costa is composed of a raw of ciliary plates called ctenes or comb plates.
9. Digestive system of branched canals, the gastrovascular cavity or coelenteron.
10. Hermaphroditic
11. Biradial symmetry
12. Size ranges from 1 to 5 cm.

Ctenophores are carnivorous on plankton but some are cannibalistic. The contractile tentacles, which lack nematocysts (except in Euchlora rubra) bear adhesive colloblasts in the epidermis. They are used in prey capture. Reproduction is both sexual and asexual. Eggs and sperms are shed to the outside through pores, fertilization occurring in the water. Asexual reproduction is by fragmentation. Small fragments shed during locomotion develop into complete individuals.

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ACOELOMATES

Phylum - Mesozoa

Approximately 50 species peculiar ciliated parasites of marine invertebrates; contain few, large cells and no organs.

Characteristics

1. Metazoan, bilaterally symmetrical of simple organisation.
2. Lack nerve cells
3. Solid body of two cellular layers but lacking endoderm and mesogloea
4. Body consists during at least part of the life-cycle of outer ciliated cells (the somatoderm) and inner reproductive cells.
5. Have fewer differentiated cell types than do sponges.
6. Life cycle complex, with both sexual and asexual generations.
7. Are less than 1 mm long.

Are parasites of body cavities of various advanced invertebrates such as cephalopod melluscs, brittle stars (Echinoderms), nemerteans, annelids, Turbellarians (flat-wors) and ascidians (chordates) Bodies of adults consist of an outer layer of ciliated jacket cells arranged in rings and surrounding contractile cells, which in turn enclose a mass of developing eggs and sperms. The eggs are fertilized and developed within the body of the female. Two layered ciliated larvae are released and enter various invertebrate hosts.

NB: Some authorities thought that mesozoans were derivatives of flatworms, secondarily simplified for parasitism. Examples Dicyema, Microcyema, Rhopalura.

Phylum - Gnathostomulida

Commonly called jaw worms. Are microscopic animals 0.5 to 3 mm long, with about 100 described species.

Characteristics.

1. Exclusively marine
2. Bilaterally symmetrical unsegmented, worm-like
3. The epidermis is single-layered, and each cell bears only one cilium.
4. The gut has a muscular pharynx, usually with pained jaws and an unpaired cuticular basal plate.
5. Have no anus
6. Have no coelom
7. Are hermaphroditic.

Gnathostomulids resemble turbellarians in being ciliated and elongated worms. A unique feature of gnathostomulids is he feeding apparatus consisting of a little pair of hard, toothed jaws and a central basal plate. With these, they scrape cyanobacteria (blue-green-algae), diatoms, fugal filaments, bacterial and other organic matter from the surfaces of sand grains, sensory structures on the head may include a halo (circle) of stiff sensory cilia, single or compound, and ciliary pits. The distribution of young and adult gnathostomulids, in or near the organically-rich anaerobic black layer of sandy sediments (sulphide-rich), suggests that their respiration is largely anaerobic. They have not been kept alive in the laboratory, partly because they are exposed to too much oxygen. sexual reproduction involves internal cross fertilization by copulation. Asexual reproduction may occur through fragmentation with anterior fragments surviving.

Phylum - Nemertea

Ribbon worms - with 900 species described. Size ranges from minute to very long (several metres).

Characteristics.

1. Most marine, some terrestrial (damp places)
2. Vermiform, bilaterally symmetrical, elongate, unsegmented and often flattened.
3. Have a proboscis
4. Possess a ciliated epithelium that covers both external (epidermal) and internal (gut) surfaces.
5. Alimentary canals has a mouth and anus.
6. Most taxa have a nephridial excretory system.
7. A blood vascular system is present.
8. Are acoelomate - body is packed with parenchymatous tissue.
9. Sexes are separate but some species are hermaphroditic.
10. Many show direct development, but 3 larval forms are known (Pilidium, desor and Iwata's larvae).

The distinctive and salient feature of all members is the possession of a large proboscis that lies in a fluid - filled cavity above the gut (Rhynchoela). The proboscis can be everted through an opening at the anterior end of the body and is used in prey capture, defence and locomotion. In some members, the proboscis is armed with one or more sharp, calcareous stylets that pierce the prey and help to introduce toxic secretions. Once the prey is captured, it is swallowed whole. Most nemerteaus have great powers of regeneration, and infact some regularly fragment and regenerate as a means of asexual reproduction.

Phylum - Gastrotricha

Gatro-belly; richa - hair, are referred to as hairy bellies and are microscopic about 450 species.

Characteristics.

1. Free living aquatic animals, both in freshwater and in marine waters.

2. Lack segmentation, worm-like, bilaterally symmmertrical
3. Possess characteristic ciliated bands.
4. Have superficial bristles, scales and spines
5. Possess pairs of adhesive tubes for anchorage.
6. Protonephridia may be present (excretion)
7. Acoelomates and are colourless.

Almost any aquatic debris that contains rotifers will also contain a few members of the phylum gstrotricha. The digestive system is a straight tube with a muscular phalynx. Bacterial, protozoal and algal cells, along with particles of orgnaic detritus are ingested by the sucking action of the pharynx or by ciliary currents. The anus opens near he posterior tip of the body or at the base of he forked tail end. There are no special circulatory or respiratory systems. Most of the marine species reproduce sexually and are hermaphroditic. All or almost all of the freshwater species reproduce asexually by parthenogenesis. Females lay unfertilized eggs that hatch into females.

PSEUDOCOELOMATES

Phylum - Kinorhyncha

Kino - motion; rhynch - snout, hence the name snout-motion. Are microscoic animals less than 1mm long. About 1230 species.

Characteristics

1. Exclusively marine, free-living
2. Bilaterally symmetrical
3. Bodyis superficially segmented with 13 or 14 segments or zonites, and possesses a cuticle.
4. Possess a retractile head bearing spines.
5. There are no cilia except in sense organs.
6. There is one pair of protonephridial tubules
7. The body cavity is pseudocoel.

Kinorhynchs are minute spiny animals with a cuticular armour. Having no locomotery appendages or external cilia, they are unable to swim and burrow through the mud by pulling themselves along by a sort of a snout (hence the name "snout motion"). The first sonite (the head) is extended during locomotion and anchored in the mud by a ring of long recurved spines, while the trunk is pulled up. they feed on other micro-organisms and detritus which are sucked by use of a muscular pharynx. They possess a protrusible mouth cone with a mouth and the anus opens at the posterior end. Sexes are separate. Fertilization is apparently internal and development is direct to a juvenile.

Phylum Priapulida

A dozen or so species described, size ranges from 1 mm to several cm in length e.g. Priapulus is 20 cm long.

Characteristics.

1. Marine, benthic, free-living.
2. Bilaterally symmetrical, unregimented, vermiform.
3. Eversible anterior head
4. Protonephridia which open with gonads by urinogential ducts to exterior
5. Nervous system not separated for mepidermis
6. Large body cavity (debatable whether it is coelom or pseudocoelom) without membranous lining.

The most conspicuous feature of priapulans is the anterior part (head). It is an introvert, which can be withdrawn into the longer, posterior trunk and serves in borrowing. The muscular body is covered with a thin cuticle of chitin. Spines surrounding the mouth and extending into the pharynx are everted to captive prey, mostly soft-bodied annelids or other priapulans which are swallowed while. The gut ends in a posteror anus. Sexes are separate and fertilization can be internal or external. Some members are parthenogenetic.

Phylum- Nematomorpha

Long thin worm-like animals (hair-worm), up to 1m long. About 250 species known.

Characteristics

1. Adults are free-living in fresh water or moist soil; juveniles are parasitic in arthropods, leeches and snails.
2. Unsegmented, bilaterally symmetrical, filiform worms.
3. Gut non-functional in adults
4. Excretory system absent but have a nerve cord.

5. Pseudocoelomate
6. Sexes separate, each with a cloaca.

Nematomorphs resemble nematodes in many ways. Unlike nematodes, nematomorphs do not show constancy of cell number and pseudocoelom becomes largely filled with mesenchyme cells. They also have a rounded anterior end and a rounded or lobbed pesterior end. Adults do not feed and depend on nutrients obtained during the parasitic larval phase. Sexes are separate and after mating, females lay eggs which hatch into larvae with a spiny eversible proboscis armed with 3 stylets. When ingested by a suitable host, they use the proboscis to pierce host's intestines into the body cavity where they grow and develop. The full-grown worm emerges when the host approaches fresh-water.

Phylum- Acanthocephala

Thorny or spiny headed worms, ranging in length from 1.5 mm to about 1 m. About 1000 species are known.

Characteristics

1. All are endoparasites as adults in vertebrates (mammals birds, fish, amphibians and reptiles).
2. The larvae develop in the female uterus and need an intermediate host (usually an arthropod) for further development.
3. Unsegmented, bilaterally symmetrical
4. Anterior extremity is formed of a protrusible proboscis armed with hooks.
5. There is no gut.
6. They are pseudocoelomate
7. Excretory system – protonephridia only in one family.

Unlike flatworms (hermaphroditic), acanthocephalans have separate sexes. After mating and fertilization taking place, fertilized eggs are extruded through the female gonopore into the host's intestine, and shed to the outside with the host's faeces. The hooked larva, called an acanthocephalans are moniliformis in rats, acqjuired from cockroaches and Macracanthorhynchus in pigs, acquired from beetle larvae. Both can occasionally be found in humans but do not cause serious disease. Acanthocephalans show strong similarities to priapulans.

Phylum – Entoprocta

Sessile "lophophorates". True lophophorates are coelomates with a row of hollow tentacles surrounding the mouth and is used in filter feeding.

Characteristics.

1. Allare marine except *urnatella* (fresh water form)
2. They are sessile, solitary or colomal
3. individuals possess a circlet of ciliated tentacles
4. The gut is U-shaped with both the mouth and anus opening within the tentacular circle.
5. Possess protonephridia
6. Ave pseudocoelomates

When first discovered in early 19^{th} century, Entoprocta were considered to be unusual beyozoans (Ectoprocta) Entoprocts are tiny (5 mm or less) and they live on the bodies or tubes of other animals such as polychetes, sipunculans, bryozoans, ascidians and sponges. Large members permanently attaché to rocks or plants. Most or all species are hermaphrodites, but usually eggs and sperms are produced at different times within an animal. Most are protandrous and sperms mature first. They also reproduce extensively by budding from specific regions near the mouth or stalk. Larger forms develop attachment stolons, and budding occurs from them also.

COELOMATES

Lower Protostomes

In the previous lesson of the module, we set out to present phyla which can be regarded as minor in terms of numbers of individuals, species and basic ecological significance. These minor phyla are not treated extensively.,The present lesson covers small phyla of lower coelomate protostomes, all of which lack segmentation and are similar in general appearance namely:**Echiura, Sipuncula**, **Priapulida, Pogonophora, Onychophora, Tardigrada and**

Pentastomida. This lesson covers four small phyla of marine wormlike animals that are **protostomes** and two of them show some evolutionary relationship to annelids:- **Priapulida** ,**Echiura Sipuncula**.

Phylum Priapulida

The priapulids are another minor group of marine , benthic animals.

They are found in the colder waters of both northen and southern hemispheres.

Only a dozen or so species have been described. Size ranges from 1mm.to several cm in Length e.g *Priapulus* is 20 cm.long.

This is a very small phylum of highly distinctive marine worms, free-living(solitary) which liveburied in marine sands in sea floors in intertidal zones and mud.

In many anatomical features, tceolomate phylum Sipuncula

and the pseudocoelomate phyla including the rotifers and acanthocephalans

Morphological structures

Bilaterally symmetrical , unsegmented, vermiform.

The most conspicuous feature of priapulans is the anterior part (head). It is an introvert which can be withdrawn into the longer posterior trunk and serves in burrowing.

The cylindrical body has proboscis-like region, posterior trunk and caudal appendage(Fig.).

The proboscis bears the mouth at the terminal end (eversible) which is surrounded with concentric toothed cuticle. This circumoral region is usually invaginated into the interior,but

when this is everted in capturing the prey, the teeth point forward.

The proboscis contains a spiny introvert which turns in on itself to force the prey into the alimentary tract.

The proboscis is separated from the trunk by a constriction.

The proboscis is as large in diam.as the trunk and sometimes may be larger.

It bears 25 longitudinal ridges each made up of papilla or spines surrounding the mouth and extending into the pharnx which are everted to capture prey.

One or two clusters of respiratory appendages are at the posterior end of the cylindrical trunk.

Scattered over the trunk region are many tubercles and spaces that give distinctly warty appearance to the animal.

The digestive system consists of mouth , pharynx, intestine and rectum.The gut is a stright tube running to a terminal anus, that is quite unlike conditions in the sipunculids At the posterior end of the trunk are two urinogenital pores and anus. There is a large (spacious) open body cavity (debatable whether it is coelom or pseudocoelum).

This is lined with a thin peritrophic (non-cellular layer) which also covers the viscera.

They are predaceous in their eating habits, feeding on other small, marine animals, mostly soft- bodied annelids or other priapulans whch are swallowed whole.

They capture slow moving prey with the circumoral spines and swallow it whole.

There is a central protonephridia made up of large numbers of nucleated solenocytes which serve to carry both excretory and genitalducts.

The protonephridia open with gonads by urinogenital ducts to exterior.

The gonads form tangled masses of tubules.

Early cleavage stages are radial not spiral

The larva which results has the cuticle in the form of plates like the rotifer and the adult form is acquired through a series of cuticular moults.

Priapulids moult throughout both juvenile and adult life,

In other ways the priapulids resemble pseudoceolomates.

Priapulida are unique in having a cuticle that contains chitin and is moulted throughout their adult life.

Classification

Phylum has three families (i) **Family –Priapulidae** –large forms e.g *Priapulis priapulopsis*

(ii) **Family- Tubiluchidae** –minute intestinal species e.g *Tubiculuchus*

found in shallow waters of tropical and sub-tropical Western Atlantic

(iii) **Family: Maccabeidae**- found in deeper waters in the Mediterranean e.g *Maccaabeus*

The phylum Priapulida is made up of two genera, *Priapulus* and *Halicryptus* and eight species seven of which belong to *Priapulus.*

Basic adaptive features

They are bottom muck residents, predaceous and are not filter feeders.

The eversible proboscis with its numerous recurved teeth surrounding the mouth and lining the pharnx is well adapted for predaceous habits.

Restricted movements make it possible for the worms to capture only slow-moving preys.

Because of their feeding habits and generally stable ecologic niche, perhaps little opportunity has existed for evolutionary diversification within the group.

The development of oxygen-transporting **hemerythrin** cells in the coelomic cavity has been of survival value in their special mode of living.

The nervous system is in very close association with the epidermis. The papillae and spines may sense organs.

The paired urinogenital organs are each made up of a gonad on one side and clusters of solenocytes on the other. A central urinogenital or protonephridial tubule serves to carry both excretory and genital products.

Sexes are separate and sex cells are shed into the sea, where fertilization occurs.

The egg hatches into a small postgastrula, the larval trunk of

which is later encased in a cuticularized lorica similar to that of rotifers.

The larvae live in the same habitat along with the adults and feed on detritus. Larvae moult in about two years and emerge as juveniles without the lorica and with the same structures as adults. In the section to follow, this lesson sets out to present phyla which can be regarded as minor in terms of numbers of individuals, numbers of species, and basic ecological significance. These phyla are not treated extensively. We will therefore consider six minor phyla which are not triploblastic coelomates.

Phylum : Echiura

This is a minor phylum which seems to have arisen from a group of ancestral to the present day Annelida, in which metameric segmentation has already evolved. Approximately 140 species have been described. Echiurans or spoonworms are sausage-shaped, marine animals that resemble sipinculans in size and general habit. Many species e.g. *Thalassema, Urechis* and *Ikeda* occupy burrows in sand and mud whereas others live in rock and rock crevices, snail shells and other secure shelters. Range in size (trunk length) from approximately 1cm. to over 50cm. (*Urech*i). A majority live in shallow water but there are also deep-water species.

May be an important food in the diet of some fishes.

External Structure

The echiuran body consists of two distinct regions, a non-segmented cylindrical trunk and a flat, spoonlike or ribbon-like, grooved anterior non-retractable proboscis which is used in feeding corresponding to the annelid prostomium.

The proboscis is quite mobile and contractile but it cannot be drawn into the body as in the case of the Sipuncula introvert.

The mouth, located at the base of the proboscis, opens into a short stomach and a long greatly coiled intestine.

The thick fleshy body wall is covered with a thin cuticle secreted by the epidermal cells. The body surface may be smooth or ornamented with annulae or papillae. A pair of short, hooked, chitinous setae occurs on the anterior ventral surface of the trunk. When they are extended from their setal sacs, they curve posteriorly and are used for digging as the animal burrows. Some Echurians e.g. *Echiurus, Urechis* have one or two circles of hooked setae at the posterior end of the trunk. These setae are used for burrow maintenance and, perhaps, for anchorage.

InternaI Structure

The body wall is annelid-like and there are setae which are chitinous and formed in sacs as in earthworms. The trunk consists of a thin poorly developed cuticle, glandular epidermis, a musculature consisting of at least three layers and a ciliated peritoneum that lines the coelom. The coelom is an open cavity without septa as in annelids. The wall of the prostomial proboscis differs from that of the trunk in being ciliated ventrally, lacking a spacious coelom and having specially well developed longitudinal and dorsoventral muscles. These muscles are responsible for retracting and flattening the proboscis respectively.

Echiurans are slow, sluggish animals whose movements are predominantly a slow peristalsis of the trunk used for burrowing and burrow ventilation, and extension/retraction movements of the prostomium, used in feeding and probably to escape predators. The digestive tract is extremely long and coiled and loosely suspended in the coelom.

Feeding

Most echiurans are deposit feeders, catching food particles on mucus secreted by the proboscis and carrying it back to the mouth along the ciliated groove. They feed by sweeping organic detrital particles into the spoon-shaped proboscis, which is richly ciliated and projects from the trunk.

A blood-vascular system is present in all echiurans. A respiratory pigment is absent and blood is colourless. It is most probable that the blood transports nutrients from the gut primarily to the proboscis nervous system and musculature.

Gas exchange occur across the general body wall of both the trunk and proboscis is by simple diffusion.

Excretion is accomplished by two specialized organs known as **anal sacs**. Each anal sac is a thin-walled hollow diverticulum from the wall of the cloaca that extends into the trunk coelon. The anal sacs produce urine and then discharge it through the cloaca and anus by contraction of their muscular walls. Have a ventral nerve cord with no definite brain or ganglionic enlargements.

Reproduction

The sexes are separate, with a single gonad developing and attached to the wall on the peritoneum in the posterior portion of the body cavity. Gametes are released into the coelomic cavity in a mature stage and after ripening they leave via the nephrida. Fertilization is external, occurs in the sea followed by spiral determinate cleavage and a trocophore larva in which both prototroch and telotroch are well developed.

Phylum Sipuncula

The spinculata sometimes called **peanut worms** is a small group comprising approximately 320 species of unsegmented, coelomate marine worms. Adults are bilaterally symmetrical. They burrow in sand and mud or live in coral or wood excavations or in old mollusc shells. All are benthic animals, the majority living in shallow water. Vary in size, the smallest being about 3mm. The cylindrical body is divided into two regions:-slender anterior section, called the introvert, and a larger, thicker posterior trunk. (Fig.) The introvert is the extensible anterior end of the body, but it can be retracted into the trunk region.

The anterior end of the introvert bears the mouth, which is located below or is surrounded by tentacles or lobes that are used in deposit and suspension feeding most of these projections are ciliated, and each bears a deep ciliated groove on its inner side. Behind the anterior end the surface of the introvert is typically covered with spines, tubercles or other cuticular elements.

The trunk is cylindrical and in some rock-boring sipunculans, the cuticle of the trunk is thickened or sometimes calcified at the anterior end to form a dorsal or collar-like anal shield. The shield is used to block the entrance of the burrow, like the closing plate on a snails shell when the introvert is invaginated. The gut is long, highly coiled and recurved so that the anus is anterior of the trunk and dorsally located on the trunk or on the posterior part of the introvert. The gut is suspended in the large coelom. Behind the mouth, the oesophagus descends into the trunk where it joins a long intestine mound into a double helix consisting of proximal descending and distal ascending coils.

Sipunculans lack a blood-vascular system and respiratory system. Coelomic fluid is the sole transport medium for gases and nutrients. The tentacles are important respiratory surfaces in rock and crevice dwellers. Sipunculans have one or two elongated saclike nephridia in the anterior part of the trunk. The nephridia function in osmoregulation and in gamete storage and maintenance prior to spawning.

The sexes are separate in most sipunculan species. Gametes are released from the gonads at an early stage and undergo maturation in the trunk coelom. When mature, they are removed from the coelomic fluid by the nephridial funnel and stored in the nephridial sacs.

Sipunculans spawn their gametes into seawater and fertilation is external. The developing zygotes undergo spiral cleavage. A few members reproduce asexually, by unequal division of the body by budding.

Phylum : Pogonophora

Thin wormlike animals without a gut. They live in tubes of their own secretion, mainly in muddy sediments in the deep sea.

Deep-water, tube-dwelling worms. A few spp. live in rotting wood on the sea floor and large forms attach their tubes to hard substrates. The phylum is named **"beard bearers**" because of the thick tuft of long anterior tentacles that are unique to this group. Some members have one or two relatively large tentacles, but the five tentacles may number to about 200 (or more than 200,000 in some called vestimeniferans).

A rear section of the body is segmented with setae.

At the anterior end of a long middle trunk-like region is a forepart bearing long ciliated tentacles.

The slender and delicate body is protected within a secreted chitinous tube, which the worm adds at both ends as it grows, so that the tube consists of a series of rings. The tubes are embedded in soft marine sediments, mostly in cold, deep water, and are usually less than a mm. in diameter.

The oddest thing about pogonophorans is that they have no mouth or digestive tract at any stage, and therefore nourish (feed) themselves by taking up sugars and amino acids from the surrounding sea water (or perhaps by pinocytosis). The integument is responsible for all intake of food from the environment.

In addition pogonophorans harbour endosymbiotic bacteria in a highly vascularized tissue in the posterior part. There is evidence that the symbionts fix carbon dioxide into organic compounds that nourish both themselves and the pogonophorans.

Fertilization occurs externally but within the tube of the female, where fertilized eggs are retained and brooded. Asexual production by transverse fission occurs in the species *Sclerolinum braetstromi*.

Phylum- Onychophora-claw bearers e.g. *Peripatus macroperipatus*

Introduction

The Onychophora are regarded by several authors as a primitive link between the two phyla Arthropoda and Annelida.

Arthropod features

(i) The presence of a cuticle shed by ecdysis

(ii) Hardened claws and jaws.

(iii) The tracheal system with spiracles

(iv) The large haemocoel

(v) A dorsal ostiate heart.

By contrast the characteristic annelid features include:-

(i) extensible body with its pseudostriated muscles

(ii) serial nephridia

(iii) the presence of cilia

(iv) simple eyes

The integument of onychophorans consist of:-

(i) an epidermis

(ii) dermal connective tissue characterized by collagen fibers

Phylum Onychophora – claw bearer e.g. *Peripatus maeroperipatus*

Characteristics

1. Prominent in discussions of arthropod evolution due to possessing a combination of soft-bodied annelids and hard-cuticle arthropod characters.
2. Commonly called **velvet worms**, length up to 15 cm and about 70 species described.
3. Free-living, all are terrestrial in damp places in tropical or south temperate areas. Live beneath stones, bark of fallen trees (logs) and leaf litter or along margins of bodies of water.
4. Possess a thin, flexible cuticle and they respire by means of a simple tracheal system with openings (spiracles) that cannot be closed like in arthropods (that is why they live in humid, damp places). The tracheal system is another trait that make onychophorans so arthropod-like and demonstrate that they arose from stocks with very similar potentialities.
5. The head is composed of three somites.
6. The first pair of appendages on the anterior end are the antennae, and each has an eye at its base. The antennae are the principal sense organs.
7. Have numerous short (stumpy), peg-like, paired appendages (14-43 pairs) with a long body, features which make them look like caterpillars.
   - The limbs are arranged ventrolaterally and each limb has a pair of claws.
   - The only real evidence of segmentation externally is provided by the paired appendages.
8. They creep about, with the body undulating as they walk.
9. Excretion is by nephridia. Each somite except the first two has a pair of nephridia, but the first pair has been converted into salivary glands. Nephridia are osmoregulatory.
10. Body wall has circular and longitudinal muscle layers.

- The space between the body wall and the gut is partitioned to form the various haemocoel sinuses.
- A dorsal diaphragm separates the pericardial sinus in which the heart lies from the much larger **perivisceral sinus** in which the gut and other organs lie.
- A ventral diaphragm partially separates a ventral sinus from the perivisceral sinus.
- Muscles associated with the appendages cut off the lateral sinuses.
- None of the partitions are complete, and blood flows freely from one compartment of the haemocoel to another.

11. The outer surface is covered by a thin flexible cuticle that is unlike the jointed exoskeleton of arthropods.The integument of onychophorans consist of:

i) a cuticle

ii) an epidermal

iii) dermal connective tissuecharacterized by collagen fibres

- It is chitinous and not segmented. The cuticle is warty (protuberances) and bumpy. Some 14 to 43 pairs of short limbs (legs) are arranged ventralaterally along the body and each bears two claws.

12. The mouth is terminal, flanked by glandular oral papillae and armed with a three-piece tooth apparatus.

- They defend themselves with adhesive secretions secreted, squirted from glands in the mouth region.

13. Onychophorans are omnivorous and feed on wood fibre, insect excreta, small molluscs and termites (at times can be cannibalistic).

14.Circulation

- The heart is a contractile tube, open to the pericardial cavity by segmentally arranged openings called **ostia**. Heart pulsations force the blood forward and down into the perivisceral sinus. Body movements as well as heart movements help to circulate the blood. The heart, ostia and the system of haemocoelic sinuses closely resemble the similar parts of arthropods, although the system is less highly organized than in the majority of arthropod groups.

15. Sexes are separate, and there is sexual dimorphism (males have fewer legs).
16. Mating has not been observed but there may be copulation and internal fertilization.They can either be oviparous (development of the egg outside of the body), ovoviviparous or viviparous (fertilized eggs are retained in the oviduct, they hatch and active larvae or nymphs deposited rather than eggs).

The onychophorans are regarded by several authors as a primitive link between the two phyla Arthropoda and Annelida.

Annelida features

1. Body wall is annelidan as a method of extensible body with its pseudostriated muscles.
2. Locomotion, unjointed legs
3. The excretory system, N.S
4. The presence of cilia and simple eyes
5. Serial nephridia
6. Simple eyes

Arthropod features

1. Haemocoelic body cavity (Large).
2. Development (the presence of cuticle shed at ecdysis).
3. Structure of jaws (hardened claws and jaws)
4. Possession of salivary glands.

5. An open circulatory system
6. Claws at the tip of the legs.
7. Dorsal ostiae heart.
8. Tracheal system with spiracles

    - Of the various group of arthropods, they resemble the myriapods most closely. **How?** Their body form is similar, **tagmosis** (i.e. the grouping of segments into functional units).
    - Midgut is similar and genital

Differences

The young of Onychophora and *Peripatus* have no resemblance to the trochophore larva of annelids. This means that Onychophora are not directly derived from annelids but have their own history.

Phylum Tardigrada _Water bears

- Are strictly aquatic invertebrates ranging from 50-1600μm in length. Their cylindrical body consists of a distinct head, four indefinite body segments and is provided with four pairs of stumpy, lateroventral legs with claws, toes or disc-shaped adhesive expansions.
- Habitats: Targigrades occur in a variety of habitats from deep sea, marine and freshwater beaches to the soil, where they live between grains of sand, amongst algae, in water films surrounding terrestrial plants and leaf litter.
- Are rarely ectoparasites and commensals.
- Tardigrades living in terrestrial habitats survive desiccation in the state of **anhydrobiosis**, in which they are resistant to extreme cold, heat and various chemicals.

- Obligate freshwater and some terrestrial species are able to form cysts which are less resistant than anhydrobiotic tardigrades.
- The Phylum Tardigrada has three recognized classes:
  (1) **Class.- Heterotardigrada** – comprise the orders Arthrotardigrada which include most of the marine forms and Echiniscoidea.
  (2) **Class-. Mesotardigrada** with only one species.
  (3) **Class-. Eutardigrada** containing the orders **Parachela** and **Apochela.**
- The integument of tardigrades consists of a simple layer of epidermal cells, which secrete a cuticle that extends into the hind-and foregut.
- The cuticular surface may be smooth or variously sculptured. Some tardigrades have lateral cuticular expansions, others, in particular many Echiniscoidea are provided with rigid dorsal, rarely ventral plates. Arthrotardigrada generally possess up to 11 cephalic appendages and spines or cirri on the dorsum and on the legs.
- The thickness (up to 5µm) and permeability of the cuticle may vary with the habitat, and thickness is correlated also with the size of the animal.

The first observation of the rounded form and stumpy, clawed limbs of these animals inspired the common name **water bears**, while Tardigrada (slow-stepper) was coined for the resemblance of their slow lumbering gait to that of a tortoise. Approximately 600 species of tardigrades have been described, many of which are cosmopolitan (world-wide distribution). The tardigrades are commonly called water bears. They are microscopic animals less than a millimetre long.

The unjointed nature of the legs and other characteristics such as the lack of chitin groups them in a separate phylum. The characteristics are such that they, like the onychophorans are evidence of a relationship between annelids and arthropods.

Habitat

A few marine interstitial tardigrades have been collected from both shallow and deep water. There are some freshwater species that live in bottom detritus or an aquatic algae and mosses. There are even species that live in the water films of soil and forest litter. Many live in the water films surrounding the leaves of terrestrial mosses and lichens, especially those that grow on stones and trees.

- Freshwater and terrestrial species respond to unfavourable environmental conditions by producing resistant stages.
- These stages together with the production of resistant eggs, underlie the broad distributions of many species of tardigrades, through dissemination by wind or on the bodies of other animals.

Tardigrades feed on detritus, bacteria, algae, protozoans, microscopic animals or the cell contents of plants. A few marine tardigrades are ectoparasitic on invertebrates such as holothurians.

Characteristics

- Microscopic animals, occurring in marine and freshwater habitats, living amongst sediments or on the surface of vegetation. Terrestrial tardigrades live in the water film surrounding soil particles, and on the surface of plants, particularly mosses, liverworts and lichens.
- Bilaterally symmetrical and some show signs of segmentation.
- An indistinctly demarcated head is followed by four segments, each bearing a pair of stumpy legs. The legs terminate in claws or toes with adhesive pads or claws. The head bears two eyespots.
- Mouth has a sucking pharynx with stylets.

- Body is covered by a cuticular exoskeleton secreted by a single-layered epithelium and undergoes periodic moults. The cuticle is delicate in the more aquatic forms and thickened into plates of armour in moss dwellers.
- Sensory bristles may stick out from various points on the body.
- Circulatory, respiratory and excretory systems are absent. Certain cells in the epidermis and gut lining in some members have excretory functions. Other members have three small glands that open into the hind gut and are thought to be excretory.
- The body cavity of tardigrades is a hasemocoel, as in orychophorans, although a dorsal heart is lacking. The coelom is restricted to the cavity of the reproductive system.
- The digestive system consists of an ectodermal foregut and hindgut and endodermal midgut. The foregut and midgut are lined with cuticle.
- Projecting into the digestive tract and capable of being protruded from the mouth are too sharp stylets, secreted by paired glands. During the feeding, the mouth is placed against the plant cells, and the stylets to puncture the cell wall of mosses are projected and algal filaments on the bodies of small animals such as rotifers and nematodes and even other tardigrades.
- Liquid contents of the cell are then sucked out with a muscular pharynx.
- The nervous system is segmentally arranged. The dorsal brain is composed of three lobes (the **protocerebrum, deutocerebrum and tritocerebrum).**
- Tardigrades are dioecious with a single dorsally located ovary or testis. Parthenogenesis is also believed to be common and hermaphroditism has been reported in some species.
- Females are usually more numerous than males and in some spp. males are not known.
- Copulation takes place, fertilization is internal and except in marine species where mating and egg laying occur at the time females moult and are often fertilized and protected within the shed cuticle.

- Development is direct and rapid. Cleavage is holoblastic but does not follow a typical spiral or radial pattern.
- Tardigrades are among the few animals that can interrupt their normal activities at almost any stage of life and enter a state of **cryptobiosis** ("hidden life"), in which all activities ceases and metabolism declines to almost undetectable levels. They can survive adverse environmental factors/conditions e.g. extremes of temperatures (1-150ºC), atmospheres with low oxygen concentration, high carbon dioxide or even almost total vacuum, immersion in alcohol and other toxic chemicals, irradiation with large doses of UV or X-rays.

They can assume normal life after unto 5 or more years of cryptobiosis and repeated cryptibiotic periods can extend a life span of only a year or so to 60-70 years.

Phylum Pentastomida –Tongue worms

The pentastomids, or tongue worms, are elongated, soft bodied wormlike parasites of the respiratory systems of vertebrates. They live mainly in reptiles and rarely in birds and mammals. Phylogenetically they are most closely aligned to the Arthropoda.

About 100 species are known. As adults, most live in respiratory system of reptiles, especially snakes, lizards, amphibians and crocodilians, but one sp. *Reighardila sternae* lives in the air sacs of sea birds and another inhabits the nasopharynx of canines and felines.

- Certain similarities with the Annelida have been pointed out, but most modern taxonomists align them with the Arthropoda.

Morphology

- The pentastomid body is elongated (Fig.), usually tapering toward the post end and often showing distinct segmentation, forming numerous annuli.

- It is indistinctly divided into an anterior forebody and a posterior hindbody which is bifurcated at its tip in some spp.
- The exoskeleton contains chitin, which is sclerotized around the mouth opening and accessory genitalia.
- A striking characteristic of the adult pentastomids is the presence of 2 pairs of sclerotized hooks in the mouth region. The hooks are single in some spp. and double in others.
- The body cuticle in some spp. also has circular rows of simple spines; the annuli may overlap enough to make the abdomen look serrated.
- The cuticle is similar to that of arthropods, although it is then, soft and untanned. The muscles are arthropodan in nature, being striated and segmentally arranged.
- The digestive system is simple and complete, with the anus opening at the post end of the abdomen.
- Pentastomids are dioecious and show sexual dimorphism in that males are usually smaller than females
- **Biology**

  Adult pentastomids feed on tissue fluids and blood cells of their host.
- They appear to stimulate a strong host immune response, but, because they are long lived, they must evade its consequences.

The gland cells and Ionocytes

- The integument of pentastomids reveals three specializations: gland cells, ionocytes and sensilla.

Ionocytes (chloride cells): All pentastomids maintain a blood concentration lower than that of the host. Since pentastomids feed on host blood or lymph, which is rich in sodium and chloride, it is conceivable that these cells excrete excess ions.

Sensilla: These are represented by diversity of papillae in the anterior region of the adult R. Sternae. They are arranged bilaterally and are concentrated in the head area of adults. The sensilla may be taken as indicating relationship between the Pentastomuda and the Arthropoda.

HIGHER PROTOSTOMES

In the previous lesson on coelomates you were introduced to Lower Protostomes. You learnt that Lower Coelomate Protostomes lack segmentation and are similar in general appearance. They are minor in number of individuals, species and basic ecological significance.

In this lecture you will study the characteristics of the major phylum **Annelida.** The lesson begins by explaining the structural organization of the annelids' body into many segments resembling rings strung together (**annullus = little ring**).

The phylum **Annelida** is the best example of animals that have used serial repetition of similar parts, or metamerism for body construction.

The chief advantage of metamerism especially of the primitive plan is that it allows controlled movement.

Metameric arrangement largely restricts the coelomic fluid to separate septal compartments. This lesson gives and extensive examination of the characteristics, reproduction and development in three main classes in the phylum Annelida and their economic importance.

Consequently, at the end of this lesson, practical activities on selected annelids have been given to selected annelids to have been given to reinforce what was learned.

Objectives

From your study of this lesson, you should be able to:-

- Give the meaning of metamerism
- State the theoretical basis of metamerism

- Describe the characteristics of the phylum Annelida
- List and describe the characteristics of the three main classes in the phylum Annelida.

Phylum Annelida

The phylum Annelida consists of bilaterally symmetrical animals that belong to the **Prostomia,** which have spiral cleavage and a schizocoelic coelom. Annelids are traditionally placed near the base of the protostome radiation and close to the molluscs.

They are commonly divided into three main classes – Polychaeta e.g Nereis, Oligochaeta e.g.earthworms and Hirudinea e.g leeches.

The polychaetes are mainly marine; the oligochaetes are chiefly terrestrial and freshwater; and the leeches (Hirudinea) are wholly or partly parasitic, chiefly in freshwater or on land.

There are more than 14,000 species of annelids or segmented worms. Many are microscopic but a few reach great length e.g AustraVassian earthworm, *Megascolides*, up to about 4 metres.

Characteristics

1. Annelid worms are coelomates, the coelom being mesodermal (schizocoelic) in origin.
2. The body is vermiform, bilaterally symmetrical and metamerically segmented. Each segment is separated from contiguous ones by a transverse septum, although this basic feature may be modified in some forms.
3. There is a hydrostatic skeleton rather like a line of water-filled bags, the animal being bounded by a thin, flexible cuticle overlying the epidermis.
4. Annelids may have chaetae; hard, bristle-like structures projecting from the body wall.
5. The group is triploblastic, and has a body wall musculature of two layers (external circular and internal longitudinal muscle).

6. The Central Nervous System (CNS) of pre-oral ganglia linked by connectives to a pair of ventral ganglionated cords.
7. Possess **nephridia** and coelomoducts typically for excretory and reproductive purposes.
8. Circulation is closed and tubular.
9. Reproduction may involve copulation; cleavage is spiral and development is determinate.
10. The larva is the trochophore.

Larval form

The larval type of the annelids is the trochophore, although this is only found in a number of polychaetes.

Oligochates usually show direct development as also do Hirudinea, with juveniles hatching in an immature adult form.

The trochophore larva (Fig.).

It is typically a spherical organism with a major ring of cilia (**the prototroch**) lying just anterior to the mouth. There are subsidiary groups of cilia of the larval surface, the most imprtant being the anterior apical plate. This is thought to be a sensory area.

Mobility is still conferred by the prototroch ring of cilia.

Metamorphosis

The adult develops from the larva by an elongation of the post-oral region.

The portion anterior to the prototroch remains as the prostomium of the adult worm.

The region behind the mouth elongates by the addition of mesodermal blocks, progressively carrying the larval anus to the rear, the guts extending in simple fashion between the larval mouth and the anus.

Metamorphosis is often accompanied by a change in larval behaviour. Newly-hatched larvae are relatively strong swimmers and may migrate to the surface of the sea. They later become negatively phototropic and positively geotropic moving towars the sea bed.

There they complete the metamorphic changes, providing the habitat is suitable.

Adult Body Form (Fig.).

Adult annelid worms are elongate animals, and show metameric segmentation. There is an internal division of the body by regularly repeated septa.

Within the segments of the body there is serial repetition of various organ systems. This is especially marked in polychaetes. The repetition of like units in the formation of an individual unit occurs in three great phyla of animals – Annelida, Arthropoda and Chordata – the bodies are serially divided into units (metameres or segments) that are generally similar.

In annelids metamerism is an adaptation for burrowing; in chordates for swimming. Hence segmental organization gave annelids a new mode of locomotion unlike platyhelminths and nematodes which mostly squeeze into existing spaces in bottom ooze,soil leaf litter, or a host's body. Annelids can create their own spaces, even in well packed substrates.

In annelids many internal organs e.g nephridia, certain blood vessels ganglionic enlargements and lateral nerves are serially arranged according to segments.

Many groups other than Annelida, Arthropoda and Chordata have some tendencies towards a segmental arrangement e.g in the cestodes, the proglottids may represent true segmentation although in these flatworms new units are added just behind the anterior scolex whereas in true metamerism new segments are formed first infront of the terminal segment.

There may be many types of metamerism for different adaptive reasons – swimming, reproduction and repetition of organs in elongated animals for effective control.

Origin of Metamerism

The fission theory: Explairn origin of metamerism as a result of an incompletely divided chain of zooids, a common method of asexual reproduction.

The Pseudometamerism theory:- Where organs were repeated and scattered through the body before segmentation occurred.

The Muscular theory: Stresses the idea that metamerism first arose as a locomotory device in the somite arrangement of muscles for undulating movements such as are found in certain larvae and in the freshwater origin of the vertebrates.

It is possible that segmentation could have arisen first from the repeated arrangement of certain body units with the later development of interseptal partitions separating these units from each other.

Summary – Annelida

1. Members of the phylum Annelida are vermiform (wormlike) segmented animals.

Segmentation or metamerism, which is a distinguishing feature, may have evolved as an adaptation for peristalisis burrowing in soft substrata.

2. The primary segmental structures are the coelomic body compartments which locally isolate pressure changes and as a result minimize muscle involvement during peristalic burrowing.
3. Segmental chaetae or setae increase traction with the substratum.
4. A fibrous cuticle covers and protects the body.
5. The nervous, circulatory and excretory systems are segmented for as maintenance systems. They accommodate the primary segmentation of the coelomic compartment.
6. The gut is typically a straight tube extending through the body between the anterior mouth and posterior anus.

It is convenient now to proceed to discuss the basic level of organization of the recognized annelid groups.

Class Polychaeta e.g Bristle worms, *Nereis, Eunice*

- Forms the largest and oldest group pf the annelida with more than 10,000 species and 1600 generations.
- The polychaetes are almost entirely marine.
- They are distinguished by possession of limb-like or paddle-like projections from the body called parapodia. usually bear bunches of stiff, hair-like chaetae or setae.
- Burrowing polychates such as *Arenicola,* much prized a fishing bait have reduced parapodia.
- Many are highly coloured (red, green etc) and some are very beautiful.
- Size range varies from less than a millimeter to about 3 metres in length.
- **Ecological classification of polychaetes**:- divides them into two subclasses, the errant or free-moving, **Errantia** and the sedentary tube dwellers or **Sedentaria.**

a) Errantia include active burrowers, pelagic forms crawlers and tube dwellers. They may leave their tubes for food or for breeding or may even carry their tubes about with them

b) The Sedentaria live in tubes and rarely expose more than the head from the end of the tube.

Tube Structure and building

The tubes are built by cementing sand grains and bits of shell together by mucus to construct more or less cylindrical tube.

Mucus secreted by special gland shields may simply harden to form a protective covering around most of the body. The parapodia are usually reduced or absent and feeding arms develop from the head.

The tube may be smooth and regular or it may be jagged and rough in appearance.

Tube may be branched and very extensive in nearly all tube-building worms part of the tube projects above the surface as a chimney.

Tube-dwelling polychaetes are carnivorous, feeding of other animals or filter feeders, straining small organisms or food particles out of the surrounding water.

Morphological Features

Polychaetes may be classified morphologically as scale-bearing, jaw-bearing or crown-bearing forms.

Typically the polychaete worm consists of an anterior head or prostomium. The head may bear eyes, antennae and a pair of palps.

The first segment, the peristomium, surrounds the mouth fornimg the lower lip and may bear setae and other complex structres such as the crown.

The peristomium may fuse with more posterior segments to form a food-gathering structre. It also often bears a pair of palps. (Fig)

The trunk consists of a definite or indefinite number of segments. Most body segments of the Sedentaria have a pair of lateral fleshy expansions called parapodia. (Fig)

Each parapodium may be single (**uniramous**) or double (**biramous**). Each parapodium may also have emergent setae. The type of setae and arrangement has taxonomic value.

Functions of Setae (chaetae)

(i) Swimming, (ii) Gripping surfaces (iii) reproduction, (iv) respiration (v) as organs of defense or offense and (vi) sensory organs

In many pelagic polychaetes the setae may be entirely absent and the parapodia function as paddles in swimming.

Sexual reproduction in the Polychaeta

Ploychaetes have separate sexes **(dioecious)** with gonads repeated along the length of the body, as also are the excretory organs.

There is external fertilization.

Hermaphroditism is widespread and may be simultaneous or sequential.

Gametogenesis

Most polychaetes that have been studied lack permanent gonads. The origin and Proliferation of gametes is poorly understood.

Usually, gametogonia or gametocytes are liberated into the coelom from cells lining the peritoneum. Gametes then mature while floating in the coelomic fluid. In a number of polychaete families however, the females have distinct ovaries and the males have definite testes. Most ovaries lie beneath the gut on the peritoneum.

The peritoneum can have cells in close contact with the developing oocytes, termed follicle cells and may also be associated with blood vessels and /or nurse cells.

When the oocytes are released into the coelom they may be solitary or continue to be associated with nurse cells.

The testes of polychaetes also lie in the peritoneum, but usually contain only spermatogonia and stem cells.

Fully differentiated sperm have not been found in testes.

In nearly all polychaetes the sperm develops in the coelom in syncytial masses (rosettes, morulae or platelets).

In most polychaetes, gametes are expelled from the body via nephridia, although in some species their release results from rupture of the body wall.

Fertilization Mechanisms

Many polychaetes shed eggs or sperms freely into the water, where fertilization takes place. There is essentially no interaction between the sexes and no parental care.

Many benthic polychaetes swarm to the surface regions of the water and spawn.

Sperms may be released in spermatophores. These may float in the water and be gathered by the female or may be transferred to the female during pseudocopulation or copulation.

Larval Development

There is a great range of larval forms and development within the polychaetae. The most notable variations upon the body type are shown by the burrowers in which the head is reduced, but which posses a proboscis that is utilized in burrowing. The coelomic space is large and uninterrupted by septa, enabling shunting of body fluid to and fro during the process.

Reproduction and development

Asexual Reproduction in the Polychaeta

Asexual reproduction takes the form of division of the body and regeneration of the missing parts (**schizotomy**).

Schizotomy has been further divided into two processes, **paratomy** and **architomy.**

Paratomy is the formation of a recognizable complete individual which then separates from the 'parent' stock.

Architomy is simple fission or fragmentation of the body with no prior individualisation.

Nearly all develop via a planktonic trochophore larva.

Summary- Polychaeta

1. The evolution of the class Polychaeta from the ancestral burrowing annelids is perhaps correlated with a shift to a crawling, surface existence. Various lines then diverged to invade other habitats and to assume other modes of existence, including burrowing in soft buttons.
2. Surface-dwelling, errant polychaetes possess well-developed parapodia and heads (prostonium) with sense organs. They crawl with the parapodia, using them as leglike appendages. Most are predaceous, but some are herbivorous or scavengers. They typically possess an eversible pharynx equipped with jaws.

3. Gallery-dwelling and burrowing polychaetes show some convergence with earthworms. The prostomium is usually small and more or less conical with poorly developed sense organs.

The parapodia are smaller than those of surface dwellers and provide anchorage in peristalic movement through the galleries. Gallery dwellers are predators or deposit feeders.

4 The more sedentary burrowers, which live in simple vertical or U-shaped excavations, move by peristalic contractions and possess parapodial gripping the mucus-lined burrow walls. Those species that are nonselective deposit feeders have a small prostomium without conspicuous sense organs; those that are selective deposit feeders usually possess head appendages, such as tentacles or long palps, which are used in feeding.

5 Tubicolous polychaetes live in tubes composed of secreted materials or of sand grains or shell fragments cemented together. The majority are selective or nonselective deposit feeders or filter feeders and are adapted much like sedentary burrowers. The few that are predatory are similar to errant surface dwellers.

6 Most large polychaetes possess gills (thin walled evaginations with an interior vascular supply), or some part of the parapodium is especially modified as a gas exchange surface.

7 Internal transport is provided by a blood vascular system by coelomic fluid or by a combination. Gas transport frequently involves respiratory pigments of which haemoglobin is the most common, but some polychaetes possess **chlorocruorin** or **hemerythrin**.

8 Most polychaetes possess paired segmental mentanephridial system, in which each nephrostome opens into the coelomic compartment that is anterior to the one housing the tubule. Polychaetes that lack a blood-vascular system possess nephridia which may be primitive for the phylum.

9 Primitively, Polychaetes have a ladder-like ventral nervous system.

10 The sexes are separate in most polychaetes. The gonads are in the connective tissue but the gametes are often released into the coelom for maturation and storage.

After matruring in the coelom, they exit by coelomoducts, coelomoducts joined to nephridia alone or rupture of the body wall.

In primitive polychaetes, gametes are associated with most segments.

11 Copulation is rare, and synchronous emission of sperms and eggs is important.

12 A trochophore larva is the basic larval stage of polychaetes.

Class Oligochaeta -earthworms

Oligochaetes are freshwater, terrestrial and marine annelids most burrowing in mud or soil. Some measure less than ½ mm; while the giant (terrestrial earthworms) Australian earthworm grows to 3m long.

They have both external and internal segmentation. Annelids with few chaetae that are not mounted on parapodia (**oligo=few**). A characteristic arrangement is a pair of ventrolateral bundles and a pair of ventral bundles on each segment (Fig) The setae are produced in setal sacs in the body wall. (Fig)

Setae are manipulated by protractor and retractor muscles and can be set at different angles for effective use.

The head is reduced, probably an adaptation to burrowing.

Oligochaetes are foraging herbivores, obtaining food (leaves and other vegetable matter) both from the surface of the ground and during the course of burrowing. The gut is adapted for varied diet and cellulases and chitinases may be among the digestive enzymes present.

They feed on dead organic matter, particularly plant material.

Earthworms and other species ingest soil as they burrow, digest any organic matter and void the rest as worm casts, which can often be found in a garden. Earthworms play a crucial rolein loosening, aerating and mixing soil.

Researchers use earthworms as model animals in experiments to test for toxic pesticides or wastes that may make their way into soil.

Unlike the polychaetes, oligochaetes are hermaphroditic; each worm has both male and female organs and fertilization is internal.

During copulation, two animals touch each other and each passes sperms to the other.

Fertilized individuals later lay eggs in a cocoon secreted by the **clitellum** (modified region of the epidermis) is the most obvious structure on the outside of an earthworm.

Parthenogenesis may also occur in a few species. Asexual reproduction is common in aquatic families, by transverse fission. Considerable regeneration occurs in oligochaetes.

The nephridiopores by which excretory (osmoreguatory) organs open to the exterior and dorsal pores are found in each segment.

The gut is notable for the presence of calciferous glands associated with the oesophagus.

Movement

Locomotion of terrestrial forms involves alternate contraction of the circular and longitudinal muscle layers upon the enclosed fluid-filled coelomic space with the resulting extension and contraction of the body.

This peristalic progressive motion is aided by the gripping and anchoring action of the setae on the substratum. Peristalic waves pass along the body, providing alternately elongating or shortening portions.

The coelom is well developed and is segmentally partitioned by septa.

The digestive tract consists of a buccal cavity, pharynx, oesophagus and intestine. The intestine is a straight tube lined with cilia, secretory cells and absorptive epithelial cells.

Respiration is chiefly a functionof the body surface in oligochaetes but some specialized areas (gills) are found in polychaetes and a few oligochaetes and leeches.

The circulatory system is a closed system. The blood is colourless in some species but usually carries haemoglobin or erythrocruorin in solution in the plasma producing a red colour.
The formed elements or amoebocytes are colourless but are phagocytic.

Development

The entire development of the egg into the juvenile worm takes place in the cocoon. There are no larval stages, for development is direct.

Summary-Oligochaeta

1. Members of the class Oligochaeta are believed to have evolved independently of the Polychaeta from the ancestral annelids.

Most move using peristalsis and have well-developed segmentation and a simple prostomium.

2. The first oligochaetes were probably inhabitants of freshwater sediments. Some lines then successively invaded drier substrata to give rise to the earthworms. Others remained aquatic but became adapted for living in loose debris and algae.
3. The digestive tract of oligochaetes is adapted for a diet of decomposing organic matter, largely plant material.
4. Gills are generally absent, but cutaneous vascular networks are well developed in larger species, especially earthworms.
5. Oligochaetes have a metanephridial system like that in polychaetes, consisting of segmentally arranged tubules.
6. Many of the adaptations of earthworms for life on land are behavioural but others include the presence of calciferous glands and enteronephric nephridia, the production of urea, encystment and egg deposition in cocoons.
7. Oligochaetes are hermaphrodites with well-developed reproductive systems limited to a few segments. There is copulation and reciprocal sperm transfer.

Fertilization and direct development occur within a cocoon secreted by the clitellum.

Class Hirudinea – The leeches

About 500 species, comprise the smallest annelid class.

Most live in fresh water, but some are marine and a few live in humid jungles.

They have a constant number of segments (34).

There are no paraodia and no chaetae.

Have a clitellum in segment 9,10 and 11 for secreting a cocoon.

Are hermaphrodite with well-defined gonads. Leeches often stay with their eggs, arching over them and moving up and down, ventilating them with fresh oxygen-bearing water. The young hatch as little leeches not larvae, and often fasten onto the parent until they find their first meal.

Have separate nephridia and coelomoducts. They lack setae.

Leeches possess an anterior and posterior sucker for attatchment and for clinging to their host to suck blood; however, some are not parasitic.

A few have become predators and feed on earthworms and insects, which they swallow whole.

Unlike other annelids, sizes vary from less than ½ inch long, and the larger ones may exceed a foot in length when fully extended.

All have great powers of extension and contraction.

Of the characteristic suckers, the anterior is usually smaller than the posterior and generally surrounds the mouth.

The body is commonly divided into five regions:

(1) a head region of four segments bearing the ventral oral sucker, mouth, eyes and jaws (in some).

(2) A preclitellum of four segments (5,6,7,8).

(3) A clitellum of three segments (9,10,11)

(4) A middle region of 15 segments (12 to 26)

(5) The terminal region of seven segments modified to form the posterior sucher.

Leeches evolved new modes of locomotion in place of annelids ancestral hydrostatic crawling and burrowing.

Their derived character include a flattened body with a sucker at each end, and loss of the setae and septa (and hence loss of internal segmentation).

In most species, a series of small coelomic channels is all that is left of the circulatory system.

Using this modified body plan some leeches move inch-worm-fashion: they attatch the posterior sucker to a surface, extend the body forward, attatch the anterior sucker, and pull the rear up to join the front end. Others are swift and graceful swimmers, throwing their long bodies up and down into a series of curves.

Summary- Hirudinea

1. The presence of a clitellum, the similarity of reproduction and the existence of a single genus with anterior setae and coelomic compartments are evidence that members of the class Hirudinea evolved from an ancestor shared in common with the oligochaetes.
2. Leeches possess an anterior and posterior sucker.

   They lack setae, extenal segmentation and septa.

   They have expanded connective tissue and a reduced coelom that is specialized into a circulatory system. All of these features are probably correlated with a change from peristaltic locomotion to movement involving extension and contraction of the entire body and attatchment with the suckers.
3. The shift in the mode of movement is probably related to the assumption of predaceous and ectoparasitic feeding habits.

4. A proboscis or jaw with a pumping pharynx is utilized in both predaceous and ectoparasitic feeding.
5. Vertebrates are the principal hosts of ectoparasitic or blood-sucking leeches. Digestion of blood, which is slow, depends on exopeptidases produced by the leech and a symbiotic bacterial flora.
6. Segmentation is reflected internally in the nephridia and the ganglia of the ventral nerve cord.
7. The blood vascular system is similar to that of other annelids or has been replaced by coelomic channels.
8. Reproduction is similar to that of oligochaetes, but some species transfer sperm by hypodermic impregnation with spematophores.

Definition of Key Words

Biramous – Having two branches or rami (of arthropod limbs)

Chaetae – Chitinous bristles of polychaete worms (sometimes called setae)

Cestodes – see Tapeworms

Cilia – Characteristic of many protozoan and metazoan cells, motile outgrowths of the cell surface that is typically short and its effective stroke is stiff and oarlike.

Cleavage – The process of repeated cell division of the fertilized egg to form a ball of cells that becomes the blastocyst. The cells do not grow between divisions and so they decrease in size.

Clitellum –saddle –like region of some Annelida (earthworms, leeches). Prominent in sexually mature animals.Contains glands which secrete a mucus sheath around copulating animals thus binding them temporarily tother, and a cocoon around fertilized eggs.

Coelom – Body cavity between the body wall and gut wall, lined by an epithelium derived from mesoderm.

Dioecious – Having male and female reproductive systems in separate individuals. An alternative term is gonochoristic.

Hermaphrodite – Having male and female reproductive systems in the same individual. An alternative term is monecious.

Metamerism – The division of an animals' body into a linear series of similar parts or segments

Metamorphosis (metamophoses) – Transformation from a larva into an adult.

Mesodermal – Embryonic germ layer that forms the tissues situated between the ectoderm and endoderm.

Nephridia – Excretory tubules that usually open to the exterior via nephridiopores. The inner ends of the tubules may be blind (protonephridia) ending in terminal cells or may open into the coelom (metanephridium) through a ciliated funnel.

Parapodia – Segmental appendages of polychaete annelids, containing an extension of the coelom.

Pelagic – Living, floating or swimming above the water column.

Proglottids– Section of the body of a eustode (tapeworm), containing a set of reproductive organs.

Prototroch – Pre-oval ring of cilia of a trochophore larva.

Schizoelic – Coelomic cavity formed by splitting within mesoderm.

Scolex (Scoleces) – Anterior head region of tapeworms that is adapted for adhering to the host.

Setae – Bristles, especially in crustaceans and other arthropods but also in polychaetes.

Septa – Transverse partitions in annelids separating the coelomic cavities of successive segments.

Triploblastic – Describing an animal in which the body is developed from three cell layers; ectoderm, mesoderm and endoderm.

Uniramous – (of arthropod limbs) having a single branch or ramus.

Zooids – Basic body units in a colonial animal.

Further Reading

1. Barnes, E.E. and Barnes, R.Dr (1991) Invertebrate Zoology. Sixth Edition. Saunders College Publishing.
2. Hickman, C.P. (1967) The biology of the invertebrates.
3. Ruppert, E.E. (1994) Invertebrate Zoology 6th edition. Saunders College Publishing.

Questions

1. List five diagnostic features of the phylum Annelida.
2. Describe the main anatomical features of the phylum Echinodermata.

a) State three distinguishing features of the phylum Mollusca.

b) Give two important classes in this phylum.

3. Outline the affinities between annelids and arthropods that reflect their common ancestry.
4. Name the main extant groups of arthropods and give the main distinguishing features.
5. Describe the main types of metamorphosis found in insects.
6. Explain briefly the economic importance of insects.
7. Give the five classes of Uniramia and explain briefly why these arthropod taxa are grouped together.
8. Give the main characteristics of upepods and briefly explain why upepods are ecologically important in the marine ecosystem.

LESSON

Phylum Arthropoda

- **Structure**
- Introduction
- Objectives
- Diversity and evolution in arthropods
- General characteristics of the phylum Arthropoda (i.e what defines an arthropod)

Arthropod classification: Subphyla

(i) Trilobita

(ii) Chelicerata

(iii) Crustacea

(iv) Uniramia

Classes:

(i) Insecta

(ii) Arachnida

(iii) Crustacea

(iv) Myriapoda

Arthropod Body Plan

- Feeding, Respiration, Internal Transport, Excretion, Response
- Ecological aspects and adaptation of arthropods
- Summary
- Definition of key words
- Further Reading
- Questions

Introduction

Welcome to Lesson-   in which you will learn a lot about arthropods. Arthropods belong to the phylum **Arthropoda**. (Gr. Arthro- joint, pod- literally means foot and refers to the appendages). The significance of this lesson is that it presents you with a chance to learn about the largest and most diverse phylum in the Animal Kingdom. You already know about many common arthropods. In fact, you have probably even eaten a number of them! All arthropods belong in a large group of animals called invertebrates – animals without backbones. Earthworms, snails, slugs and clams are invertebrates but are arthropods because they do not have jointed legs.

Arthropods and vertebrates are the only two animal groups with many members fully adapted for life on land. Arthropods have evolved fundamentally different patterns of living e.g parasitic habits have invaded land independently with varying degrees of success.

A number of terrestrial groups have successfully adapted to living in the water for all or a part of their lives. This remarkable adaptive plasticity has led to the earth's land or water formations where temperatures rise above freezing for a long enough time to permit breeding.

Objectives:

- After reading this lesson you should be able to :
- Describe the subphyla and classes of arthropods.
- Explain how arthropods perform their essential life functions.
- Describe the evolution of arthropods
- Distinguish between members of this phylum
- Discuss growth and development in arthropods.
- Explain special adaptations of arthropods
- Explain why there are so many different arthropods
- Know the ecological aspects if importance, habitats and biology.

The diversity of arthropods is described in this lesson with emphasis on evolution from the sea to freshwater and land.

Why are there many kinds of arthropods?

One reason is that they have been evolving on Earth for a long time.

The first arthropods appeared in the sea more than 600 million years ago. Since that time, arthropods have experienced several adaptive radiations. Some arthropods have remained in the water, where they have colonized all parts of the sea and most freshwater habitats. Others were among the very first members of the animal kingdom to colonize the land.

The descendants of those pioneers were on hand when the first flowering plants appeared millions of years later.

Arthropod Ancestors

The roots of the arthropod family tree were soft-bodied animals that left few fossils. But by studying both living and fossil invertebrates, clues to arthropod evolutionary history have accumulated.

Diversity and evolution in arthropods

More than one million of arthropod species have been described, the class insecta accounting for the most majority.

There are many more in the tropics that have not been found. Arthropods occupy about every conceivable ecologic niche from low ocean depths to very high altitudes and from tropics far into both north and south Polar Regions.

They occur in the sea, in fresh water, brackish water, on land, in air and the bodies of other animals.

As you will soon learn, members of this phylum vary enormously in size, shape and habits. They range from microscopic mites to giant Japanese spider crab (10-12ft).

Arthropods are protostomes and are clearly related to annelids, but whether arthropods arose from annelids or both from some common ancestor is uncertain. The annelidan ancestry of arthropods is reflected in their segmentation, the plan of their nervous system, and their determinate cleavage.

Insects, centipedes and millipedes seem to have evolved from ancestors closely related to the ancestors of modern annelid worms.

Evidence for this line of descent can be found in the form of wormlike animals that live today in the moist tropics only. Other arthropods, including crustaceans, spiders and the extinct trilobites, evolved from more ancient and more distantly related ancestors.

The body form of the earliest arthropods is thought to be similar to that of trilobites.

A typical Trilobite (Fig)

- Head: a thick, tough outer covering
- Body composed of many segments: an anterior cephalon, a middle trunk region of unfused segments and a posterior **pygidium.** Each of the segments bore a pair of appendages. The cephalon carried a pair of antennae and a pair of dorsolateral eyes.
- Each appendage was divided to form two branches, one a walking leg and one a feather-like gill.
- Variations in form indicated that there was some diversity in trilobite living habits, such as burrowing epibenthic crawling, planktonic and swimming.

Evolutionary trends of most living arthropods

- Many have far fewer body segments.
- Many segments found in their embryos fuse into larger segments during development.
- Appendages have become lost or increasingly specialized for feeding, locomotion and other functions.

The Evolutionary Relationships between Arthropods and Annelids

1. Arthropods, like annelids are segmented.

Segmentation is evident in the embryonic development of all arthropods and is a conspicuous feature of many adults, especially the more primitive species.

2. The nervous systems of annelids and arthropods are constructed on the same basic plan. In both, a dorsal anterior brain is followed by a ventral nerve cord containing ganglionic swellings in each segment.
3. The embryonic development of a few arthropods still display some degree of spiral determinate cleavage, with the mesoderm in these forms arising from the 4d blastomere.
4. In the primitive condition, each arthropod segment bears a pair of appendages. This same condition is displayed by the polychaetes, in which each segment has a pair of parapodia.

General characteristics for the classification of arthropods

- are extremely well developed: rapid and precisely controlled movement is The arthropods include crustaceans, insects, centipedes, millipedes, symphylans, pauropodans and the extinct trilobites.
- All arthropods are characterized by a segmented body. The body segments are of ten grouped into regions, for example, head, thorax and abdomen. The body is covered by a jointed external tough skeleton (chitin-protein exoskeleton), with paired jointed appendages on each segment.
- A complex nervous system with a brain located in dorsal part of the head, connective nerves passing around the anterior end of the digestive tract and a ventral nerve cord with a ganglion in each body segment. The brain, sense organs and nervous system characteristic of arthropods and complex behaviour has evolved.

- An open circulatory system (hamocoel) powered by a single dorsal heart into which blood flows through paired openings (ostia).
- A greatly reduced body cavity (coelom) is a haemocoel containing the blood which enters the dorsally placed heart through holes (ostia). Because the jointed exoskeleton blocks growth of the organism, it must be shed periodically. The cuticle is moulted at intervals when the young animals increase in size. This phenomenon called moulting or ecdysis, is a characteristic feature of the phylum and is controlled by hormones.
  It permits rapid growth in size and significant change in body form until the new exoskeleton, secreted by the animal has hardened.
- Arthropods are mainly terrestrial, but aquatic representatives are well known.

Respiratory and excretory organs differ greatly between aquatic and land forms. Malpighian tubules, a second type of excretory organ, are associated with the gut and are a new development in many terrestrial arthropods.

- Typically sexes are separate (dioecious).
- The need for external fertilization is often associated with complex behaviour.

The body plan and physiology of arthropods

- The exoskeleton (**exo-** means outside) is a system of external supporting structures that are made primarily of the protein chitin.
- Some of the exoskeletons, such as those of most insects, are leathery and flexible.
- Exoskeletons of ticks, crabs and lobsters are extremely hard.
- These tough exoskeletons protect body from physical damage.
- The exoskeleton of many terrestrial arthropods are waterproof. This adaptation restricts the loss of water from the body and enables arthropods to live in extremely dry environments such as deserts.

- The exoskeleton also helps arthropods move efficiently and adapt to their environment in many ways.

Disadvantages of exoskeletons

- Because an exoskeleton is a solid coating, not a living tissue, it cannot grow as the animal grows.
- Movement can occur only at the joints of the armour.

Feeding

The appendages of arthropods have evolved in ways that enable these animals to eat almost any food you can imagine.

Every mode of feeding is seen in arthropods, herbivores, carnivores, parasites, filter-feeders and destritus feeders.

Although some herbivores, such as locusts eat just about anything green, other herbivores are more selective.

Some herbivores are specialized to eat specific parts of plants. Others feed exclusively on a particular kind of plant.

Some carnivores, such as spiders, praying mantises, centipedes and king crabs, catch and eat other animals. Other carnivores, such as many crabs and crayfish, feed primarily on animals that are already dead.

External parasites-such as ticks, fleas, and lice drink blood and body fluids or nibble on the skin of other animals, including humans.

Some internal parasites passively absorb nutrients through the body wall whereas others eat away the host from inside. Many marine arthropods are filter feeders that use comblike bristles on their mouthparts or legs to filter tiny plants & animals from the water.

Respiration

Arthropods have evolved four basic types of respiratory structure

- 1.) Gills
- 2.) Book gills
- 3.) Book lungs
- 4.) Tracheal tubes

Some species completely lack specialized respiratory organs.

Many aquatic arthropods such as crabs and shrimps, have gills that look like a row of feathers located first under cover of the skeleton.

These gills are formed from part of same appendages that form mouthparts and legs.

Movement of the mouthparts and other appendages keeps on steady stream of water moving over the gills.

Book gills (which are formed in horseshoe crabs) and **booklungs** (which are found in spiders and their relatives) are unique to these arthropods. In both these structures, several sheets of tissue are layered like pages in a book.

The many tissue layers increase surface area for gas exchange.

A horse-shoe crabs book gills are carried beneath its body, whereas a spiders' book lungs are contained inside a sac within the body.

An opening called the spiracle connects the sac containing the book lungs with the fresh air outside.

Most terrestrial arthropods – insects, some spiders and millipedes, for example: - have another respiratory device found in no other animals.

From spiracles, long branching tracheal tubes reach deep into the animals' tissues. The network of tracheal tubes supplies oxygen by diffusion to all body tissues.

As the arthropods walk, fly or crawl, the movements of their body muscles cause the tracheae to shrink and expand, pumping fresh air in and out of the spiracles.

Tracheae tubes work well only in small animals; large animals require a more efficient way to deliver oxygen and remove carbon dioxide.

Internal Transport

In arthropods, a well-developed heart pumps blood through an open circulatory system.

In spiders and some insects, the heart is long and narrow and stretches along the abdomen.

In lobsters and crayfish, the heart is smaller and lies about halfway down the body.

When the heart contracts, it pumps blood through arteries that branch into smaller vessels and enter the tissues.

The blood leaves the vessels and moves through spaces in the tissue called sinuses.

Eventually, the blood collects in a large cavity surrounding the heart from which it re-enters the heart through small openings and is pumped around again.

Excretion

In arthropods, as in many other animals, undigested food becomes solid waste that leaves through the anus. The nitrogen-containing wastes that result from cellular metabolism are removed in different ways in different arthropods. Most terrestrial arthropods, such as insects and spiders dispose of nitrogen-containing wastes by using **malphigian tubules.** (Fig)

Malphigian tubules like other arthropod organs are bathed in blood inside the body sinuses.

The tubules remove wastes from this blood, concentrate them, and then add them to undigested food before it leaves the anus.

Terrestrial arthropods may have small excretory glands at the bases of their legs in addition to, or instead of malphigian tubules. In aquatic arthropods, cellular wastes diffuse from the body into the surrounding water at unarmored places such as the gills.

Many aquatic arthropods, such as lobsters, also eliminate nitrogen-containing wastes through a pair of green glands located near the base of the antennae. These wastes are emptied to the outside through a pair of openings on the head.

Response

Most arthropods have well-developed nervous systems. All have a brain that consists of a pair of ganglia in the head. These ganglia serve as central switchboards for incoming information and outgoing instructions to muscles. From the brain, a pair of nerves runs around the oesophagus and connects the brain to a nerve cord that runs along the ventral part of the body.

Along this nerve cord are several more ganglia, usually one for each original body segment. These ganglia serve as local command centres to coordinate the movement of legs and wings. (That is why many insects can still walk or flap their wings after their heads are cut off). Where many body segments have fused together, as in insects, there are several ganglia for each major body part.

Arthropods have simple sense organs such as statocysts and chemical receptors.

Most arthropods also have sophisticated sense organs such as compound eyes may have more than 2000 separate senses and can detect colour and motion very well. Many insects can see ultraviolet light, which is invisible to humans.

Both crustaceans and insects have a well-developed sense of taste, although their taste receptors are located in strange places

The chemical receptors associated with the sense of taste and smell are located on the mouthparts, antennae and legs.

Flies, for example, know immediately whether a drop of water they step in contains salt or sugar. Crustaceans and insects have sensory hairs that detect movement in the air or water. Many insects have well-developed ears that hear sounds above the human range. An arthropods' well-developed sense organs help it detect and escape predators.

The combination of these sense organs and a tough exoskeleton is enough to protect many arthropods.

Some arthropods have additional means of protection. Scorpions, bees and some ants have venomous stings, and many spiders and centipedes have venomous bites.

Lobsters and crabs can attack potential enemies with powerful claws. Many insects and millipedes fight back using nasty chemicals. Some arthropods trick predators by creating a divesion e.g some crabs can drop a claw or leg. This body part keeps on moving to distract predators while the rest of the animal scurries away.

The crab then grows back the lost limb.

Some hide through camouflage-matching the colour and texture of their surroundings so closely that they seem to disappear.

Others imitate the warning colouration of poisonous or dangerous species-a phenomenon called

Movement

Arthropods have well-developed muscle systems that are co-ordinated by the nervous system. Muscles generate force by contracting, then transfer that force to the exoskeleton.

At each body joint, some muscles are positioned to flex the joint and other muscle to extend it.

The pull of muscles against the exoskeleton allows arthropods to beat their wings against the air to fly, push their legs against the fround to walk, or beat their flippers against the water to swim.

Reproduction

Reproduction in most arthropods is simple. Males and females produce sperm and eggs respectively, and fertilization usually takes place inside the body of the female. In spiders and some crustaceans, the male deposits a small packet of sperm that the female picks up. In most insects and crustaceans however, the male uses a special reproductive organ to deposit sperm inside the female.

Growth and development in arthropods

Exoskeletons, as useful as they are, present a problem in terms of growth. Arthropods must replace their exoskeletons with larger ones in order to allow their bodies to increase in size as they mature.

The exoskeleton not only covers all appendages and sense organs, but also lines the gut as far down as the stomach.

In order to grow, all arthropods must moult, or shed, their exoskeletons. The moulting process is controlled by several important hormones, the most important of which is called **moulting hormone.**

Moulting Process

When moulting time is near, the epidermis (the layer that covers the outside of the body) digests the inner part of the exoskeleton and absorbs much of the chitin in order to recycle the chemicals in it.

After it secretes a new exoskeleton inside the old one an arthropod pulls completely out of its old exoskeleton.

Arthropods often eat what is left of the old exoskeleton.

The animal then expands to its new larger size and the new exoskeleton (which is still soft) stretches to cover it. The animal must then wait for the new exoskeleton to harden, a process that can take from a few hours to a few days.

Thus arthropods moult several times between hatching and adulthood. In most cases the process of growth and development involves metamorphosis or a dramatic change in form.

Some arthropods e.g grasshoppers, mites and crustaceans, hatch from eggs int young animals that look much like the adults. However, these young animals lack functioning sexual organs and often lack other adult structures such as wings. As the young grow, they keep moulting and getting larger until they reach adult size.

They gradually acquire characteristics of adults. In insects, this kind of gradual change during development is called **incomplete metamorphosis**.

Metamorphosis

Many insects, such as bees, moths and beetles, undergo a four-stage process of development called **complete metamorphosis**. (Fig)

The eggs of insects that undergo complete metamorphosis hatch into larvae that look nothing like their parents.

As these larvae grow they moult repeatedly, growing larger each time but changing little in appearance. When a larva reaches a certain age, it sheds its larval skin one last time and becomes a pupa.

During the pupal stage, the insects' body is totally rearranged – adult structures grow from tiny buds and larval structures are broken down to supply the raw materials for the adult structures.

When metamorphosis is complete, the animal emerges as a fully grown adult with both internal and external body parts that are completely different from what it had before.

Metamorphosis is controlled by a complicated interaction of several hormones, including moulting hormones.

In insects that undergo complete metamorphosis, the levels of juvenile hormone (JH) help regulate the stages of development.

High levels of juvenile hormone keep an insect in its larval form each time it moults.

As the insect matures, its JH decreases.

At some point, the level of JH drops below a certain critical point. The next time the insects moults, it becomes a pupa. And when no JH is produced, the insect undergoes a pupa-to-adult moult.

Functional Significance of Hormones

Because the balance of JH, moulting hormone, and other hormones is critical in arthropod development, it is possible to combat insects by tampering with their hormone levels.

Certain plants defend themselves against herbivorous insects by producing chemicals that prevent moulting, cause insects to develop at the wrong rate, or keep insects from becoming functional adults.

Researchers have developed chemicals that act in a similar manner. These chemicals may eventually enable people to control crop-eating insects without using dangerous poisons.

Arthropod classification

Arthropods are the real rulers of the earth and have colonized various habitats. Since many arthropod species remain undiscovered and undocumented the true number and diversity is undiscovered.

Today most biologists divide arthropods into four subphyla:

1. **Trilobita**: This is thought to be the oldest subphylum of arthropods. Trilobites were dwellers in ancient seas. They are now all extinct.
2. **Chelicerata**: Chelicerates include spiders, ticks, mites, scorpions and horseshoe crabs.
3. **Crustacea:** Crustaceans include such familiar (and edible) organisms as crabs, lobsters and shrimps.
4. **Uniramia**: Containing the centipedes, millipedes and insects.

The name uniramia refers to the fact that the appendages are basically unbranched and have been thought to be derived from an unbranched and have been thought to be derived from an unbranched condition. Uniramian's include most arthropods.

The diversity of arthropods described in the following sections will emphasise on morphology of members distinguished under four classes: Chelicerata, Arachnida, Myriapoda and Insecta.

Class Chelicerata

Class Chelicerata includes spiders, ticks, mites, scorpions and horseshoe crabs.

Chelicerates are arthropods that are characterized by a two-part body and mouthparts called chelicerae.

These arthropods lack antennae.

All chelicerates have a body that is divided into two parts: the cephalothorax (**prosoma)** and the abdomen (**opisthosoma).**

The anterior end of the cephalothorax contains the brain, eyes, mouth and mouthparts and oesophagus.

The posterior end of the cephalothorax carries the front part of the digestive system and several pairs of walking legs.

The abdomen contains most of the internal organs. (Fig)

All chelicerates have two pairs of appendages attatched near the mouth that are adapted as mouthparts.

The first pair of food-handling appendages, are the **chelicerae**,the second pair which is longer than the chelicerae, are called **pedipalps** and four pairs of legs.

Class Arachnida

The most familiar chelicerates are the arachnids, which includes spiders, scorpions, ticks and mites.

Members of the class Arachnida are terrestrial that lack book gills. All adult arachnids have four pairs of walking legs on their cephalothorax.

Arachnids are largely predatory chelicerates that have pedipalps adapted for capturing and holding prey and chlicerae adapted for biting and sucking out their soft parts. Other arthropods form the principal prey.

Spiders are predators that usually feed on insects. A few large tropical spiders are capable of catching and eating small vertebrates, such as hummingbirds.

Spiders capture their prey in webs. Others stalk and then pounce on the prey. Other spiders lie in wait beneath the lid of a camouflaged underground burrow, leaping out to grab unlucky insects that venture too near. Spider uses it hollow fanglike chelicerae to inject paralyzing venom into it.

When the prey is paralyzed, spider introduces enzymes into the wounds made by the chelicerae.

The enzymes break down the prey's liquified tissues into the digestive sustem.

Mites and Ticks – Are small arachnids, many of which are parasites on humans, on farm animals and on important agricultural plants. In many mites and ticks, the chelicerae are needlelike structures that are used to pierce the skin of their hosts.

Question: What are chelicerates? How are chelicerae modified for feeding in ticks?

These chelicerae may also have large teeth to help the parasite keep firm hold on the host.

The pedipalps are often equipped with claws for digging in aid holding on.

Large arachnids (scorpions, some spiders) possess book lungs as gas exchange organs; small forms (psedoscorpions, some spiders, mites) possess tracheae.

The heart is most highly developed in large species with book lungs, and blood contains **hemocyanin.**

Excretory organs are coxal glands and malphigian tubules. A waxy epicuticle is an important adaptation for water conservation and has certainly contributed to the success of arachnids as terrestrial arthropods.

Many are secretive or nocturnal in habit. Leaf mold is the habitat for many small species, especially pseudoscorpions, mites and spiders.

The primtive mode of sperm transfer is indirect by **spermatophores** (scorpions, pseudoscorpions and some mites).

The unique indirect sperm transfer of spiders is probably derived from handling of spermatophores by the male with the pedipalps. Sperm transfer is direct in many mites.

Class Crustacea

Crustaceans include such familiar (and edible) organisms as crabs, lobsters and shrimps. (Fig) Contains over 35,000 species. Crustaceans are primarily acquatic, although there are some terrestrial species. Most are marine but there are many freshwater species. Crustaceans vary in form and size ranging in size from microscopic water fleas less than 0.25mm long to Japanese Spider Crabs that grow up to 6mm and lobsters that have a mass of more than 20 Kgs.

Structural Characteristics

Crustaceans are characterized by a hard exoskeleton.

Crustaceans are unique among arthropods in having two pairs of antennae.

Have two pair of mouthparts called **mandibles** and two pairs of **maxillae**. The mandibles are short heavy structures designed for biting and grinding food. Also they are used in filter feeding, finding and picking up detritus or are needlelike structures used to suck blood from a host.

The main crustacean body parts are the head and thorax fused into a cephalothorax that is covered by a tough shell called the **carapace**. Unlike most other arthropods, many crustaceans have calcium carbonate (limestone) in the exoskeleton which makes the shells hard and stony.

The two pairs of antennae serve as sense organs and are used in filter-feeding in some other crustaceans.

Water fleas use antennae as oars to push them through water.

Questions: What is a cephalothorax?

Describe the types of appendages and their functions.

Appendages on the thorax and abdomen are branched and vary in structure and fuction.

On a crayfish's thorax and abdomen appendages are adapted for several different functions: -

A pair of large claws, which are used to catch prey and pick up, crush and cut food are located on the thorax.

Four pairs of walking legs are also attatched to the thorax.

Flipperlike appendages called **swimmerets**, which are used for swimming are located on the abdomen.

A large pair of paddlelike appendages are found on the second-to-last abdominal segment.

The paddlelike appendages and the final abdominal segment together form a large tail (**uropod**). This tail provides a powerful swimming stroke that rapidly pulls the animal backward.

Excretory organs are a pair of blind sacs in the haemocoel of the head that open onto the bases of the second pair of antennae (antennal glands),maxillae (**maxillary glands**).

Crustacean sense organs include two types of eyes: a pair of **compound eyes** and a small, median, dorsal **naupliar** eye composed of three or four closely placed **ocelli**. Some groups lack compound eyes, and the naupliar eye, characteristic of the crustacean larva does not persist in the adult of many groups.

Copulation is typical of most crustaceans, and egg brooding is common.

The earliest hatching stage is a naupliar eye, which possesses a median naupliar eye ond only the first three pairs of body appendages (a pair of claws, walking legs, swimmerets).

Summary – Crustacea

The 38,000 species of crustaceans constitute the only major group of aquatic arthropods. Most are marine, but there are many freshwater species and there have been a number of invasions of the terrestrial environment.

Crustaceans are extremely diverse in structure and habit, but they are unique among arthropods in having two pairs of antennae. Other characteristic head appendages are one pair of mandibles and two pairs of maxillae. The trunk specialization varies greatly, but a carapace that covers all or part of the body is common.

Crustacean appendages are typically biramous and depending on the group, have become adapted for many different functions.

Gills, which are absent only in species of minute sizes, are typically associated with the appendages, but the location, number and form vary greatly.

Crustacean classification

The diversity of the large groups of crustaceans include:

Branchiopoda: Brachiopods are small crustaceans e.g water flea Daphnia that are almost entirely restricted to freshwater.

Some are found in temporary pools, mostly absent from running water.

They have flattened, polyramous thoracic limbs (called phyllopods) that are used in swimming and filter-feeding. Abdominal limbs are reduced in number or absent. Most branchiopods are suspension feeders and collect food particles with fine setae on the trunk appendages.

The excretory organs are maxillary glands.

Copepoda: Largest class of small crustaceans 8500 species described. Characterized by the presence of a head shield and the absence of a carapace.

They have a single median, sessile eye, lack abdominal appendages and have well-developed caudal rami. Copepods occupy a wide range of aquatic and many terrestrial habitats and include a number of parasitic forms on various marine and freshwater animals particularly fish.

A major part of the diet of many marine animals is composed of copepods.

Both the thoracic appendages and second antennae are used in rapid swimming.

There are no gills in free-living copepods. The excretory organs are maxillary glands. Male copepods are smaller than females. Copepods are among the few small crustaceans that form spermatophores.

The spermatophores are transferred to the female by the thoracic appendages of the male. The eggs are shed singly in water in some species. Other copepods enclose eggs (50) in an ovisac, remain attatched to the female genital segment, where it functions as a brood chamber. The eggs typically hatch as **naupli**. After 5 or 6 naupliar instars the larva passes into the first copepodid stage. The adult structure is attained typically after six noupliar and five copepdid stages.

Some copepods are ectoparasites on fish and attach to the gill filaments, fins or the general integument. Other copepods are commensal or endoparasitic within polychaete worms, the intestine of echinoderms (particularly crinoids) and in tunicates and bivalves.

Cnidarians, especially anthozoans are the hosts for many species of copepods.

In most parasitic copepods, the adults are adapted for parasitism.

Ostracoda- (Mussel or seed shrimp)

Are small crustaceans, widely distributed in the sea and in all types of freshwater habitats.

Body is completely enclosed in a bivalve carapace.

Most ostracods are minute.

The majority live near the bottom, where they swim intermittently or crawl over or plough through upper layer of mud and detritus. Some are commensal on other animals, living, for example, among the leg setae of crayfish.

One or both pairs of antennae are the principal locomotor appendages and are variously modified.

Exhibit diverse feeding habits; some are carnivores and others herbivores, scavangers or filter feeders.

Algae are a common food plants and the prey of carnivorous species includes other crustaceans, small snails and annelids.

Gills are lacking and gas exchange is integumentary.

The locomotor and feeding currents provide ventilation.

Some ostracods possess antennal glands, some have maxillary glands and some possess both types of excretory organs in the adult.

Some ostracods use bioluminescence as a sexual attractant, much like fireflies. Ostracods were the first crustaceans in which bioluminescence was observed. Eggs are shed freely in the water or are attached singly or in groups to vegetation and other objects on the bottom but the eggs are brooded in the dorsal part of the shell cavity in some ostracods.

On hatching the juveniles are enclosed in a bivalve carapace. The valve skeleton is shed at each moult but moulting ceases at adulthood. Parthenogenesis occurs in some freshwater species.

Crustacean classification

The three major classes of small crustaceans are the Branchiopoda, the Ostracoda, and the Copepoda.

Brachiopods are distinguished by their foliaceous appendages, which in many species are adapted for suspension feeding.

Water fleas (cladocerans) have a carapace that encloses the trunk but not the head.

Clam shrimp (conchostracans) have a bivalve carapace that encloses the entire body.

Fairy shrimp and brine shrimp (anostracans) lack carapace.

Brachiopods are largely inhabitants of freshwater, especially temporary pools and ditches.

Only cladocerans are also represented in permanent fresh, and to a lesser extent, marine waters.

Ostracods have a bivalve carapace impregnated with calcium carbonate that encloses the entire body.

Most ostracods are less than 2mm long. They are mostly marine benthic animals, but some species live in fresh water.

Ostracods have an extensive fossil record.

Copepods possess more or less cylindrical, tapered bodies.

Long, laterally projecting first antennae and a persistent naupliar eye are distinctive features of many species. The trunk appendages are markedly biramous.

Most copepods are less than 2mm long.

Most copepods are marine, but some species are common in freshwater lakes and pools.

There are planktonic, epibnthic and interstitial species.

There are many parasitic forms.

Feeding habits among these classes of small crustaceans are diverse.

Suspension-feeding, planktonic copepods are of great importance in marine food chains.

Many freshwater forms, especially branchiopods, undergo parthenogenesis, produce both dormant and rapidly hatching eggs, and encyst.

Barnacles, members of the class Cirripedia, are unique among crustaceans, indeed most arthropods, in being sessile.

A number of peculiarities of barnacles such as a carapace covered with calcareous plates, suspension-feeding, cirri, hermaphroditism, long tubular penis, and dwarf males, may be correlated with their sessility.

Free-living barnacles are attatched by a peduncle (stalk) or are sessile (stalkless). The peduncle represents the pre-oral part of the body and contains the cement glands.

The oldest known fossil barnacles are pedunculate.

Lepadids, which are extant pedunculate barnacles attatch to floating, inanimate objects, such as wood, or to pelagic animals.

Scalpellids attach to rocks and, in addition to the five large principal plates, have many small plates covering the peduncle and capitulum.

The sessile barnacles, which are believed to have evolved from the scalpellids, are stalkless, the pre-oral region being represented by the basis, which contains the cement glands. Only the paired terga and scuta are movable.

The other capitular plates form a circular wall around the barnacle Sesile barnacles are especially adapted for intertidal life on hard substrates that are subjected to waves, surge and currents.

Commensalism, which has evolved in all three major lines of barnacles, has resulted in reduction of the protective calcareous plates.

Commensalism has undoubtedly been the avenue of parastism which is characteristic of one third of the species of cirripeds.

Subphylum Uniramia

Contains more species and includes centipedes, millipedes and insects. Have one pair of antennae. Appendages are not branched.

They inhabit terrestrial habitats, some live in freshwater and a few species live in marine environments.

The gas exchange organs are tracheae.

The excretory organs are malpighian tubules.

Myriapoda

Include centipedes and millipedes.

The nervous system is not concentrated, and the ventral nerve cord contains a ganglion in each segment.

Indirect sperm transfer by spermatophore is highly developed.

Have approximately 3000 species of centipedes and 7500 species of millipedes

They are characterized by a long, wormlike body composed of many leg- bearing segments.

Gas exchange is typically by a tracheal system in which the spiracles cannot be closed (another path of water loss).

Live beneath rocks, stones, logs of trees, in soil and humus or other moist areas.

The head bears a pair of antennae and usually ocelli, but except in certain centipedes, true compound eyes are never present.

Excretion takes place through malphigian tubules.

Chilopoda- Include centipedes, are canivorous.

In addition to other mouthparts, they have a pair of poison claws in their head region.

Eat other arthropods, earthworms, small snakes, mice, etc.

Have many legs from 15 to 170 pairs of legs, depending on the species.

Each body segment bears one pair of legs except for the first segment (which bears the poison claws) and the last three segments (which are legless).

Distributed throughout the world in both temperate and tropical regions.

They live in soil and humus and beneath stones, bark and logs for protection not only from possible predators but also from dessication. At night they emerge to hunt for food or new living places.

Glands on the abdomen secrete was which is used to build storage chambers for food and other structures within a beehive.

Movement

As you have just read, insects have three pairs of walking legs.

These legs are often equipped with spines and hooks for holding onto things and for defence.

In addition to being used for walking, the legs may be adapted for functions such as jumping (e.g grasshoppers and fleas) or capturing and holding prey (as in praying matises).

Along with birds and bats, insects are the only living organisms that are capable of unassisted flight. Butterflies fly quite slowly and have limited maneuvarability, certain flies, bees and noths can hover, change direction rapidly and dart off at speeds up to 53 Km an hour.

In flying insects, most of the space in the thorax is taken up by large muscles that operate the wings.

The ernomous amount of energy required by these muscles during flight is supplied by oversized mitochondria (which are about half the size of a human red blood cell).

The wing muscles in many insects also have a special blood supply that helps retain heat produced by muscle activity e.g bees can maintain a wing muscle temperature of up to $35^{o}C$.

This means that bees can keep their flying muscles warmer than the outside temperature and operate efficiently even when it is cold outside.

Summary

Members of Subphylum Uniramia include the centipedes, millipedes and insects.

They are terrestrial arthropods with appendages that are primitively unbranched.

They possess one pair of antennae and the mouthparts include a pair of mandibles.

The Uniramians are believed to have evolved from terrestrial ancestors, which may have resembled members of the phylum Onychophora.

The myriapodous arthropods include the centipedes (class Chilopoda) and millipedes (class Diplopoda), plus two other small classes (Symphyla and Pauropoda).

All have long trunks with many segments and appendages. Tracheae provide for gas exchange and malpighian tubules for excretion.

Myriapods live in leaf litter and beneath stones, logs and bark. Many of their structural features are adaptations for locomotion.

Centipedes possess one pair of legs per segment. In many groups the trunk has been strengthened for a running gout by overlapping tergites or tergites of unequal size the larger extending into adjacent segments.

Centipedes are largely predacious, and prey (mostly other small arthropods) are caught and killed with a pair of anterior forcipules.

Millipedes possess two pairs of legs per segment a condition derived from the fusion of two original segments.

The millipede diplosegments appear to be and adaptation for pushing gait. The trunk is strengthened to withstand the pushing force generated by a large number of legs.

Most millipedes feed on decomposing vegetation.

Depending on the group, protection is gained from **repugnatorial** glands, coiling and rolling up.

The gonopores are located at the posterior end of the trunk in centipedes and at the anterior end of the trunk (third trunk segment) in millipedes.

The 70 species of onychophorans are terrestrial, caterpillar-like animals of the tropics and Southern hemisphere.

The soft body, which is covered by a thin, flexible, chitinous cuticle, is adapted for squeezing beneath stones, logs and other objects.

Onychophorans possess a pair of antennae, a pair of clawlike mandibles and many pairs of non-jointed, peglike legs.

Intenally, there is a combination of arthropod and annelid.

Features: Body wall of circular and longitudinal muscles segmented nephridia, reduced coelom, haemocoel and tubular heart and tracheae. The chitinous cuticle is periodically moulted.

In some species, sperms are transferred as spermatophores.

Some onychophorans are oviparous, but many breed their eggs internally and give birth to their young.

Diplopoda

Include millipedes or thousands-leggers

Each body segment is formed from the fusion or two segments in the embryo and this bears two pairs of legs.

They live in damp places under rocks and in decaying logs. Cosmopolitan in distribution.

They are herbivores, feed on dead and decaying plant material (vegetation).

When disturbed they roll up into a ball to protect their soft undersides.

Some defend themselves by secreting unpleasant toxic chemicals. The calcareous exoskeleton offers some protection to upper and lateral sides of the body.

Crawl slowly over the ground.

Eyes may be totally lacking, or there may be 2-80 ocelli arranged about the antennae.

Antennae contain tactile hours & chemoreceptors. Animal continually taps the substratum with antennae as it moves along.

Class Insecta

The class Insecta or **Hexapoda** is described as a successful group. A reasonably good estimate of the number of described species of metazoan animals is about 1,035,250, of which three-quarters are insects.

The great success of insects in terms of number of species and individuals and the extent of their geographical distribution might therefore, be attributed to the evolution of five significant features:

1.) A waxy epicuticle, reducing desiccation.

2.) Wings for flight enhancing access to food or optimum environmental conditions and other resources, as well as helping evading predators.

3.) Wing folding at rest, permitting utilization of confined microhabitats.

4.) A resistant eggshell, permitting exposure to more extreme environmental conditions.

5.) A development that includes a larva, permitting the juvenile insect to utilize different resources than the adult.

The class Insecta, containing more than 750,000 described species, is the largest group of animals. It is three times larger than all the other animals groups combined.

Insects have occupied every environmental niche on land, acquatic habitats and are absent only from subtidal waters of the sea.

Distinguishing characteristics

Insects are distinguished from other arthropods for having the following morphopgical characteristics: -

1.) Three pairs of legs attatched to the thorax.

2.) One or two pairs of wings on the thorax.

3.) The body is divided into three parts: - head, thorax and abdomen.

4.) The head typically bears a single pair of antennae.

5.) The head bears a pair of compound eyes.

6.) A tracheal system provides for gas exchange.

7.) Thorax composed of three segments: - prothorax, mesothorax and metathorax.

Insects get their name from the Latin word insectum, meaning notched, which refers to the division of their body into three main parts: head, thorax and abdomen.

Factors contributing to the success of insects

The powers of flight also evolved in reptiles, birds and mammals, but the first flying animals were insects.

Ecological Significance of Insects in the Terrestrial Environment

1.) Two thirds of all flowering plants depend on insects for pollination.

2.) Importance to humans e.g mosquitoes, lice, fleas, bedbugs, etc contribute directly to human misery.

Affect us indirectly as vectors of human disease or of diseases of domesticated animals.

Examples of diseases: -

Malaria, encephalitis and yellow fever.

Tsetsefly (sleeping sickness)

Lice (typhus and relapsing fever)

Fleas (bubonic plague) and the housefly (typhoid fever & desentry)

3.) Some Insects destroy crops and greatly redu the high agricultural yield necessary to support large human populations.

4.) Vast sums of money are expended to control insect pests.

5.) The overzealous use of pesticides can in turn be hazardous to the environment and to human health.

External Morphology

The exoskeleton of an arthropod segment is composed of **a tergum, a sternum and two pleura**. Some parts are more highly sclerotized and are called sclerites. They are often separated by sutures. Insects are extremely varied in body shape and habits.

Insect Body Regions

The heads of most insects are oriented so that the mouthparts are directed downward (**hypognathous**). A more specialized anteriorly directed position (prognathous) is found in some predaceous species e.g beetles; a posteriorly directed position (opisthognathous) is present in the hemipterans and homopterans which have sucking mouthparts. (FIG)

Head

Head forms a complete, external capsule surrounding the soft, inner tissues.

The head bears one pair of compound eyes and one pair of antennae. Between the eyes and the antennae are usually three ocelli.

Three pairs of appendages contribute to themouthparts. One pair of mandibles is located anteriorly followed by a pair of maxillae and then the labium (representing a fused pair of second maxillae).

Anteriorly, the mandible is covered by a shelf like extension of the head, forming an upper lip or **labrum.**

From the floor of the prebuccal cavity projects a median lobelike process called the **hypopharynx**. The hypopharynx arises behind the maxillae near the base of the labium.

Thorax

Forms the middle region of the insect body.

Composed of three segments: **a prothorax, a mesothorax and a metathorax**.

A pair of jointed legs articulates withthe pleura on each of the three thoracic segments.

The thoracic terga of insects are called nota and it is with the nota and pleural processes of the mesothorax and metathorax that the two pairs of wings articulate.

Feeding

Mouthparts can take on ernomous variety of shapes on species adapted to feed on different foods e.g grasshoppers' mouthparts are designed to cut and chew plant tissues into a fine pulp.

A female mosquito's mouthparts form a sharp tube that is used to perce skin and sucks blood.

A butterfly's mouthparts that are fused together to form a long tube that is used for sipping nectar.

A bee has mouthparts that are used for chewing and gathering nectar.

A fly has a spongy mouthpart that is used to soak mop up food.

Many insects produce saliva that contains digestive enzymes and helps break down food. The saliva of female mosquitoes, which is injected when the mosquito bites, contains chemicals that prevent blood from clotting.

Honeybees have a number of adaptations for gathering, processing and storing food. The legs and bodies of worker bees are covered with hairs that collect pollen. Chemicals in bee saliva help change nectar into a more digestible form-honey.

The basal section of the leg articulating with sclerites in the pleural area is the **coxa,** which is followed by a short **trochanter.**

The remaining sections consist of **a femur, a tibia, a tarsus, and a pretarsus**.

The pretarsus is represented chiefly by a pair of claws.

The legs of insects are generally adapted for walking or running. Others are modified for grasping prey, jumping, swimming and digging. (FIGS)

Abdomen

The abdomen is composed of 9-11 segments plus a telson, but the telson is complete only in theprimitive proturans and in the embryos.

The only abdominal appendages in the adult are a terminal pair of sensory cerci borne on the 11^{th} segment.

A variety of abdominal appendages, serving different functions are present in many insects larvae.

Insect Wings

Wings are characteristic features of insects, but a wingless condition occurs in a number of groups. In some e.g ants and termites, the absence of wings is secondary.

Some parasitic insect orders e.g lice and fleas have lost wings completely.

Primitive apterygote insects in the orders Protura, Thysanura, collembola arose from ancestral wingless insects, winged insects and those that are secondarily wingless are referred to as **pterygotes.**

Because wings are evaginations, or folds of the integument they are composed of two sheets of cuticle separated by tubular thickenings called veins. Veins effectively support the wing and contain circulating blood and common tracheae and sense organs.

There is great variation in the wings of insects. In many insects the two wings on each side are coupled by interlocking devices so that the wings operate together as a simple unit.

In other insects the hind wings serve as hard protective plates, called elytra.

Members of the order Dipteral (flies and mosquitoes) have the second pair of wings reduced to knobs called **halteres**. They beat with the same frequency as the forewings and function as gyroscopes to offset flight instability.

Internal Anatomy and Physiology
Nutrition

Insects have adapted to all types of diets. Primitive mouthparts are adapted for chewing. The mandibles are heavy and capable of cutting, tearing and crushing and the maxillae and labium function in food handling.

The hypopharynx aids in swallowing. Insects with chewing mouthparts include dragonflies, crickets, locusts, and apterygotes.

The larvae of moths and butterflies also have chewing mouthparts. The diets of chewing insects may be herbivorous or carnivorous.

The specialization of mouthparts has been primarily in modification for piercing and sucking. Mouthparts may be adapted for more than one function: chewing and sucking, cutting and sucking, or piercing and sucking.

The mouthparts of moths and butterflies are adapted for sucking liquid food, such as nectar from flowers.

A part (the galea) of each of the two greatly modified maxillae forms a long tube through which food is sucked. (FIG)

When the insect is not feeding, the tube is coiled. The other mouthparts are absent or are vestigial.

Piercing mouthparts are characteristic of herbivorous insects such as aphids and leafhoppers, which feed on plant juices.

Some predaceous insects, such as assassin bugs and mosquitoes, which utilize the body fluids of other animals as food, also have piercing mouthparts.

In all these insects the mouthparts are elongated and are organized in various ways to form a beak e.g the beak of the plant-feeding and predaceous bugs (Hemiptera and Homoptera) consists of a stylet composed of the mandibles and the maxillae that lie in a groove on the heavier labium.

The stylet contains one lumen for the outward passage of salivary secretions and another for sucking in fluids. Other parts of the beak do not penetrate.

Bees and wasps have mouthparts adapted for both chewing and sucking.

Food taken into the mouth passes into the foregut, which is commonly subdivided into an anterior **pharynx, an oesophagus, a crop, and a narrower proventriculus**.

The pharynx is highly modified as a pump in sucking insects. The crop, when present is a storage chamber. The proventriculus in insects that eat solid food bears teeth or hard protuberances for triturating food.

Most insects possess a pair of salivary glands whose fuction varies, but usually secrete saliva, which moistens the mouthparts and may be a solvent for the food. The salivary glands may also produce digestive enzyes which are mixed with the food mass before swallowing.

The insect midgut (ventriculus or stomach) is the principal site of enzyme production, digestion and absorption.

Most insects possess out pocketings of the midgut called gastric caeca. Gastric caecae are the final site of digestion of principal sites of food absorption.

The hindgut or proctodeum cosists of an anterior intestine and a posterior rectum, both of which are lined by cuticle.

The hindgut functions in the egestion of waste and in water and salt balance.

In most insects, rectal pads or glands, occur in the epithelium. These organs are the pricipal sites of water reabsorption.

Digestion of cellulose by some termites, certain wood-eating roaches and cockroaches is made possible by the action of enzymes produced by protozoa or bacteria (cockroaches) that inhabit the hindgut of these insects.

Acetic acid formed by the breakdown of wood is actively absorbed by the hindgut epithelium in these insects.

Higher termites secrete cellulase but possess nitrogen-fixing gut bacteria.

A tracheal system provides for gas exchange. Spiracles are located along the sides of the thorax and abdomen but vary in number depending on the species.

The nitrogenous wastes of insects is uric acid, which is excreted through malpighian tubules. Insects are capable of producing a hyperosmotic urine, which together with the waxy epicuticle is an important adaptation for reducing water loss and has contributed to the success of insects as terrestrial animals.

The tubular heart is located in the lorsal part of the abdomen and propels blood anteriorly through a short aorta. The remainder of the blood vascular system is a haemocoel.

The insect nervous system is basically like that of other arthropods.

The brain is composed of a protocerebrum with eyes, deutocerebrum with antennae and a tritocerebrum. The ventral nerve cord forms a chain of median segmented ganglia. As in other arthropods the ventral segmental ganglia, both thracic and abdominal, are often fused. The greatest number of free ganglia is three in the thorax and eight in the abdomen.

The sub oesophageal ganglion is always composed of three pairs of fusal ganglia which control the mouthparts, the salivary glands and some of the cervical muscles.

Associated with a hypocerebral ganglion lying over the foregut and put beneath the brain are two pairs of glandular bodies: the **corpora cardiaca** and the **corpora allata**.

These two bodies, together with the **prothoracic glands** and certain neurosecretory cells in the protocerebrum, are the principal endocrine centres of insects. They produce hormones that control growth and development.

Other endocrine functions include regulation of water reabsorption heartbeat and certain metabolic processes.

A variety of sensilla located over the body surface, especially on the antennae and legs exhibit a range of receptor functions e.g orientation, food and mate detection, etc.

Questions

1.) In what ways are arthropods related to annelids?

2.) Describe mechanisms of survival that have enabled insects to be so abundant.

3.) Give characteristics common to both members of the group Myriapoda and Insecta.

4.) What were the advantagea to the arthropods moving to land?

5.) Distingish between apterygote and pterygote insects.

6.) Explain briefly the types of metamophosis found in pterygote insects.

7.) Give the characteristic features of Chelicerata.

8.) Explain the main structural features of insecta that have contributed to their biological success in the terrestial environment.

9.) Discuss briefly excreetion and water balance in a named crustacean indicating the main excretory wastes.

10.) Tabulate the external morphological features that distinguish diplopods from chilopods.

Phylum Mollusca

Structure

- Introduction
- Objectives
- Basic characteristics of the phylum Mollusca
- Ecological aspects and adaptations of Mollusca.
- Classification and morphology of molluscs.
- Summary
- Definition of key words.
- Further reading
- Questions

Introduction

Welcome to Lesson 7 in which you will learn about higher protostomes. Most of the animals in the groups covered thus far have been of relatively small size. One of the problems presented by increase in size is adequate distribution of materials to all parts of the body. With the appearance of the pseudocoelom, some distribution of materials was possible by way of the pseudocoelomic fluids.

However, the full development of complex animals seems to stem from the appearance of a true coelom and the consequent development of two distinct layers of mesoderm with greater potential for differentiation of organ systems, including distribution mechanisms.

The two main lines of evolutionary development of more complex animals are distinguished in several ways, one, which is the method of coelom formation.

In protostome coelomates of the annelid – arthropod line, a solid mass of mesoderm originates from a single cell which can be identified early in development. This mesoderm splits to form the coelom, called **a schizocoel (Gk. Schizo, plit; koilos, cavity**).

In this group the blastopore becomes the mouth. Many of the schizocoels have a similar larval form, the trochophore, whose mouth region is derived from the blastopore.

You will find this lesson extremely interesting and important due to the widespread and conspicuous nature of these invertebrates.

In this lesson we will put emphasis on molluscs. The Mollusca are most diverse of all invertebrate phyla and include a whole range of animals, living (over 50,000 species described), and fossils (some 5000 fossil species are known).

At first sight they seem to be so different kinds as to be related. In abundance of described species, the large phylum Mollusca is second to the phylum Arthropoda (about 110,000 to the Arthropoda 850,000). Secondly its members are among the most conspicuous invertebrate group to casual collectors, gardeners, etc and lastly their ecology is very diversified if its members have been able to occupy nearly all kinds of habitats.

They include such forms as the curios plated **chitons (amphineurans**), **the slugs and snails (gastropods), the tooth-shells (scaphopods), bivalves,their possible progenitors (the extinct rostroconchs**) and finally the most complex and highly organized of all molluscs: **the cephalopods**, which include the modern **squids, octopuses, and pearly nautilus and the fossil ammonoids and belemnites**.

The significance of this lesson is that : (i) it presents the extreme variety of mollusks(ii) their morphology and(iii) adaptive traits which have contributed to the success of the group.

Due to the numbers of individuals, number of species, the sort of habitats they occupy and their hefty shells, they are among the most obvious of the worlds' invertebrates.

Objectives

After reading this lesson you should be able to: -

- Explain the diversity of molluscs.
- Outline the classification of main divisions within phylum Mollusca.
- Distinguish between members of the phylum Mollusca.
- Describe the morphology and ecology of molluscs.

Fundamental Organization

A generalized mollusc is an aquatic animal that moves over and grazes on the surface of a hard substratum.

The group has enormous diversity, both in shape and size, ranging all the way from the twisted, wormlike solenogastrids to the baglike octopuses and from minute snails to the giant squids, which are the largest of all invertebrates.

To understand the basic design and to understand the body organization of the major classes of molluscs, we begin by examining the characteristics of a generalized mollusc.

Characteristics

1. Molluscs inhabit marine, freshwater and terrestrial.
2. The typical larva is a modified trochophore type or direct.
3. The body is bilaterally symmetrical with segmentation reduced.

4.The mollusc typically has a head, a visceral mass, a mantle, a shell and a muscular foot. A mantle is an extensive evagination of the body wall, which covers all or part of the body and usually secretes a protective shell. The shell encloses a cavity into which open the anus and kidneys.

5.Molluscs are coelomate but the coelom is often greatly reduced to the pericardial cavity, lumen of gonads and nephridia.

6.The alimentary canal has a muscular buccal mass, a characteristic rasping organ, called **a radula**, salivary glands and stomach into which opens the digestive gland. There may be a crystalline style.

7.There is an open circulatory system.

(Fig. Molluscan body plan)

8.The nervous system is a circumoesophageal ring (which may be condensed into a central mass of nervous tissue, containing cerebral and pleural ganglia), pedal cords, and visceral loops.

9.Gas exchange by gills or lining of the mantle cavity.

10.Excretion by nephridia.

General morphological features

Although each class of molluscs has its own distinctive features of morphology, the basic molluscan plan makes it possible to present certain descriptions of their characteristic organs.

External Morphology

The basic molluscan body plan consists of **a muscular head-foot**, with the part of the body containing the internal organs on top of it. This in turn is covered by **a mantle**, a flattened piece of tissue, which in many species secretes a calcareous shell- i.e contains calcium carbonate.

Between part of the mantle and body lies a mantle cavity, used for gas exchange and in some species also adapted for feeding and locomotion.

The epidermis of molluscs, especially outside the mantle, contains mucigenous cells or mucus-secreting glands. These cells and glands produce slime for locomotion and food-collecting strings. Some molluscs, like the cephalopods, have many **chromatophores,** whereby they can produce the fastest colour changes in the animal kingdom.

The shell is one of the most characeristic structures of molluscs, although it may be reduced or absent altogether in some.

The process of the origin of the shell and development involves many problems of crystal formation and deposition, calcification, and synthesis of components.

There are many variations in its structural organizations among the different taxa.

The formation of the shell involves three major processes: -

(i) The synthesis of the organic matrix of conchiolin (an albuminoid material) and calcium carbonate formation.

(ii) The secretory activity of the mantle in forming the shell components.

(iii) The arrangement of the crystalline layers and the deposition of crystals.

The shell reaches its greatest development in the gastropods and the bivalves.

The molluscan foot is adapted for attatchment to a substratum or for locomotion, or for both of these functions.

The foot has been exploited in various ways and is found in diverse forms and shapes among molluscs.

Its primitive form has been generally an elongated, broad, sole-like structure with successive waves of contraction for creeping locomotion.

Modifications include the attachment disk of the limpets, the compressed hatchet foot of the bivalves and the many divisions of the anterior foot of the cephalopods.

The foot is the only organ of locomotion and corresponds to the ventral body wall of most other animals.

The foot in most molluscs has great powers of expansion and contraction, using the hemoskeleton of blood turgidity together with an extensive system of muscles to change the shape of the foot for adaptive purposes. The action of the foot is aided by the abundant mucus, which can act either as an adhesive or as a mucus sheet over which the foot can glide by ciliary action.

Methods of Locomotion

1. **Creeping movement** brought about by waves of contraction and relaxation.
2. **Bipedal locomotion** involving an alternate extension and contraction of first one half of the foot and then the other.
3. **Leaping movements** by vigorous muscle contractions.
4. **Swimming** by means of lobes or parapodia of the foot, by the whole body, by mantle appendages such as fins or by the highly efficient jet propulsion of cephalopods with lateral fins acting as keels or gliding surfaces

Internal Anatomy

In general, the alimentary canal is composed of :

(i) An anterior buccal cavity and

(ii) Oesophagus in which are the radula and jaws

(iii) The midgut of stomach and liver

(iv) The hind gut or intestine

Feeding Habits

Molluscs have different feeding habits: -

(i)Those living on coarser food use the radula as a rasp. The radula is covered with chitinous teeth used to scrape up food, such as algae on rocks.

(ii)Carnivorous molluscs, such as the cephalopods have powerful jaws for seizing and tearng their prey and the radula to aid in the process.

(iii)Bivalves which have no mandibles, radula or pharyngeal musculature have efficient ciliary-feeding devices.

As in annelids, the excretory organs are nephridia.

The dorsal heart is enclosed in a pericardium.

Usually two auricles receive blood from the gills, and the single muscular ventricle pumps the blood through the body.

The blood vessels are made up of arteries, veins and sinuses.

The formed elements of the colourless blood are represented by a variety of amebocytes and corpuscles.

Many molluscs have copper-bearing haemocyanin as a respiratory pigment although a few have haemoglobin.

Respiration takes place through the surface of the mantle by means of gills or ctenidia, or by a form of lung.

Reproduction

Most molluscs are dioecius. A few are hermaphroditic e.g land snails, nudibranch and some bivalves.

Pulmonates, like earthworms, usuallly practice cross-fertilization.

Fertilization may be external or internal.

In general, gametes are shed into the water where fertilization occurs and the young develop via a trochophore larva.

In others the eggs are incubated in the parents' body.

Certain types of viviparity occur.

Parthenogenesis is non-existent.

Embryologic development resembles that of polychaete worms. Cleavage is spiral and total.

Molluscan larva is compared with the trochophore of the annelids.

However, some species have direct development: the egg hatches as a miniature version of the adult, with no distinct larval stage.

The chitons and scaphopods have large, yolky eggs that form embryos with a prominent prototroch ring of cilia.

Bivalves form a glochidium that is a small version of the adult, but with a ciliated velum and showing much active swimming.

Cephalopods demonstrate direct development to a juvenile within the egg sac.

The phylum Mollusca is made up of seven classes having the same basic plan of organization: -

1. **Class: Amphineura (Polyplacophora)**
2. **Class: Gastropoda**
3. **Class: Bivalvia (Lamellibranchia)**
4. **Class: Cephalopoda**

5. **Class: Monoplacophora**
6. **Class: Aplacophora**
7. **Class: Scaphopoda**

Class: Amphineura (Polyplacophora) e.g Chitons

Basic amphinuran features

Amphineurans (chitons) are bilaterally symmetrical marine molluscs. The oval body is greatly flattened dorso-ventrally. The mouth and anus lie at opposite ends of the body. The body is covered by a calcareous shell composed of eight curved slightly overlapping plates. This is the most distinctive characteristic of chitons. From the nature of the shell is derived the name of the **class Polyplacophora, meaning "bearer of many plates."**

Chitons have become highly adapted for adhering to rocks and shells.

Except for the overlapping posterior edge, the margins of each plate are covered by mantle tissue, but the degree of coverage varies among different species.

The peripheral area of the mantle called **the girdle**, is thick and stiff and extends a considerable distance beyond the lateral margins of the plates.

The girdle surface is covered by a thin cuticle and may be smooth or bear scales, bristles or calcareous spicules.

Chitons possess a simple nervous system which reflects overall bilaterally symmetry of the body and an emphasis on the foot rather than the head. They have no cephalic eyes or tentacles and the head is indistinct.

The mantle is thick and the foot is broad and flat to facilitate adhesion to hard, rocky shores where they are well adapted to withstand the buffeting of waves.

If they are dislodged, they can curl up in a ball like a woodlouse and suffer little damage from being thrown about by the currents.

There are approximately 800 existing species of chitons.

The Kenyan coast has chitons commonly seen on rocky cliffs.

Chitons range in size from 3mm to 40cm, and are commmonly drab shades of red, brown, yellow or green.

Foot and locomotion

The broad, flat foot occupies most of the ventral surface and functions in adhesion as well as locomotion.

Chitons creep very slowly by pedal muscular waves of contraction passing forward in the same manner as snails.

The division of the shell into transverse and their articulation with one another enable the chitons to move over and adhere to a sharply curved surface.

Chitons are common rocky intertidal inhabitants and lie motionless at low tide.

When the rock surface is submerged or splashed, they move about to feed.

They are usually negatively phototactic and thus tend to locate themselves under rocks and ledges.

They are most active at night if they are submerged by the tide.

Nutrition

Most chitons are microphagous feeders of fine algae and other organisms. They scrape from the surface of rocks and shells with their radula. Radula is long and bears rows of 17 recurved chitinous teeth. Radula functions as a scraper during feeding.

The mantle cavity is divided into two channels, one on either side of the foot. Each channel contains a few to many bipectinate ctenidia.

Sexes are usually separate.

Reproduction

Fertilization either external or within the mantle cavity.

Hermaphroditism is found in some members.

Development proceeds via spiral cleavage, a planktonic trochophore larva and direct metamorphosis.

Brooding within the mantle cavity is common.

Class Gastropoda

The **class Gastropoda (“Stomach foot**”) contains about 30,000 existing described species. The class includes marine, freshwater and terrestrial snails, slugs as well as limpets, nudibranchs, cowries, cones, etc.

Gastropods form the largest and most diverse molluscan class and the only one to have successfully colonized terrestral habitat.

This group has had an amazing adaptive radiation, which has enabled them to move into most kinds of ecologic niches and into nearly every area of the earth. They are found in marine and fresh water and on land. The pulmonate snails and several other groups have conquered land by eliminating the gills and converting the mantle cavity into a lung.

Some are adapted for living in desert regions, although most landforms prefer fairly moist habitats. They can occupy such specialized habitats as trees, exposed limestones and coral formations or they can live a parasitic existence. This class therefore shows well how diffent habitats and ways of life can be exploited by animals of one basic design.

Typically the sole is ciliated and provided with numerous gland cells or in the pulmonates, with a large pedal gland. The glands of the foot elaborate a mucous trail over which the animal moves.

Marine species have become adapted to life on all types of bottoms as well as to a pelagic existence and swim with ease.

The body of gastropods like that of other molluscs is composed of four parts – **the visceral sac, the mantle, the head and the foot**. The foot is a flat creeping sole, but it has become adapted for locomotion over a variety of substrataThey have well-defined heads (scanty or absent altogether in many other molluscs,usually with eyes located on the tips of tentacles or at the base and one or two pairs of tentacles and most have elongated, flattened foot by which they creep around.

The basic plan of the molluscan body is bilaterally symmetrical with the mouth at one end and the anus at the other. This bilateral symmetry is modified by a torsion and coiling of the visceral hump. This has resulted in reduction and asymmetry of the vascular system. The class is usually divided into three subclasses –**Prosobranchia, Opisthobranchia, Pulmonata.**

The coiled visceral sac is covered by a mantle, which projects as a free fold at the anterior end and hangs down like a skirt around the head and foot.

Their single shell, which usually assumes many forms, is spirally coiled or a flattened cone is one of their most distinguishing characteristics. The shell is reduced or absent in forms such as terrestrial slugs. (Fig)

The shell covers the main organs of the body, and a head- and-foot forms a base by which the animal moves and feeds. Compared with the chitons (Fig.), the evolution of gastropods has involved incease in size of the soft parts; in limpets the result is a tent-shaped animal (Fig.) while in others the dorsal parts of the body projects upwards and are thrown into a coil.

Shelled gastropods start life by secreting a tiny shell and then add more material around the opening, enlarging the shell as they grow.

When in danger from predator or desiccation, some snails can withdrawn entirely into the shell and close the “door” in a horny operculum located on the upper surface of the foot.

The gastropods have a unique, asymmetric organization brought about by a torsion or twisting process during their embroyonic development (that is found nowhere else among the molluscs).

Viewed from the top, most of the body behind the head twists through 180 degrees counterclockwise over the foot, bring the anus and nephridia around to where they empty on top of the head.

(Torsion is not the same as the spiral coiling of the shell, a separate and earlier evolutionary development).

Several theories have been advanced to explain the relative advantages or disadvantages of torsion to the gastropods.

i.) Some authorites feel that the main advantage of must have been to the larva in its adaptation to pelagic life. When the mantle cavity is twisted anteriorly, the larva is able to retract its head and its ciliary mechanism, the velum, into its mantle cavity and thus drop away from potential predators.

ii.) According to another theory, the main advantage of torsionl must be to the adult. Certainly torsion promotes stability in the adulat by placing the bulky mass of the animal nearer the substratum.

Problems of torsion: Torsion has caused a tendency for the incurrent water to sweep the faeces onto the gills, thus introducing a sanitation problem.

In primitive gastropods this was solved by a slit in the mantle and shell near the location of the anus.

In pectinibranchs the problem is met by an arrangement of a single gill with a row of filaments that produce a lateral through current which carries the faeces away from the gill. Some have lost the gills altogether and use an adaptation of the mantle for respiration, such as the lung in pulmonates.

Most gastropods have gills inside the mantle cavity. However, in some forms the mantle cavity acts as a lung, and the animal obtains oxygen from air instead of from water.

Some gastropods have both gills and lungs.

Like chitons, many intertidal gastropods, such as limpets and periwinkles, always return to the same “home” on a rock when the tide is out. Here they pull the shell tightly down against a groove they have worn in the rock, holding on by a suction of the foot.

Land-living slugs and snails have adaptations of structure and behaviour that conserve water. The mantle cavity, which must be kept moist, has a small external opening, and the animals tend to crawl into the shade or under rocks during the heat of the day.

Internal Anatomy

The alimentary canal varies somewhat with the food habits of gastropods, which are extremely varied. There are herbivores, predators, scavengers, deposit feeders and suspension feeders and even parasites.

The alimentary canal usually consists of:

a) **A mouth** at the snout like end of the head;

b) **A buccal or pharyngeal cavity** that is nearly always provided with a radula and often with cuticular mandibles;

c) A long, slender **oesophagus**, much folded within and some times provided with a muscular **gizzard;**

d) A thin -walled **stomach,** which may have a cuticular lining, and a liver that opens with the stomach by two ducts;

e) An **intestine** that may be long (herbivorous forms) or short (carnivorous forms) and bears a typhlosole ridge; and

f) An **anus** that is usually found on the right side of the body near the head.

Because of torsion the stomach is turned around so that the oesophagus enters at the posterior end and the intestine leaves at the anterior end.

Digestion is mostly extracellular. Digestive enzymes are produced by the digestive glands and oesophageal pouches.

Salivary glands, which open on each side of the radula, mostly secrete mucus for lubricating the radula and for easing food passage.

The stomach has no digestive function of its own but serves to hold the food that is being digested by enzymes from other sources. The products of digestion pass into the ducts of the digestive gland for absorption.

In carnivorous gastropods (many prosobranchs and opisthobranchs) the radula has fewer but larger teeth. Proteolytic enzymes are discharged into the stomach, where digestion takes place rather than in the liver cells.

A few gastropods are parasitic on molluscs and echinoderms. Many of these ectoparasites or endoparasites belong to the Pyramidellacea of the opisthobranchs.

The coelom in gastropods is restricted to the pericardium, kidneys and to some extent, the reproductive gland. The large body cavity is the haemocoel.

Nutrition

Virtually every type of feeding habit is exhibited by gastropods. There are herbivores, carnivores, scavengers, deposit feeders, suspension feeders and parasites. A few generalizations in feeding habits are as follows: -

A radula is usually employed in feeding.

Digestion is always at least partly extracellular.

With few exceptions, the enzymes for extracellular digestion are produced by the salivary glands, oesophageal pouches, the digestive diverticular or a combination of these structures.

The stomach is the siyte of absorption and of intracellular digestion if such digestion takes place.

As a result of torsion, the stomach has been rotated 180 degrees, so the oesophagus enters the stomach posteriorly, and the intestine leaves anteriorly. In the higher gastropods there has been a tendency for the oesophageal opening to migrate forward again toward the more usual anterior position.

In most gastropods the radula has become a highly developed feeding organ, acting as a grater, rasp, brush, cutter, grasper or conveyour.

The total number of teeth varies from ten to a thousand.

The character and form of the radula teeth are often important in classifivaltion.

Herbivores

The many herbivorous gastropods include some marine prosobranchs, the freshwatr prosobranchs, the operculae land snails, a variety of opisthobranchs and a majority of the pulmonates.

Most marine species feed of five algae that can be rasped from a rock or other surfaces or on large algae, such as kelps, that can carry the cot of the snail.

Freshwater and landforms also consume the tender parts of aquatic and terrestrial vascular plants, decaying vegetation, or fungi. A few terrestrial snails and slugs are serious agricultural pests.

Carnivores

A few of the many carnivores' gastropods are pulmonates, feeding on earthworms or other snails and slugs. The majority of carnivorous families are prosobranchs or opisthobranchs. The radula of these marine familes is variously modified for cutting, grasping, tearing, scraping or conveying. Consists of an elongated cartilatigenous base, the odontophore and around its anterior is a stretched membranous belt, the radula proper, which bears rows of recurved chitinous teeth. The most common adaptation of canivorous prosobranchs is a highly xtensible proboscis, which enables the animal to reach and penetrate vulnerable areas of the prey.

Suspension Feeders

The gill filaments have been tremendously lengthened, increasing the surface are that traps plankton on a mucous sheet. Others secrete a net or veil of mucous threads in the vicinity of the shell opening visceral hump. This has resulted in reduction and symmetry of the vascular system. The class is usually divided into three subclasses as follows: -

Gastropod classification:

The morphology and arrangement of the ctenidia and nervous system, the structure of the head, the shape of the foot, and radular morphology have all been used to designate higher groupings within the Gastropoda.

With this background of possible gastropod evolutionary origins, we must now consider the manner in which existing gastropods are classified.

Gastropods are divided into **three subclasses**. The first known as the **Prosobranchia** includes all gastropods that respire by gills and in which the mantle cavity, gills and anus are located at the anterior of the body. In other words, those gastropods in which torsion is clearly evident. Generally aquatic, mainly marine. Separate sexes; often with operculum. With pronounced torsion, crossed visceral loop in the nervous system.Mantle cavity opening. The primitive forms are Archaegastropoda e.g. *Haliotis*. More advanced genera are in Megastropoda e.g *Littorina.* From the Prosobranchia evolved two other subclasses: **the Pulmonata and Opisthobranchia.**

a) **Subclass Opisthobranchia**: Has some 2000 highly diverse species.

Are marine and hermaphrodite.

Characterized by the placement of the mantle cavity and its organs on the right side of the body.

The shell, massive in prosobranchs, is much reduced or even lost and many species have become secondarily bilaterally symmetrical..

Detorsion may take place and the visceral loop of the nervous system is uncrossed.

They may exhibit external bilateral symmetry.

The sea hares and the sea slugs (nudibranchs) are perhaps the most familiar members of this subclass.

Characteristic of most opisthobranchs is a second pair of tentacles called **rhinophores** located behind the first pair rare surrounded at the base by a collar-like fold.

b) **Subclass Pulmonata** – This subclass contains the highly successful land snails as well as many freshwater forms.

This group inhabits freshwater, sea-margin and terrestrial habitats. The more than 16,000 species described are widely distributed in both tropical and temperate regions.

They are hermaphrodite, exhibit torsion and many have a heavy shell, but bear no operculum.

The ctenidium is lost, the mantle cavity has been converted into a lung which opens to the exterior via a small, contractile, pallial aperture, the pneumstome._The gill has disappeared, and the roof of the mantle cavity has been highly vascularized.

The nervous system is symmetrical due to central concentration. The sense organs of gastropods include eyes, tentacles, osphradia and statocysts located in the foot.

The group includes the snails and slugs (which lose the shell) in the terrestrial order.

Stylommatophora (higher pulmonates) – larger group than that containing aquatic forms in which the eyes are at the tip of the posterior, retractile tentacles e.g Helix, etc.

Predominantly freshwater snails are found in the order Basommatophora (lower pulmonates) with tentacles e.g *Lymnae, Biomphalaria, Bulinus, and Planorbis*. Secondarily gills may develop. Gas exchange by diffusion through the pneumostome is important in most pulmonates.

Shell and mantle

The typical gastropod shell is a conical spire composed of tubular whorls and containing the visceral mass of the animal.

Starting at the apex which contains the smallest and the olders whorls are coiled about a central axis (the **columella**); the last and largest whorl (the **body whorl**), eventually terminates at the opening or **aperture**, from which the head and foot of the living animal protude.

The whorls above the body whorl constitute the **spire.**

A shell may be spiraled clockwise or counterclockwise or as it is more frequently stated, displays a right-handed (**dextral**) or left-handed (**sinitral**) spiral. A spiral is right-handed when the aperture opens to the right of the columella (if the shell is held with the spire up and the aperture facing the observer) and left-handed when it opens to the left.

Most gastropods are right-handed, a few are left-handed, and some species have both right-handed and left-handed individuals.

A gastropod shell typically consists of four layers. The outer **periostracum** is composed of a quinone-tanned, horny protein material called **conchiolin**, or **conchin**. (FIG)

Although usually thin, this periostracim may be absent, as in the cowries, or thick and hairy, as in some whelks.

The inner shell layers consist of calcium and carbonate. The outermost calcareous layer is generally prismatic; that is, the mineral is deposited as vertical crystals, each surrounded by a thin protein matrix.

The inner calcareous layers, usually two but sometimes more, are laid down as sheets **(lamellae),** over a thin organic matrix. The sheets are usually oriented perpendicular to the surface, but in some gastropods the sheet of the innermost layer are parallel **nacreous,** or **nacre**, that makes the inner surface smooth and lustrous.

The colour of the shell results from pigments in the periostracum or in the calcareous layers.

The shell is enlarged by the addition of mineral from the outer edge of the mantle to the lips of the aperture.

A constant difference in the rate of mineral deposition along the inner and outer lips results in the characteristic (logarithmic) spiral of the shell.

Growth is usually not continuous and the interval growth lines, as in bivalves (FIG) and by the sculpturing of the shell surface. In most gastropods shell growth declines with age.

Gastropod shells display an infinite variety of colours, patterns, shapes, and sculpturing.

Addendum – Opisthobranchs

The cerata, as well as other parts of the nudibranch body, are usually brilliantly coloured and commonly are red, yellow orange, blue, green or a combination of colours.

The sea slugs, as well as some other opithobranchs, have evolved other defenses in the absence of well-developed shells.

- Escape swimming is a common ability.
- Many have skin glands that produce sulphuric acid or a nonacidic noxious substance that repels potential predators, especially fish.
- Some utilize nematocysts from the prey on which they feed.
- Some have spicules embedded in the mantle.
- The flamboyant colouration of some species of sea slugs probably represents warning colouration; for other it may be camouflage.

Two Modifications in Shell Form are as follows: -

In a considerable number of gastropods, the shell is conspicuously spiralled only in the juvenile stages.

(i) The coiled nature disappears with growth and the adult shell represents a single, large, expanded body whorl. In the abalone, *Haliotis* and in the slipper snails crepidula, the shell remains asymmetrical, but in the limpets, of which there ar a number of unrelated groups the shell has become secondarily symmetrical and looks like a chines straw hat. (Fig)

Bilateralism has been derived in a very different way in the beautiful cowrie shells. Here the last whorl of the shell has completely overgrown the previous whorls and the aperture is greatly narrowed.

(ii) The second modification is shell reduction and shell loss, a condition that has occured many times in the history of gastropods. When the shell is greatly reduced, it often becomes buried within the mantle tissues.

Various species of freshwater pulmonate snails are important hosts for certain human parasites. The African genus *Bulinus,* for example, is the principal host for trematodes causing schistosomiasis.

Shell reduction or loss has occurred independently a number of times within the higher pulmonates and such naked species are called slugs (*Arion, Philomycus, Limax*). The shell is generally absent or reduced and buried within the mantle, but in Testacella a little shell is perched on the back.

The evolution of the slug form is perhaps an adaptive response to low availability of calcium because their original centres of distribution as restricted to areas of high humidity and low soil calcium.

Note: Not all land snails are pulmonates.

Class Scaphopoda:

Contains about 350 species of burrowing marine molluscs popularly known as tusk or tooth shells. These names are derived from the shape of the shell, which is an elongated cylindrical tube usually shaped like an elephants’ tusk. These animals exhibit bilateral symmetry and possess a tubular shell which opens at both ends. The shell may resemble a miniature elephants’ tusk or canine tooth of a carnivore, with a gradual reduction in diameter posteriorly.

The head and foot are a functional unit with a prehensile proboscis used for exploration and digging in the substrate. The animal lives buried in soft bottoms, with the larger anterior end downward and the small, posterior end, through which the ventilating current enters and leaves near the surface of the substratum. They burrrow like protobranch bivalves.

Sense organs are much reduced.

There is a radula in a buccal mass, but there are no ctenidia in the mantle cavity (there being instead a ciliated, respiratory epithelial area). They feed on intersitial organisms collected by means of small tentacles (captacula) and ingested with a radula.

There is no pericaridium, heart or blood vessels.

Scaphopods are dioecious, fertilization is external and development is planktonic. There is both a trochophore and a veliger larva.

Extensively marine, are mostly sedentary animals. They burrow their way through the mud or sand with foot.

Class Bivalvia

The general characteristic of the class Bivalvia, a bivalved shell and a reduced head, largely reflect adaptations for burrowing in soft substrata, although many species have secondarily colonized other epibenthic habitats.

A primary feature, to which many others are related, is lateral compression of the body.

A radula is absent and there is no buccal mass.

The body is flattened between the two valves joined by a dorsal ligament which constitutes a hinge and held together by two large abductor muscles.

Most bivalves can protect themselves against predators or desiccation by retreating into the shell and closing it tightly, but some are too large to fit.

The body is bilaterally symmetrical and strongly compressed laterally.

The edge of the mantle is drawn out, forming two siphons, one for water leaving.

The head is reduced or absent lacking cephalic tentacles, eyes and radula with the mouth located beneath the anterior shell muscle.

Sense organs are located on labial palps, siphons and mantle edge.

The class is one of suspension feeders i.e ciliary filter feeders: cilia on the fills beat and draw a water current in through the incurrent siphon and across the gills, where food becomes trapped in strands of mucus.

The gill cilia then moves these strands to the mouth. The gills are usually very large, having assumed in most species a food-collecting function in addition to that of gas exchange.

Some bivalves supplement filter feeding by deposit feeding, sucking up sediments and digesting the organic matter so obtained, or by housing photosynthetic symbionts in the mantle.

The foot is flattened laterally being mostly used in burrowing and rarely creeping. It is protuded by blood pressure. Foot is reduced, vestigial or absent in forms cemented or fixed by abyssus to the substratum.

Many bivalves are sedentary animals, adhering by the secretion of a foot gland **(byssus)**.

Clams live buried in sand or mud, mussels attached to some substrate by adhesive threads.

A few species are more mobile and can swim for brief periods by clapping the valves of their shells together. Others burrow or creep around using the muscular foot.

The visceral mass is centrally placed and extends into the foot with no coiling or torsion.

The gonad lies in the foot.

The mantle forms two lobes that enclose the body and secrete the shell.

In most bivalves the mantle edges are fused posteriorly to form two siphons which carry water into and out of the mantle cavity.

The ctenidia are large, well-vascularized (facilitating respiration), greatly ciliated, and involved in feeding processes.

Many bivalves are important as “shellfish” for human food, and some are specially cultivated for this purpose.

Pearls are also cultivated, by placing a rough bit of foreign matter, between the mantle and shell of a pearly oyster and waiting several years while the animal covers this core with layers of smooth inner shell material.

Analysis of the adhesives by which bivalves attatch to solid surfaces has led to the invention of better glues for underwater use, including surgery and dentistry.

A number of bivalves are very destructive. “Shipworms” are marine bivalves that use the edges of their shells to burrow into wood.

They eat the sawdust produced, and their symbiotic, nitrogen-fixing gut bacteria digest it, providing both partners with food.

One exotic species, the tiny European zebra clam (zebra mussel), has caused extensive clamage throughout the Eastern United States, clogging cooling systems and water intake pipes and outcompeting native species in many freshwater habitats.

The class Bivalvia contains three major groups, distinguished by the nature of their gills; protobranchs, lamellibranchs and septibranchs. The protobranchs are generally believed to be the most primitive of existing bivalves. The septibranchs are highly specialized. The lamellibranchs include the majority of the bivalve species.

Bivalves are dioecious. Other species are hermaphroditic with a single gonad situated near the digestive gland.

Development: In most bivalves fertilization occurs in the surrounding water.

Fertilization is wholly external or takes place within the mantle cavity. The gametes are shed into the suprabranchial cavity and then swept out with exhalant current. Some bivalves brood their eggs within the suprabranchial cavity, as in some shipworms, or within gills, as in *Ostrea edulus*. Brooded eggs are fertilized by sperm brought in with the ventilating current. The larvae are mostly planktonic and pass through a free-swimming trochophore succeeded by larval veliger stage before is typical of marine bivalves. (FIG)

The veliger is bilaterally symmetrical and eventually becomes a free-swimming trochophore enclosed within the two valves characteristic of bivalves.

Like gastropods, some marine bivalves have long-lived**, planktotrophic** (feeding) **veligers**; others have short-lived, **lecithotrophic** (non-feeding) veligers.

Life Span

Ages of 20 to 3years are known to be common for bivalves, and for certain species there are records of 150-year-old individuals.

The growth stages of commmercial species are well known e.g Oysters (Ostreidae) reach marketable size in 1-3 years depending on the species, latitude and various environmental conditions.

Class Cephalopoda

Cephalopods are primarily swimming molluscs. The cephalopods represent the most specialized of the molluscan group and in many ways are the most advanced of all the invertebrate groups.

Although they possess the basic molluscan pattern of structure, they surpass all other molluscs in: -

(i) Organization

(ii) Cerebral development

(iii) More precise sensory information

(iv) Learning processes, and

(v) Locomotion

The exclusively marine class Cephalopoda contains the **squids, octopuses, cuttlefish** and **nautiluses**. These are specialized for life as active predators and are among the largest, fastest and most intelligent invertebrates.

Cephalolopods appear quite closely related to gastropods, but the body has been rearranged and streamlined.

Characteristics

- The cephalopods are bilaterally symmetrical animals but they have undergone profound rearrangement of their body axes in comparison with other animals.

The anteroposterior axis has become greatly shortened and the dorsoventral axis lengthened. This causes the latter axis to become the dominant functional anteroposterior axis.

As a result of this shift of axes the former ventral surface is dorsal the dorsal surface is psterior, and the posteriour surface is ventral.

- The posterior mantle cavity has become ventral. The mouth is now located in the middle of the foot, the edges of which are modified into grasping arms and tentacles lined with rows of suckers.
- Most existing cephalopods have a very reduced internal shell (squids) or one that is vestigial or none at all (octopuses).

The relict *Nautilus* has a completely developed shell filled with gas that give the animal neutral buoyancy is also represented by fossil cephalopods.

The evolutionary trends of the class have emphasized a vigorous, aggresive mode of life, fast locomotion and predaceous habits.

Among the modifications have been the elimination of the heavy calcareous shell and the formation of fins as an aid to active locomotion.

- The head is well developed and integral with the foot, with complex eyes, cerebral ganglia organized as discrete brain enclosed in a cartiliginous casing.

Sense organs are numerous and large. There is a large buccal mass and powerful radula.

The foot, comprising tentacles and siphon, is associated with the head, and is used for walking (in Octopods) or prey capture (decapods and octopods).

The visceral mass is bilaterally symmetrical, there being no coiling, and the enveloping mantle is muscular and pumps water in and out of the mantle cavity.

- Most cephalopods swim by jerky jet propulsion. They do this by taking water into the mantle cavity when the mantle circular muscles are relaxed and long-muscles contracted . When the mantle cavity is filled by inhalant current the circular muscles relax, the longitudinal muscles contract ,draw the mantle margin tightly around the neck and expel the water through the funnel and squirt it out through a siphon, which the animal can point in various directions to determine which way it will move. As the water is expelled in one direction from the ventral funnel, the animal is propelled in the opposite direction. Water enters the mantle cavity in the neck region between the mantle and the head. When the mantle cavity is filled by inhalant current the muscles relax.

Some squids hold the swimming speed record for invertebrates at 40km per hour.

Octopuses are more adapted for crawling and walking than for active swimming, usually lying in wait for prey near the entrances of their dens in rock crevices.

Cephalopods grasp their fish or invertebrate prey with their arms and tentacles and tear it apart with beak-like jaws.

Some octopuses can inject poison into their prey through their beaks.k

Cephalopods have gills in the mantle cavity and the forceful pumping of water into and out of the cavity during locomotion supplies them with plenty of oxygen. The gills are feather-like, consisting of a central axis with numerous side filaments. Respiration may occur through the body surface in some cephalopods with vestigial gills.

Unlike other molluscs, cephalopods have a closed circulatory system, which transports food and oxygen rapidly and supports their high metabolic rate.

Large eyes are the main sense organs.

Cephalopod and vertebrate eyes show striking convergent evolution; both have a cornea, iris diaphragm, and lens.

The brains' large size correlates with cephalopod's keen senses complex behaviour, control of many appendages and ability to learn and remember.

The digestive system in a U –shaped tube that runs from the mouth to the anus, which opens into the mantle cavity near the siphon.

The vast majority of living cephalopods are members of the subclass Coleoidea. These are dibranchiate (one pair of kidneys and ctenidia).

Have an internal shell.

Eight or ten suckered tentacles.

A funnel, which is a closed tube, eyes with lenses and an ink sac.

Cephalopods are entirely carnivorous, feeding upon crustaceans, fishes and even their own kind. The prey is seized by the suckers or clawlike hooks of the arms. The food is then bitten into pieces by the jaws and passed down the digestive tract to the stomach.

Most cephalopods have pigment cells called chromatophores in their skin and can change colour rapidly by expansion and contraction of different-coloured chromatophores. This ability is used in camouflage, courtship, and other communication. Most cephalopods also have a large ink sac and when attacked eject a cloud of dark ink and shoot off in another direction.

Ecology

Cephalopods are exclusively marine forms and occupy most ecologic habitats, such as littoral, surface pelagic and abyssal.

They are absent from brackish and fresh water and do not tolerate water of reduced salinity.

The active squids are found mostly in the open sea, while the octopuses live in crevices along rocky shores.

They have excellent protective devices against their potential enemies mostly sperm whales and certain fish.

They have remarkable powers to change colours.

The pigment cells are of two kinds, chromatophores and iricocytes which are found in the dermis.

The ability of a chromatophore to change colour is very rapid because of its quickness to contract and expand. Patterns of colour often match the animals' surroundings in a highly effective way.

The sexes are separate and fertilization is internal.

During copulation, the male uses a specialized tentacle to transfer spermatophores from his own mantle cavity to the females.

The female then lays one or a string of fertilized eggs.

Development is direct and the young hatch as miniature versions of the adults. Female octopuses tend their eggs removing debris and squirting water from the siphon over the nest until the eggs hatch

Questions.

1. Describe the structure and function of the following:

a. Radula

b. Pedicellaria

2. Illustrate the structure of a hypothetical ancentral mollusc.
3. Describe torsion in gastropods
4. Write explanatory notes on the gastropod shell.

LOPHOPHORATES

Introduction

In the previous lesson, we dealt with three major (higher) protostomes:- **Annelida, Mollusca** and **Arthropoda**.We now have to deal, in the present lesson, with a group of three even more "independently assorted" minor phyla, united only in being tropoblastic, coelomate and not metameric. The three phyla – **Phoronida, Brozoa(Ectoprocta**) and **Brachiopoda** – possess in common **a lophophore**, which is a variously-shaped distal fold surrounding the mouth, specialized as a food collecting mechanism also used for respiration. The lophophore is characterized by a crown of ciliated tentacles. The tentacles are hollow, each one containing a branch of the coelom derived from the mesocoel. These animals present a mixture of body characteristics of protosomes and deuterostomes i.e. they appear to be near the junction of protostome and deuterostome lines. Adults are all sessile with pooly developed heads, U-shaped guts and a protective outer covering over what is most often a colony of zooids.

Objectives

After reading this lesson you should be able to:

- Explain the diversity of lophophorates
- Identify the shared features of lophophorates
- Outline the characteristics of protostomes and deuterostomes

Like protostomes, the blastophore becomes the mouth, yet like deuterostomes cleavage is radial, early development is regulative

and the coelom has three primary divisions – **protocoel, mesocoel and metacoel**, but owing to the degeneration or reduction of the head associated with a sessile, sedentary existence, the protocoel has disappeared, leaving only the mesocoel and metacoel divisions of the coelomic cavity. Most lophophorates have a septum separating the mesocoel from the metacoel.

Further protostome – like characters are the protonephridial excretory tubules in the larva, the nervous system inside the epidermis and the tube made of chitin.

The mesoderm arises from the gut wall as in deuterostomes but the coelom is formed by splitting as in protostomes. Other shared anatomical features that lophophorates possess are a U-shaped gut, protective coverings and embryology of both protostome and deuterostome characteristics.

Having learned that the common lophophore leaves no doubt of their relationship, each phylum has developed a life style of its own and has followed distinctive lines of specialization. Let us look at the basic morphological characteristics distinctive of each phylum, feeding diversity, reproduction and ecology.

Phylum Phoronida

The phoronids belong to the coelomate protostomia.

They live in an organic closed tube which constitutes an integument with or without adherent elastic particles depending on whether the species burrow into unconsolidated sediment or bore into hard substrates.

Their trochophore type of larva and the origin of the mouth from the blastophore place the phylum among the protostome.

Both bryozoans and phoronids have the same coelomic cavity pattern of mesocoel and metacoel, the same horseshoe arrangement of the lophophore, the same pattern of body wall musculature and the same type of main nervous system.,

Differences are such as the well – developed vascular system and the metanephridia of phoronids..

The **Lophorata** as a group seen to stand somewhere between the Protostoma and the Deuterostomes.

Characteristics

Phoronids are a small group consisting of two genera and 14 species of wormlike animals.

Distributed in tropical and temperate shores.

All members are marine and live within a chitinous tube that is either buried in sand or attached to rocks, shells and other objects in shallow waters.

Structure and Function

- The body of the phoronid is cylindrical, less than 20 cm long.
- They are sedentary animals.
- Lack appendages and regional differentiation except the conspicuous lophophore anteriorly and a bulbous ampulla at the posterior end.
- They are elongate, cylindrical, coelomate animals exhibiting bilateral symmetry.
- The gut is U-shaped, mouth and anus being close together. The lophophore bearing end bears both mouth and anus.
- There is a pair of metanephridia doubling as gonoducts.
- A closed circulatory system is present.
- Respiratory exchange occurs through the surface of the lophophores and oxygen is transported to the body bound to the haemoglobin in the vascular system.

- Phoronid blood contains both colourless blood cells and red blood cells with haemoglobin.
- The tentacles of the lophophores are filled with coelomic fluid that serves as a hydrostatic skeleton holding them upright.
- Like all lophophorates, phoronids are filter feeders.
- The tentacular cilia beat downward creating a water current from which plankton and suspended detritus are collected and entangled in mucous on contact with the tentacles.
- When the animal feeds, the lophophore is projected from the end of the tube.
- Cilia convey the food particles into the mouth. Digestion occurs extracellularly within the oesophagus and stomach.
- The nervous system is not well developed – expected of sedentary animals.
- Movement is sluggish and is usually restricted to travelling down the tube, although when removed from their tube some can burrow into soft sediments and secrete a new tube.
- Phoronids reproduce sexually and asexually, by budding and by transverse fission.
- Approximately half of the species are hermaphrodites. Other species have separate sexes (dioecious).
- The gonad is situated around the stomach in the ampulla.
- Fertilisation is internal and the zygotes are either released into the sea water or protected during early development among coils of the lophophore.
- Cleavage is radial and the blastophore becomes the mouth.
- A free swimming ciliated larva called an **antinotrocha** develops and feeds upon plankton and swims by the action of using a ring of cilia around the anus, the **telotroch.**

Phylum Bryozoa (Ectoprocta)

The ectoproct-bryozoans are coelomate moss-animals with about 4000 living species and about 20,000 extinct forms. It is the largest and most common of the lophophorate phyla.

Characteristics

They occur in fresh water habitats and the majority of species are marine.

They are sessile, solitary or colonial.

Individual possess a circlet of ciliated hollow tentacles surrounding the mouth used in filter feeding. The gut is U-shaped with the mouth and anus opening within the tentacular circle.

The Bryozoan Colony Structure

Most live in colonies that vary greatly in shape, size and organization due to the variety of growth forms.

All ectoproct colonies start as a single zooid, which buds repeatedly to form a colony of interconnected genetic, though not morphological replicates.

Some colonies are small microscopic patches and others are huge masses of zooids. The basic unit of organization in all bryozoans colonies is the **zooid**, which consists of soft tissues within some type of protective exoskeleton that also holds the colony together and helps attach it to the substrate. The exoskeleton is made largely of chitin, protein and calcium carbonate.

Zooids live in extensive, sessile colonies each fastened within a secreted exoskeletal box, or **cooecium**, and feeding by means of an extensile lophophore bearing ciliated tentacles (FIG).

Many genera like *Electra*, form creeping mats while others produce upright branching systems reminiscent of hydroid coelenterates.

A zooid, or the single individual of a colony is composed of two parts, the body wall and the visceral mass.

Live on the bodies or tubes of other animals such as polychaetes, sipunculans, ascidians ,bryozoans, ascidians and sponges. Large members permanently attach to rocks or plants.

Internal Anatomy

- The digestive tract has the typical U-shape of all lophophorates, with the mouth and anus both on the dorsal side (FIG).
- The mouth is located in the centre of the lophophore and opens into a ciliated pharynx which may be followed by an oesophagus.
- The oesophagus opens into the stomach, which is divided into a tubular cardia, a posteriorly elongated caecum, and a narrow pylorus.

  The stomach empties into the intestine or rectum, which opens at the anus just outside the circle of tentacles.

Feeding

Feeding in ectoprocts is carried out by the lophophore, which may be horse-shoe-shaped or circular.

When fully expanded, the lophophore forms a funnel of diverging tentacles surrounding the mouth and leading into it.

Each tentacle is ciliated with two lateral tracts of cilia (facing those of other tentacles) and a third tract of shorter cilia on the median inner surface.

Cilia on the tentacles pass food to the mouth at their base and hence into U-shaped digestive tract.

Digestion

Food particles may be carried down the digestive tract singly or in mucous cords.

Both intracellular and extracellular digestion may occur in the marine species, but only extracellular digestion occurs in the freshwater forms.

Circulation and excretion

The excretory, vascular and respiratory organs are absent in ectoprocts.

Gaseous exchange occurs through the body surface and the coelomic fluid in the spacious coelom and its divisions circulate the products of digestion.

Waste materials are picked up by coelomocytes and other cells and are discharged to the outside. They possess protonephridia (for osmoregulation)

Reproduction

Colonies are hermaphrodite, but produce eggs and sperms at different times within an animal and release into the water, the majority have internal fertilization and brood their eggs into the body cavity.

Asexual reproduction also occurs, chiefly budding from specific region near the mouth or stalk. Larger forms develop attachment stolons and budding occurs from them also.

Not only hermaphroditism, but dioecism, polyembryony, protandry (male then female) and body regeneration. Most are **protandrous** and sperms mature first.

The lophophore and digestive tract may degenerate to make room for one large egg to develop.

The embryo develops into a trochophore larva which is released into the water and rapidly settles onto any other suitable substrate.

Here it finds a new colony, producing other individuals by **budding.**

Phylum Brachiopoda

The phylum contains more than 300 marine lophophorates with two calcareous valves and a shell enclosing the body.

Commonly known as lamp shells and in contrast to other lophophorates and benthic, most brachiopods are large (1-6cm).

All are marine and are found from the intertidial to the deep sea, the majority inhabiting shallow water.

These animals superficially resemble bivalve molluscs in possessing a mantle and a calcareous shell of tow valves (lobes). This encloses nearly all the body and approximate that of small molluscs in size. However, the valves of brachiopods enclose the body dorsoventrally instead of laterally as in bivalve molluscs and the ventral valve is larger than the dorsal one.

The brachiopods are further distinguished by the flexible stalk which protrudes from one valve near the hinge.

Similar to other lophophorates, brachiopods are sessile and most species live attached to rocks or other firm substrates flexible by a cylindrical extension of the ventral body wall called the pedicle. Some forms e.g. lingual, live in vertical burrows in sand and mud bottoms. When a pedicle is absent, they attach themselves by the ventral valve. The lophophore is horseshoe-shaped with the arms often looped or spiralled.

General Morphological features
External Anatomy

The valves are separate but may be hinged at the posterior end.

Each of the two valves is bilaterally symmetrical and is usually convex.

The smaller dorsal shell fits over the large ventral shell, the apex of which in some groups contains a hole like that in a Roman lamp – hence the name “lamp shell”.

The external surface of the valves may be smooth or variously ornamented with folds, growth lines, ribs or spines.

Some bright colours as well as dull ones may be present.

The shell consists of calcium carbonate, calcium phosphate and chitin in various combinations.

The shell is composed of three layers :-

- The outermost **perostracum** of chitin
- The middle **laminated layer** of calcite
- The inner **prismatic** layer of fibres.

Both the internal and external shell characters are of taxonomic importance.

Internal Anatomy

Animals are coelomate.

The coelomic cavity consists of the embryonic mesocoel and metacoel.

The lophophore is an anterior outgrowth of the body wall and lies in the space between the mantle lobes, filling the greater part of that cavity.

The lophophore is fringed with long, ciliated tentacles that occur in a row along the entire edge of the lophophore.

A respiratory system is absent in branchiopods and gaseous exchange may occur principally in the mantle and lophophore.

Oxygen transport is probably provided by the coelomic fluid through the mantle channels and oxygen is carried at least in part by hemerythrin in the coelomocytes.

Brachiopods have a poorly-developed blood-vascular system, but the blood is colourless and contains few cells.

The circulatory system is an open one.

There is a contractile vesicle (heart) located over the stomach.

All brachiopods have one or two pairs of metanephridia located on each side of the digestive tract.

These open internally in the coelom and externally as a nephridiopore.

Waste is picked up by coelomocytes from the coelomic cavity and ejected via the metanephridium into the mantle cavity.

The nervous system consists of an oesophageal nerve ring with a small ganglion on its dorsal side and a larger ganglion on the ventral side.

The mantle margin of brachiopods is probably the most important site of sensory perception.

Reproduction

Brachiopods are dioecious. Gametes mature in the coelom.When ripe, the gametes pass into the coelom and are discharged to the exterior by way of the nephridia.

Eggs are shed into the seawater and fertilization is external at the time of spawning.

Cleavage is radial and nearly equal and leads to a coeloblastula that undergoes gastrulation and invagination.

The blastopore becomes the anus and the mouth forms secondarily, as in deuterostomes i.e. it arises as an outpocket of the archenteron.

The embryo eventually develops into a free-swimming larva.

The inarticulate larva is planktotrophic and resembles a minute brachiopod.

Feeding:

As in other lophophorates, the branchiopod lophophore is basically a crown of hollow tentacles surrounding the mouth. In order to increase surface area, the lophophore projects anteriorly as two arms or branchia, from which the name branchiopoda is derived. In simplest form the lophophore is horse-shoe-shaped, and each arm, or branchium, projects anteriorly.

The arms may be further looped or spiralled in complex ways, greatly increasing the collecting surface area of the lophophore.

Each branchium bears a row of tentacles and the tentacle-bearing ridge is flanked by a branchial groove at its base.

When a branchiopod are ciliary feeders, water with food particles entering along the gaping valves flows over the lophophore.

Summary – Lophophorates

1. Lophophorates are a group of three phyla that possess an anterior, ciliated, tentacular structure, called **a lophophore**. Each tentacle contains an extension of the coelom. The lophophore is used in suspension feeding.
2. The phylum Bryozoa contains minute sessile animals that form colonies of a cm. or more in length and diameter. The majority of the more than 5000 species are marine, but there are some freshwater species.
3. Many of the peculiarities of the bryozoans are related to their miniaturation, colonial organization, and sessile life habit.
4. The body of most species (marine forms) is covered by a chitinous cuticle or a cuticle overlying calcium carbonate. The exoskeleton account for the great fossil record of bryozoans.
5. The lophophore can be retracted within the body encasement. It is protruded by elevation of coelomic fluid pressure, usually produced by compression of some area of the body. The skeletal covering, as well as the colonial organization has restricted the area of possible compression to the frontal surface of most bryozoans.
6. Particles collected by the lophophore are passed into the large stomach, which makes up the greater part of the U-shaped gut. Digestion is both extracellular and intracellular.
7. As might be expected from the minute size of individual zooids, bryozoans lack typical systems for gas exchange and circulation.
8. Some bryozoan colonies are stoloniferous, but most species have erect or encrusting colonies of contiguous zooids.
9. Polymorphism is common, and physiological exchange occurs among zooids of a colony via pores in their walls through which extends a nutrient carrying mesenchymal cord, called the **funiculus.**

10. Bryozoans are hermaphrodites. Their gonads lack gonoducts and gametes break into the coelom and exit by way of tentacular pores (sperm) or by an elevated intertentacular organ (eggs).
11. Some bryozoans are oviparous but most brood their eggs, which undergo biradial cleavage. A lecithotrophic larva is typically present.

Definition of key words

Lophophore_ A circle or horse-shaped fold of the body wall encircling the moutth (but not the anus), bearing hollow tentacles, each containing a branch of the mesocoel. Cilia beat food – bearing water current from the tip of each tentacle to the base, where the water leaves the animal.

Mesocoel- coelomic cavity, the middle of the three divisions of the coelom typically occurring in deuterostomes. Provides the hydraulic system of a lophophore.

Ampulla- a small vesicle containing liquid, for example those associated with the tube feet in many echinoderms.

Zooid- individual within a colony which are joined together, applied to bryoans and a few other animals.

Protonephridia- nephridium with a closed end, as in flame cells. Occurs in platyhelmiths, nemertines, a few polychaetes and larvae.

Hermaphrodite- having both male and female sex organs in one individual, which may be cross-fertilized, or more rarely self-fertilized.

Stolons- root-like part of a colony of animals which fixes it to the substrate (e.g hydroids, Cnidaria, Hemichorda), or as in some annelids and Urochordata) an outgrowth of the body involved in budding.

Protandrous- hermaphrodite in which the male organs develop and function first , then eggs in hermaphrodite animals.

Benthic-living on the seabed, a lake bedd or a stream.

Sedentary- sluggish lifestyle, involving little movement from place to place.

Proboscis- any tubular projection at the anterior end of an animal, whether it is part of the gut or separate from it.

Lecithotrotrophic- feeding on yolk reserves.

Periostracum- external organic layer of calcereous shell.

Archenteron-intrnal cavity, lined at least in part by endorderm, developed in an embryo at the gastrula stage, later giving rise to thegut cavity.

Haemerythrin- rare iron-containing respiratory pigment of some annelids.

Coeloblastula- hollow blastula in which the exteernalwall of cells encloses a fluid- filled cavity, the blastocoel.

Nephridiopore- external opening of a nephridium.

Metanephridium- with an open end, occurring in some worm phyla.Combined with coelomoducts, make the excretory organs of most annelids.

Dioecious- having male and female reproductive systems.

Further Reading

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Questions

1) What are the advantages of parthenogenesis and of asexual reproduction?

2) What groups of invertebrates constitute the lophophorates?

LOWER DEUTEROSTOMES

Phylum: Chaetognatha

The chaetognaths, known as **arrowworm*s*,** are common animals found in marine plankton surface to several hundred metres. They look like tiny transparent arrow as they sprint after their planktonic prey.

The phylum consists of less than 100 species. Most abundant in warm, shallow seas and are influential planktonic predators, feeding on any small plankters from diatoms to juvenile fish. Chaetognaths, in turn, form a major part of the diet of many fish. They are highly sensitive to changes in salinity and temperature, and for this reason are useful indicators of the movements of particular water bodies in the ocean.

External structure

Chaetognaths are all much a like. Differences in the position and number of lateral fins, the shape of the caudal fin and detailed differences in head construction and in the seminal vesicle are the most important taxonomic characters.

The body of an arrowworm is shaped like a feathered dart and ranges in length from 1-10cm (Fig.). The body comprises three regions : a head, trunk and a postanal tail region, a narrowed neck separates the head and trunk.

The trunk and tail bear distinctive lateral and caudal fins, which are a diagnostic feature of the phylum. The fins are used to stabilise the animal and to maintain its position in the water column. The body of a chaetognath is bilaterally symmetrical.

On the ventral side of the rounded head is a large chamber (the *vestibule*) that leads into the mouth. Along the edges of the head hang large, curved chitinous hooks or bristles or grasping/rasping spines used for catching and holding prey:

It is this feature that gives the chaetognaths their name, which means *'bristle-jawed'*. A pair of eyes is located posterioly on the dorsal surface.

Most species have anterior and posterior rows of teeth.

In the neck region is a peculiar fold of the body wall, the *hood*, that can be pulled forward to enclose the entire head between feeding periods. It reduces friction and protects the teeth and bristle (spines) when they are not in use and reduce water resistance during swimming.

During feeding , the hood is retracted, the mouthparts extruded, and the teeth and bristle spread. The snap shut with statling speed to capture prey.

The remainder of the body is composed of the elongated trunk and the tail.

Internal Anatomy

Besides the epidermis and cutile the body wall contains four bands of longitudinal muscles (two dorsolateral and two ventrolateral) and basement membrane on which the epidermis rests and which also forms special plates in the head. No circular muscles.

Coelom is divided into compartments represented by one coelomic space in the head, paired coelomic sacs in the trunk, and one or two coelomic sacs in the tail region. The coelomic spaces are separated by septa.

Chaetognaths do not swim much, but drift passively most of the time. The fins are merely for flotation but play no role in swimming. Swimming is accomplished by movements of the caudal fin and the contraction of the longitudinal body muscles.

The digestive tract is simple. The mouth leads into the bulbous, muscular pharynx that penetrate the head/trunk septum to join a straight intestine with a pair of lateral diverticula at its anterior end and an anus opening ventrally at the septum separating the trunk and tail.

Chaetognaths: They are mostly carnivorous, living on copepods, small worms, larva, eggs, crustaceans and occasionally detrital particles. Prey is captured with the aid of the grasping spines or hooks and pushed into the mouth where it is lubricated by pharyngeal secretions and then passed to the posterior of the intestine.

Here the food is rotated and often moved back and forth until it is broken down. Digestion is probably extracellular. There are no gas exchange or excretory organs, and the coelomic fluid acts as a circulatory medium. Soluble wastes probably escape through the body surface.

The nervous system consists of two principal ganglia, a dorsal ganglion in the head and a large ventral ganglion in the trunk connected by circumoesophageal commissures and nerves.

The chief part of the nervous system is a circumetric ring surrounding the pharynx.

Respiration apparently occurs at the body surface.

Reproduction

Coelomate with an anterocoelic development.

All chaetognaths are *hermaphroditic* with a pair of tubular ovaries in the trunk just anterior to the trunk-tail septum and a bandlike testis on each side of the tail, just behind the septum.

Self -or cross-fertilization occurs via spermatophore. Development is planktonic and direct.

Chaetognathous have radial and equal holoblastic cleavage and form a coeloblastula (a blastula with a small). Species of chaetognaths are taken as indicative of various oceanic and coastal water masses. In British waters, *Sagitta elegans* is typical of phosphate-rich water stemming from the Atlantic and containing some substance that actively encourages larval growth.

Sagitta setosa occurs in phosphate poor water, without the larval-promoting substance and is more typical of coastal waters.

The two species are readily identified by the shape and position of the seminal vesicle.

Phylum Hemichordata

This phylum contains large solitary worms (Enteropneusta) or hydroid-like colonies of tiny zooids (Pterobranchia) inhabiting the sea.

The body and its cavity (coelom) are divided into three regions –an anterior proboscis, a short trunk-reflecting an underlying tricoelomate organization.

The occurrence , or rudimentary expression of three chordate features-notochord, dorsal hollow nerve cord, and gill slits –was formerly the basis for including the hemichordates. This phylum contains animals that resemble echinoderms in their embryology, lophophorates in the possession of hollow tentacles, and chordates in the possession of pharyngeal gill slits.They are exclusively marine and the phylum is made up of three classes –**Enteropneusta,Pterobranchia and Planctosphaerida**

Class-Enteropneusta.

The 70 species of Enteropeusta, or acorn-worms, are primarily benthic ,inhabitants of shallow water.

Some occur in the deep sea and in association with hydrothermal vents. Some live under stones and amongst seaweeds and shells and many common species, including those of *Saccoglosssus* and *Balanoglossus*, construct mucus-lined burrows in mud and sand.

External organization

Acorn-worms are relativelyy large solitary animals.

The majority range between 9 and 45cm in length.The common *Balanoglossus aurantiacus* of Southern-eastre coast of the USA reach approximately 1m in length.

The cylindrical and rather flaccid body is composed of a blunt anterior acorn shaped, *proboscis,* a middle *collar* and a long posterior *trunk*. The trunk bears on each side a longitudinal row of gill slits. These regions correspond to the typical deuterostome body divisions, **protosome, mesosome and metasome**. The proboscis is usually short and more or less conical and is connected to the collar by a slender but sturdy stalk.

The collar is a short cylinder that anteriorly overlaps the proboscis stalk, embraces the posterior end of the proboscis, and contains the mouth on the ventral side.

The trunk constitues the major part of the body and is divisible from anterior to posterior into three regions. The branchiogenital region, the hepatic region and the *intestinal* region.

The branchiogenital region bears a longitudinal row of gil pores on each side of the dorsal midline but no atrium (gill cover). A stream of water enters the mouth, passes into the pharynx, and then exits through the gill pores, driven by cilia. Gas exchange occurs as water passes through the pharyngeal clifts.

Lateral to the two rows of gill pores, the trunk bears either two low gential ridges (*Saccoglossus, Schizocardium*) or two large flaplike genital wings (*Balanoglossus*, ptychodera) which arch over the gill region and enclose a water channel.

A specialized hepatic region follows the branchiogenital region posteriorly and is characterized by numerous fingerlike hepatic sacs (except saccoglossus) that project dorsally from the surface of the body.

The long cyclindrical intestinal region completes the trunk and extends rearward to the termianl anus.

A corn worms are tricoloemate, like most lophophorates and echinoderms. A single unpaired protocoel occupies the proboscis, a pair of mesocoels is found in the collar, and a pair of metacoels is found in the trunk.

Feeding.

The enteropneusts may be deposit feeders, suspension feeders, or both. Burrowing species are predominantly deposit feeders, consuming sand and mud, extract out and digest out the organic matter, a few species protrude the proboscis from the burrow opening and collect suspended material from the water.

Non-burrowing species are primarily suspension feeders but may also ingest deposited material. Food in the form of detritus and plankton, or sediment in burrowing species, is trapped in mucus on the surface of the proboscis and is driven by cilia and transported posteriorly over its surface into the mouth. Within the pharynx, the water is separated from the ingested food, which passes into the posterior part of the gut particles that are not ingested are transported rearward by cilia over the collar and trunk.

The digestive tract is a straight tube that is histologically differentiated into a number of regions.

The blood - vascular system consists of an anterior heart and two main contractile vessels, one middorsal carrying the colourless blood forward, and one midventral, carrying blood posteriorly. Smaller vessels supply the blood wall and major organs.

They have no nephridia.

Acorn worms are dioecious animals that release eggs and sperm into the surrounding seawater where fertilization occurs (i.e. fertilization is external).

The zygotes undergo holoblastic, equal, radial cleavage as shown below.

Gastrulation is by invaginatron, and the blastopore becomes the anus. Development follows the typical deuterostome pattern and leads in many species to **a tornaria** (meaning "turner") larva of which is strikingly like that of echinoderms. This provides the main evidence for assuming a close evolutionary relationship between the hemichordates and echinoderms.

After a planktonic existence, the tornaria larva lengthens and settles to the bottom as a young worm. The hemichordate integument fulfils several functions. (I) forms the outer protective layer of the body, the epidermis, with its highly developed ciliary and mucous cells, forms an essential part of the food capturing and processing devices, which are controlled by the intraepidermal nervous system.

- The intergument can secrete the tubes in which pterobranchs live.

- The integument of pterobranchs is colonized by bacteria, the functions of which are unknown.

HIGHER DEUTEROSTOMES

At the top of the invertebrates is an assortment of small animals, the main ones are the **echinoderms** and the **protochordates**. These ,together with a number of minor groups , are apparently all quite closely related, and most assume a rather passive mode of life on thr sea-floor. Among these lay the group which eventually gave rise to chordates. The protochordates betray their closeness to the chordate line because at some time in their life they possess a rod of turgid cells, the **notochord**. Notochord supports the body and this rod acts in the same way as, and is a precursor to thespinal column of the vertebrates.

Phylum Echinodermata

Structure

Introduction

Objectives

- General Characteristics of the Phylum Echinodermata
- The unique characteristics of the phylum Echinodermata.
- Ecological aspects and adaptations of Echinodermata
- Internal skeleton
- The water vascular system
- Classification of the phylum Echinodermata
- Class Asteroidea
- Class Ophuroidea
- Class Echinoidea
- Lass Crinoidea

- Summary
- Definition of key words
- Further reading
- Questions

Phylum Echinodermata

Structure

Objectives

After studying this topic, the student should be able to: -

- Determine the position of the phylum Echinodermata in the animal kingdom and its biological contributions.
- Describe the characteristics of echinoderms.
- Know the classification of this phylum – the characteristics to the class level.
- Know the structure and functional morphology of the echinoderms.
- Distinguish between members of this phylum.
- Describe briefly the classes of echinoderms.
- Explain special adaptations of echinoderms and their larval biology.
- Know the feeding habits, habitat and reproductive system.
- Know the ecological aspects of importance, habitats and biology.

Introduction

The echinoderms (**echinos- hedgehogt, derma-skin**) are all marine forms and include sea stars, brittle stars, sea urchins, sea cucumbers and sea lilies. Their structure is highly characteristic, with radial symmetry in the adults of most of the representatives and bilateral symmetry in the larvae.

In most echinderms the radial symmetry is **pentamerous** or star-shaped. No cephalisation is found and the body can be divided into an oral and an aboral side. The oral side is usually closest to the substrate.

General Chracteristics of the Phylum Echinodermata

The characteristics of the echinoderms can be summarised under the following headings: i) **Form and size**: Body unsegmented with pentamerous body plan.

Body rounded, cylindrical or star-shaped, with five or more radiating areas or ambulacra, alternating with interambulacral areas. Size varies.

FIG. SEASTAR)ii)**Symmetry:** Primarily bilaterally symmetrical (larva forms), secondarily radially symmetrical (adults).Universal display of 5-rayed symmetry.

(iii) **Germ layers**: Triploblastic, coelom cavity lined with mesoderm.

(iv) **Level of Organization**: Organ level organization.

(v) **Locomotion**: Movement by means of hydraulic organs, the tube feet as the main locomotory type. Other types are movement by spines, crawling, swimming and active movement of arms. Some are sessile.

(vi) **Mode of living:** All are free living, no parasitic forms occur, some are predators or commensals.

(vii) **Body covering and/or skeleton**: Endoskeleton (the name means spiny-skinned) of dermal calcareous ossicles with spines or spiculas. Body covered with epidermis (often ciliated) sometimes with pedicellariae.

viii) **Nutrition /digestion**. Digestive system usually complete, anus absent in ophiuroids.

ix) **Nervous system**. Consists of a circumoral ring and radial nerves.

Usually two or three systems of networks located at different levels in the body.

No brain, sensory system consisting of tactile receptors and chemoreceptors, photoreceptors and statocysts.

x) Excretion. No excretory organs.

xi) Transport system. Blood-vascular system (hemal system) much reduced and included in a unique water-vascular system.

xii) Respiratory system. By means of dermal branchiae, tube feet, respiratory tree(holothuroids) and bursae (ophiuroids) and bursae (ophiuroids).

xiii) Reproduction. Sexes are nearly always separate fertilization being external.

Development is radial through free-swimming,bilateral larval stages which undergo metamorphosis. Follows deuterostomatous type in which the anus is derived from the blastopore.

xv) Body cavity.True enterocoelous coelom, divided in two parts – the perivisceral cavity and the water-vascular system.

xvi) The skeleton is internal and dermal, composed of ossicles of calcium carbonate.

Unique features:

- Tube feet for locomotion through the water-vascular system
- Pedicellariae

The Unique Characteristics

- The system of coelomic channels comprising the water-vascular system.
- The calcereous dermal endoskeleton, the first indication of an endoskeleton.
- Metamorphosis from bilateral larva to radial adult
- The hemal system.

The embryological pattern of development, which is similar to that of the chordata in the following respects:

- The embryonic mouth develops as an ectodermal stomodeum which is connected with the endodermal oesophagus.
- In deuterostomia the anus is formed by the blastopore.

- The coelom is formed as an evagination of the enteron (enterocoela).
- Cleavage is radial and indeterminate.
- The nervous system develops in close contact with the ectoderm.

The Ecological Aspects and Adaptations of Echinoderms Important to Know:

1.) With the exception of some ophiuroid larvae, no parasitic forms occur, some are commensals.

2.) Some are important predators such as the sea stars, which perform an important ecological role as top predators of molluscs and crustaceans.

3.) Echinoderms are an important food source for some animals and serve as a delicacy for humans.

Internal Skeleton

Echinoderms have an internal skeleton (endoskeleton) made of bony plates (ossicles) of calcium carbonate (bicarbonate) ions from seawater.

In some species e.g sea urchin, plates of the skeleton are locked together to form a rigid structure.

On the other hand, most sea stars and brittle stars can flex their arms indicating that the skeleton has gaps and flexible plate junctions.

Sea cucumbers have no real skeleton, only tiny remnant ossicles.

Many echinoderms have spines. In fact, the word Echinodermata means, "**spiny-skinned**."

This refers to the calcareous spines and plates that form a skeleton just under the skin.

The spines also are part of the internal skeleton and are covered by epidermis.

Spines, such s the 40-centimetre long needle-like tropical urchin spines of *Diadema*, are actually internal structures.

The water –vascular system

Water vascular system is another phylum-level characteristic. It has a central coelomic cavity that transports seawater within the animal and bears the tube feet or podia, thin-walled extensions of the epidermis lined with mesoderm and connected with radial canals.

The water vascular system functions primarily in movement.

Oxygen exchange (respiration), molecular nutrient uptake, excretion and sensory perception are also known functions of this system.

The walking tube feet of starfish and sea urchins are connected to an 'ampulla,' a seawater reservoir that can be squeezed to extend the tube feet.

Not all can be extended at once as there would not be enough water inside the animal, nor can all be retracted.

The tips of the tube feet adhere to the substrate by suction in a sea urchin, but in a sarfish by a chemical reaction.

Secretion by the tube feet give rapidly variable adhesion as the starfish walks, since adhesive acidic carbohydrates and proteins alternate with substances of different ionic content that detach them.

Classification of the Phylum Echinodermata

The Phylum Echinodermata consists of 6,000 living species, all of which are marine and free-living.

For symmetrical beauty, the phylum Echinodermata is without match. It contains the familiar sea stars (starfish), brittle stars, sea urchins, worm-like sea-cucumbers, feather stars and sea lilies.

You must know the five major classes of living echinoderms in terms of basic functional morphology, evolution and their characteristics.

It is important to study the form and function, biology and ecological importance of these classes. The following is a brief summary of the classes. Detailed micro and macrophotography reveal special adaptations of echinoderms and their larval biology. Please note that there are three living subphyla of which one is attatched or sessile (Crinozoa).

The two unattached subphyla can easily be distinguished in that one subphylum represents star-shaped echinoderms (Asterozoa) whereas the second one represents globoid, discoid or cylindrical animals without arms (Echinozoa).

The Main Characteristics of Representatives of the Phylum

It is important to study the form and function, biology and ecological importance of these classes.

Class Asteroidea (Sears, see Fig.)

Introduction

This class is represented by the sea stars and demonstrates the basic features of the echinoderm structure and function very well and is therefore often used as an example to demonstrate the features of the Echinodermata.

Skeleton

Water vascular system

Tube- feet function

Feeding

Digestion

Spawning

Larval development and diversity

You are required to study the aspect of form and functions carefully.

Class Asteroidea: -the sea stars, contains about 2000 living species of echinoderms.

They are familiar at the shores either burrowing in sand and gravel or living among rocks. The name ‘starfish’ commonly used in the past is being replaced by the name’sea stars’ recognizing the lack of any close affinity of this group to fishes.

Skeleton

A dried sea star with skin removed shows a meshwork of bony plates and spines, creating the hardened endoskeleton. (Note the calcareous plates or ossicles).

Holes in the endoskeleton reveal where projections of the body wall can be extended out into the seawater to function as oxygen and molecular exchange organs.

These finger-like projections, the papillae increase the surface area, improving gas exchange with the surrounding seawater.

Water-vascular system (note the evolutionary development of this unique system)_

Sea star movement involves hundreds of tube feet, small hollow tubes each tipped with a suction disc, and powered by the water vascular system.

The intake to water vascular system, the madreporite, is a stony sieve plate on the sea stars’ arboral surface. It acts as a screen to keep out detritus and parasites. Animation shows that the water vascular system has a central ring canal and connecting tubes running out into each ring.

Each tube foot is connected to the radial canal and also to a balloon-like ampula.

Tube-feet function

The muscular ampula works like a squeeze bulb, forcing water into the tube foot, causing it to straighten and extend.

The tube foot can supplement its disc suction with a chemical cement that significantly increases holding power.

The tube foot can supplement its disc suction with a chemical cement that significantly increases holding power.

Some sea stars e.g Pacific Coast *Pisaster ochraceus* live in intertidal rocky habitats that experience phenomenal forces equivalent to a wind blowing across the land at 1000km/hour.

- Amazingly, the holding cement applied by the tube foot is reversible, through a biochemical process deployed by the sea star as it moves across the rock.
- Although asteroids (and other echinoderms) have a relatively simple nervous system, lacking a brain, they have the ability to coordinate hundreds or even thousands of tube feet for directional movement.

The Pacific Coast sunflower star, *Pycnopodia*, can have more than 40,000 tube feet.

Sea stars have simple eyespots at the end of each arm, probably used for orientation to light.

They use chemical reception to orient to food resources.

The Ecology of the Pacific coast Common Ochie Star

Having a water vascular system it is relatively intolerant of exposure by low tides of the daily tiday cycles.

Therefore, it generally lives lower down in the intertidal region and moves upward during periods of high tide to feed on its preferred prey species, mussels and barnacles.

Feeding and digestion

Many sea stars are carnivorous and feed on molluscs, crustaceans, polychaetes, other invertebrates and fish. Some are destructive predators of commercially important clams and oysters.

They are predators or scavangers.

Some species are suspension feeders using mucus and cilia.

Sea stars feed by forcing their stomachs from the mouth out of their bodies (eversion) onto, or into, the prey e.g bivalve. The mouth is in the centre of the underside and leading to it are grooves in the underside of each arm. These grooves are avenued by hundreds of tube feet. Then they secrete powerful digestive enzymes to break down the prey in its own body.

Animation reveals that partially-digested nutrients are sucked back into the stomach region or they are brought into the body through active transport.

Once inside, nutrients are transported by ciliary action into digestive glands in each ray.

The food molecules pass into the body fluid for distribution to the skin and other parts of the sea stars' body.

In most sea stars, undigested material is pushed out the oral opening, while some material is eliminated through a small anus located on the top (aboral) surface and hidden by spines except when voiding faeces.

Spawning

Also on top are five gonopores, openings where eggs or sperm are shed directly into the sea.

Echinoderms are usually dioecious, male and female reproductive systems in separate individuals. Have simple reproductive organs with simple gonoducts.

Gametes are shed into water where the eggs are fertilized and grow into transparent larvae.

Larval development and diversity

Larval development goes through several stages on the way to becoming a baby sea star.

Development primitively involves holoblastic, radial cleavage and a series of ciliated larval stages which metamorphose to become radially symmetrical.

After gastrulation, the larval series (including a bipinnaria larva) has a bilateral symmetry of the adult.

(see details of development in section of class Echinoidea)

Animal diversity from sponges to mindworms is discussed with reference to the evolution of multicellular tissue layers, bilateral symmetry and the coelom.

- List the characteristics of animals.
- Draw a phylogenetic free that shows the nine major phyla.
- Describe the way of life and the anatomical features of sponges.

- Describe comb jellies in general and cnidaria (hydras) in particular, including their body forms and other anatomical features.

Question: Tabulate the differences between the ophiuroides and the asteroids.

Class Ophiuroidea (Brittle Stars, see Fig.)

- Class Ophiuroidea
- Regeneration
- Diversity
- Feeding

Contains the brittle stars (also called serpent stars) and the basket stars. Judged on the criterion of numbers of individuals, this is the most successful class of the echinoderms in numbers of species.

The brittle stars constitute the largest group of echinoderms (2000 species) of rather smaller starfish.

Ophiuroidea means snake-like, referring to the form and motion of the arms. Most of the arm consists of central muscle surrounded by ossicles which are able to move on each other and give the arms great flexibility (hence the name 'serpent stars' or 'snake-like').

They have a very much harder skeleton than do asteroids.

The arms of brittle stars are slender and clearly marked off from the disc and are easily broken off, but will regenerate in a few days to weeks.

No pedicellariae or populae.

Sea stars can also regenerate arms that are broken off,but for most species, the central region of the body must remain intact.

Only a few species such as tropical species of Linkia can actually regenerate an entire new body from a piece of arm that has broken off.

The tubefeet of the brittle stars are pointed, so this group does not use suction-mediated movement i.e. tube feet group without suckers used in feeding.

The animal moves by using the arms in a kind of rowing stroke.

The tube feet push materials aside, and also chemically detect food and move it to the mouth on the undersurface (oral side) of the disc.

Brittle stars and basket stars are detritus-feeders and opportunistic scavengers.

They usually collect small pieces of decaying matter that has fallen to the bottom or is drigting near the sea floor.

Ophiuroids are the most abundant class of echinoderms with over 2000 species living in almost every ocean floor habitat.

They have been dredged up from the deepest areas of the ocean.

Class Echinoidea

Is the only one in which the skeleton forms a rigid capsule.

Class Echinoidea includes the sea urchins, heart urchins and sand dollars. (FIG.)

Unlike starfish and brittle stars, sea urchins are slow-moving browsing herbivores.

The basic body-plan of this group involves a rigid endoskeleton with a covering of outward-pointed spines. This is not a shell, but a test, because it is formed inside.

The body is encased in a fixed lattice of ossicles, which may be fused to form a rigid 'test'.

They lack arms. In the place of arms there are five anbulacral grooves, where tube feet emerge between small skeletal plates.

Sea urchins are generally more spherical or egg-shaped and the spines can vary from the extraordinary long and thin spines of tropical diadema, to the stout spines of pencil urchins used for wedging into coral pockets, to intertidal species with flattened spines (for deflecting waves) to other species with short thin spines.

One species of tropical urchin, *Toxopneusta*, has only a few spines but its surface is covered with another defensive structure highly poisonous pedicellaria. (Poison from this pedicllaria has resulted in death for a few people who have handled or encountered these tropical urchins).

Urchin pedicellaria are three-jawed pincer claws (whereas those of Asteroids have two jaws).

Pedicellaria seem to have two main fuctions: -

1.) To discourage small larva from settling on the surface of an echinoderm, and

2.) For defense against predators, (such as the Pacific sunflower star Pycnapodia) red, green or purple sea urchins will move their spines aside and deploy ther multiple pedicellaria to pich the predator. Usually, this will deter the soft-skinned predator.

In cases of attack from these predators, the urchin will direct its spines (which can be aimed outward in any direction), toward the predator.

Urchins are primarily herbivores or detritus feeders, although some actively feed on attached animals such as sponges or sea squirt.

Sea urchins' mouth is located on its underside.

Inside the mouth is a unique jaw structure called the Aristotles Lantern, here revealed through dissection.

The structure is suspended inside the spherical endoskeleton (test) by a rigid frame that also provides attachment for muscles to control the biting and ripping functions of the jaws.

Looking at the urchin test, one can see the rows of holes where the tube feet extend outward from the internal water vascular system, through the endoskeleton and out into the environment.

In many urchin species, the tube feet can be extended outward to distances greater than the length of the spines. This allows the urchin to hold onto food that drifts against the spines and move it to the mouth.

Sand dollar, heart urchins and sea biscuits are echinoids where the endoskeleton is modified with various degrees of flattening.

These represent adaptations for burrowing into different substrates- the highly flattened sand dollar lives in coarse sand, while the more bulbouus heart urchin burrows in soft muds.

In these cases, the animals have also retained a portion of the bilateral symmetry and there is a forward end which leads the burrowing. These burrowing echinoids are detritus-feeders.

Sea urchin development

Forms an excellent one for classroom observation.

Gametes are produced by carefully injecting the fertile animals with potassium ion solutions.

The eggs and sperm are collected in sea water, and the entire process of fertilization and development through larval stages can be observed through the microscope at intervals over several weeks (i.e fertilization, blastula, gastrula and several stages of larval development, remarkable development of larval spines and their transformation at the time of metamorphosis).

Class Echinoidea

Form and function

Note the following : -The skeleton of plates with spines

- The five pairs of ambulacral rows, homologous to the arms of the asteroid with pores through which the tube feet extend.
- The mechanism of spinal movement
- Kinds of pedicellariae
- Mouth with converging teeth, sometimes with branched gills.
- Bilateral symmetry which can be recognized by the position of the mouth and anus.
- Aristotle's lantern or chewing mechanism.

- Ciliated siphon.
- Feeding habits
- Closed ambulacral grooves.
- Peristomial gills and petaloids.
- Reproductive system and echinopleteus larvae.

Knowledge of these features is important for the practical.

Class Holothuroidea

Another group of common echinoderms with over 900 species worldwide.

Like brittle stars, these animals are detritivores, performing the important ecological function of mopping-up the dead and decaying material that falls to the ocean floor.

The common name sea cucumber is appropriate.

Many holuthuroids bear a strong resemblance to our garden cucumbers.

Holothuroidea are greatly elongated in the oral-aboral axis, without arms.

Their worm-like body suggests that they are bilateral creatures, not radially symmetrical.

Their relationship to other echinoderms is shown clearly by an animation that transforms the horizontal sea cucumber body into the vertically oriented body of the sea urchin.

All sea cucumbers have tube feet on the rest of the body either suckered, like those of sea stars and sea urchins or papillae for burrowing (although highly modified). The characteristic feathery features or tentacles around the mouth represent modified tube feet which pick up detritus from the substratum or pick food from the surrounding water.

In many species, the three double-rows tube feet on the upper surface have been lost through evolutionary adaptation. Their main type of locomotion is crawling.

Sea cucumbers have also lost the bulky endoskeleton prominent in other echinoderm groups.

They are the only echinoderms that are able to burrow. This is chiefly due to the fact that they are soft-bodied, with a muscular body wall and reduced and widely spaced ossicles.

The skin is leathery, and remnants of the endoskeleton in the form of microscopic bony plates are embedded inside the skin tissue.

Sea cucumber defenses include powerful toxins secreted by the skin and toxic, and sticky tubes that shoot out the anus of some tropical species (tubules of Cuvier).

Cloacal respiratory trees are unique for this animal group i.e respiration is via the anus, through which water is drawn into special organs within the body called respiratory trees.

In those, which lack such trees, the body wall is generally thin enough for gaseous exchange to take place across it.

Form and Function

Please note the following features:

The characteristics of the body wall and its ossicles.

The presence (or absence) and distribution of tube feet in this group (Note the absence of tube feet in burrowing forms). How is secondary bilaterally shown in this group and how is it shown in the echinoids?

The oral tentacles (modified tube feet).

The importance of the coelomic cavity as a hydrostatic cavity. (Why?)

Respiration is through respiratory trees, skin and tube feet.

The well developed hemal system and the water-vascular system.

The reproductive system. Note the single gonad and the auricularia (free-swimming larvae).

Would you be able to compare the echinoderm classes with regard to their skeleton, respiratory system, type of locomotion, feeding habits, etc?

Biology

Note:-

1.) The feeding behaviour.

2.) Mode of defense by self-mutilation (autotomy).

3.) Commensal interaction with fish.

Class Crinoidea

This group includes the most primitive species of the echinoderms.

The body consists of a tiny central disk which carries the main organs, and a number of long, feathery arms usually subdivided into forks and many branches and is the only one which has retained an upwardly directed mouth.

They are completely or partly sessile and are attatched to the aboral side with or without a stalk.

Crinoids commonly called feather stars, sea lilies are mostly found in warm tropical seas where they attach to corals and other surfaces.

In the first crinoids the disk was anchored to the sea floor by a stem, but during evolution there has been a tendency to lose the stem, and today there are far more of free- living crinoids than stemmed forms.

Unlike other echinoderm groups, the crinoids' mouth and anus are both on the topside of the animal, facing up and they are surrounded by five sets of branching, tentacle-bearing arms that trap suspended detritus and plankton.

When sufficiently disturbed, crinoids may swim by an awkward movement involving alternating flapping of the arms.

The digestive tract has a short oesophagus , an intestine with one mojor loop and a short rectum.

When sufficiently disturbed, crinoids may swim by an awkward movement involving alternating flapping of the arms.

Feather stars are a conspicuous aand colourful component of the fauna of many coral reefs.

Echinoderms evolved during the earliest proliferation of animal life, well over 510 million years ago.

Fossil sea cucumbers have been found in the Burgess Shales and fossil crinoids are common in sediments dated to Devonian times and older.

The class Crinoidea obviously survived until present times, but other classes did not. Paleontologists recognize at least 17 extinct classes of echinoderms.

Form and Function

Please note the following:-

(1.) The body of a crinoid, the tergmen, is supported by a cup-like calcerous skeleton called calyx.

2.) Five flexible arms, branching to form many arms with numerous lateral branchea called pinnules.

3.) Stalk with cirri.

4.) No madreporite, spines or pedicellariae.

5.) Mouth and anus both on the oral side, tube feet are minute and assist in suspension feeding.

6**.) Feeding habits**- a group which feeds on small organisms caught in ambulacral grooves. Ambulacral cilia carry the mucus – entrapped particles down the arms into the mouth. The arms are held as a funnel or, when in a current, as a planar or circular fan. The multiple arms and pinnules provide the necessary area for this mode of feeding.

8.) Nervous and reproductive systems, doliolaria larvae.

How do we know that echinoderms were derived from an ancestor with bilateral symmetry?

Questions

1) Define radial symmetry in Echinoderata.
2) Describe the basic characteristics of the Echinodermata.

3) Enumerate the distinctive features of echinoderms and name four classes of the group giving an example of each.
4) Describe the main structrural features on the oral and aboral surfaces of the class Asteroidea.
5) Describe tha water vascular system of echinoderms.

Phylum Urochordata (Tunicata)

Introduction

The Hemichordates , Cephalochordates and Urochordates are lumped together under the informal heading of protochordates or "first chordates". This lesson is devoted to a consideration of two of the three phyla recognized under protochordates : Urochordata and Cephalochordata. The protochordates are a heterogenous grou of animals possessing chordate characteristics and related to the vertebrates, which they resemble in possessing gill slits, dorsal hollow nerve cord and a notochord or at least traces of it. They do not have a vertebral column and are therefore invertebrates.

Adult urochordates, commonly known as tunicates are common and conspicuous marine invertebrates throughout the world.

The chordate features are distinct only in the free-swimming larvae which resemble minute tadpoles.

Most are sessile and the body is covered by a complex exoskeletal tunic unique in containing cellulose.

There is a highly developed, perforated pharynx, but the notochord and nerve cord are absent in the adult.

The large perforated pharynx is adapted for filter-feeding. A mucous film (the filter) is produced in the endostyle and carried across the inner surface of the pharynx by frontal cilia.

The tunicates include three classes: the Ascidiacea, the Thaliacea and the Larvacea.(We should be aware that other classifications are equally justified e.g those which give the Larvacea more independent status)

The ascidians or sea-squirts contain the majority species, some 1300 species and are the most common tunicates.

The common name (sea squirts) derives from their habit of squirting water and excretory matter through the exhalent siphon.

The bodies of solitary species range from spherical or cylindrical to irregular in shape. One surface is attatched to the substratum, and the opposite bears two openings, the buccal and atrial siphons. All colours are found in ascidians and some colonial species are strikingly beautiful.

Colonial ascidians have very small zooids, although the colony itself may reach considerable size, often a meter or more.

In the simplest colonies, the zooids are spaced apart but are united by stolons.

Adult has no coelom, shows no segmentation and possess no bony tissue.

Dorsal notochord is restricted to the tail and is found only in the larva. Larvae also have a hollow dorsal nerve cord. However, the notochord disappears during metamorphosis to the adult form.

The anterior mouth is connected to the pharynx that is perforated by paired lateral clefts. The clefts lead into another chamber, the **atrium.**

The urochordates are of special interest to biologists, biochemists and morphologists.They constitute a group of primitive chordates which togother with Cephalochordata (**Acrania**) form the Protochordates i.e animals which potentially fore- shadow structures and developments in the more advanced evolutionary level of the higher Chordata.

The integument of urochordates contains cellulose-like compound, a material which is found in the peculiar exracellular layer, the tunic (test).This overlies the ascidian epidermal cells like a manle, and has given rise to the synonym Tunicata.

Phylum: Cephalochordata

The cephalochordates or Acrania are fishlike animals that are commonly called lancelets or individually called amphioxus.

The amphioxus, *Branchiostoma* is the main example.

All cephalochordates are marine and free-living. Approximately 25 species have been found in tropical and temperate oceans worldwide.

They usually lie buried in sand with only their anterior end protruding, but they can also swim very rapidly.

External Anatomy

The cephalochordates are slender, tapering at each end and compressed laterally. The body tapers acutely to a point at both anterior and posterior ends, and it is from this characteristic meaning "opposite points" was coined. The tail is more pointed than the anterior end.

Adult lancelets range in size from 4 to 8cm.

In life, most of the internal organs and tissues can be seen through the pink translucent and iridescent skin.

The body can be divided into a poorly developed head, a trunk, and a tail. (FIG)

The head terminates in a short blunt snout or rostrum, which helps the animal push aside the sand while burrowing.

Immediately posterior to the rostrum, the head bears a ventral mouth that is surrounded by a veil of finger-like sensory projections, the oral cirri.

The bulk of the trunk is occupied by the pharynx and gonads compromising the branchiogenital region.

The body is segmented into V- shaped myomeres, which produce the characteristic lateral swimming undulations resulting from contraction of the muscle blocks or myomeres that are easily visible.

A muscular notochord that extends further anteriorly than in any other chordate prevents longitudinal shortening of the body and controls the lateral flexibility of the body.

Internal Anatomy

The notochord is located directly below the tip of the tail forward into the rostrum. The name cephalochordate, meaning “head chordate” refers to the unique continuation of the notochord anterior to the brain in amphioxus.

The entire structure is enclosed in a fibrous extracellular matrix, the notochord sheath which helps to stiffen the notochord and control its shape. Each notochord cell is a specialized muscle cell in which the contractile filaments run transversely from left to right.

Thus the amphioxus notochord is a muscle that is designed for a skeletal, rather than a motile function.

The excretory organs are segmentally arranged ciliated protonephridia that lie dorsal to certain gill bars and open into the atrium.

The nervous system of amphioxus consists of a tubular nerve cord located dorsal to the notochord. Its anterior end is differentiated slightly but is not expanded to form a brain.

Amphioxus is sensitive to light and to chemical and tactile stumuli, but no elaborate sense organs are present.

The coeloms are numerous and highly specialized to form the fins, myomeres, notochord and other structures.

Three basic chordate characteristics are (1.) dorsal hollow nerve cord, (2.) notochord, and (3.) well-developed pharyngeal gill slits which are a typical in some respects.

Feeding

Lancelets filter feed in a manner similar to that of ascidians using a pharyngeal mucous net. Cilia that cover gill bars pull a current of water into the mouth. Any food or suspended micro-organisms in the current is filtered out by a mucous net secreted by the endostyle, ventral groove in the pharynx.

The endostyle also secretes iodine-containing molecules and is homologous to the vertebrate thyroid gland, which secretes ion-containjng hormone.

Mucus-containing trapped food particles pass down the pharynx into the gut, where the food is digested and the water current is pushed out through the gill slits.

The gill slits do not lead directly from the pharynx to the exterior as in fish, but open into a large atrium, which in turn has an opening to the exterior called atriopore.

Absorbed nutrients are distributed by a circulatory system.

A blood-vascular system is present, but the blood is colourless.

No muscular heart is present; blood is propelled by the contraction of the arteries.

The sexes are separate in amphioxus, and numerous testes or ovaries bulge into the atrial cavities.

The gametes are discharged into the atrium upon the rupture of the gonad walls.

Fertilization and development are external and the zygote hatches early as a small free-swimming version of the adult.

Development occurs in the water. Amphioxus has the deuterostome pattern of embryology involving radial cleavage.

One class of tunicates contains sessile animals, the sea squirts, which are common around the Coast.

The other two classes (**Larvacea and Thaliacea**) contain planktonic forms that are rarely seen because, although common, they are fragile and transparent.

Excretion

Ascidians are ammonotelic animals but have unusual excretory organs that fulfill other excretory functions.

There are no special organs for excretion of nitrogenous wastes. Most escape by diffusion, some e.g uric acid and calcium oxalate (urates) accumulate as inert pigments and crystals and released only on the death of the zooid. This phenomenon is known as **storage excretion**. Pericardia, epicardia and renal sacs can function in storage excretion.

Reproduction

Most tunicates are capable of asexual reproduction by budding. A tunicate bud is called a **blastozooid**, and it arises from the body of the oozoid (the zooid developing from the fertilized egg).

All species are hermaphroditic with a single abdominal ovary and testes. The oviduct and sperm duct open into the atrium.

The tunic, filtering pharynx and hermaphroditism are all correlated with sessility.

Fertilization may be external with planktonic development, especially in solitary species.

Alternatively, fertilization may take place within the atrium and the eggs may be brooded there through early embryonic stages.

Development eventually leads to a tadpole larva (FIG) which possesses all the chordate characteristics.

Following a free-swimming existence of varying duration, the larva settles and attatches by the anterior end.

Metamorphosis involves degeneration of the tail, containing the notochord and dorsal nerve cord.

Differential growth results in rotation of the species to the end opposite the point of attachment.

Corset bearers

Phylum : Loricifera

This is the most recently discovered (1983) phylum of animals. The animals are believed to adhere so tightly to the gravel substrate that they cannot be extracted by conventional methods, which probably accounts for their late discovery.

The major part of the body e.g. in the species *Nanaloricus mysticus* called the abdomen, is encased within a cuticular **lorica** composed of a dorsal, a ventral and two lateral plates. It is to the lorica that the name of the phylum refers (**lorica bearer**).

The cone-shaped anterior end, or introvert, of the animal bears many recurved spines (scalids) on its lateral surface. The field of spines is continuous with that of the neck-like thorax, which joins the introvert with the abdomen.

Both introvert and thorax can be retracted into the anterior end of the lorica. Eight stylets surround the telescopic mouth cone at the end of the introvert, nut their tips open to the side of the mouth and not within it. A terminal rectum opens to the outside through a posterior anal cone. A large brain lies within the introvert, but little is known about the remainder of the nervous system.

The sexes are separate, each with a pair of gonads. A protonephridium is associated with each gonad, reminiscent of the priapulids and it empties into the urogenital duct.

A larval stage, called a Higgins larva, is somewhat similar to the adult except that the buccal cone lacks stylets and the thorax lacks spines.

The thorax can enclose the introvert but cannot itself be retracted into the lorica. The posterior end of the larval trunk carries a pair of toes which are believed to function in swimming.

Questions

1) Which of the following is not a true chordate?

2) a.) Hemichordata

3) b.) Urochordata

4) c) Cephalochordata

5) d.) Vertebrata

6) 2) In Amphioxus excretion is carried out by :

7) a.) Kidneys

8) b.) Nephrocytes

9) c.) Nephridia

10) 3.) State the only definitive characteristic exhibited by adult Ascidia.

11) 4.) State three ways in which protostomes differ from deuterostomes.

Summary

Invertebrates range from tiny protozoans to protochordates. At several points in their long evolutionary history they have reached remarkable heights of achievement: the social organization seen among the arthropods, the extraordinary adaptations of the cephalopod molluscs which so closely parallel those of some vertebrates, and the adoption of a notochord by the protochordates are particularly momentous milestones of invertebrate success. Above all, so many of them display not only fascinating beauty but also remarkable adaptations in their way of life.